

Shoulder girdle and arm

The upper limb is differentiated to achieve complex patterns of stereotactic, non-stereotactic and gestural movements (each with a distinct neurological basis) that enable hand function. The combined motion of the shoulder and elbow brings objects in the hand into the visual field, while the great range of the shoulder and pectoral girdle enable reaching into a wide external environment. Oculospinal afferents appear to be the most important modulators of shoulder joint stability necessary for *reaching*. An object is first reached and then grasped (increasing both superficial and deep peripheral afferent inputs that enhance proximal shoulder girdle stability (Alizadehkhahiyat et al 2011)), before being *retrieved* into the visual field. The requirements of attention to the visual field define the need for stability in the shoulder girdle: the closer the attention to an object in the centre of the field of vision ('macular vision'), the finer is the control of the motion and stability of the shoulder girdle.

The joints of the upper limb become more complex proximodistally. The greatest potential for stability is gained through 'close-packing' (the state of a joint where the greatest stability is achieved for the least energy) in the carpus of the hand, while the range of available motion in each joint becomes more restricted. Close-packing occurs through a lesser, more specific, range of motion in the elbow; the flexed elbow with a supinated forearm is the position for carrying the greatest load with the least energy expenditure or the greatest resistance to fatigue. Close-packing occurs in the glenohumeral joint, so providing for stability, but it is almost entirely gained through muscular activity; at the extremes of range, through capsular fibre spiralling, it is costly in terms of muscular energy expenditure, especially on a background of age- and gene-dependent musculotendinous degeneration. The least intrinsically stable joint in the body, the scapulothoracic, is entirely dependent on muscular activity for stable position and motion around the fulcrum of the acromioclavicular joint.

For the purposes of this chapter, the junction between the 'arm' and the 'elbow joint' will be taken to be the level at which the radial nerve perforates the lateral intermuscular septum to enter the anterior compartment of the arm, and the ulnar nerve perforates the medial intermuscular septum to gain the posterior compartment of the arm at the elbow.

SKIN AND SOFT TISSUES

SKIN

Cutaneous vascular supply

The skin over the lateral end of the clavicle is supplied by the supraclavicular artery, which pierces the deep fascia superior to the clavicle and anterior to trapezius. In the majority of cases, this artery arises from the superficial cervical/transverse cervical artery, but it occasionally arises from the suprascapular artery. The area over deltoid is supplied by the anterior and posterior circumflex humeral arteries. The deltoid branch of the thoraco-acromial axis contributes to the blood supply of the anterior aspect of the shoulder via musculocutaneous perforators through deltoid.

The blood supply to the skin of the upper arm may be divided into three regions with separate supplies. The deltoid region is supplied by musculocutaneous perforators, and the medial and lateral regions are supplied by fasciocutaneous perforators (Cormack and Lamberty 1994; Salmon, in Taylor and Razaboni 1994).

The deltoid region is supplied by the posterior circumflex humeral artery via musculocutaneous perforators. After exiting from the quadrangular space, it gives off a descending branch that runs down to the insertion of deltoid and the overlying skin, and an ascending branch

that passes superiorly towards the acromion and pierces the edge of deltoid and the deep fascia to fan out and supply the overlying skin.

The medial side of the upper arm is supplied by five or six fasciocutaneous perforators that arise from the brachial artery, the superior ulnar collateral artery and, if present, the single artery to biceps brachii. These perforators pass along the medial intermuscular septum to spread out in the deep fascia and anastomose with perforating vessels superiorly and inferiorly, and from the lateral side. There are virtually no musculocutaneous perforators through biceps or triceps.

The lateral side of the upper arm below deltoid is supplied by perforating vessels from the middle collateral and radial collateral arteries (the terminating bifurcation of the profunda brachii). The middle collateral artery sends perforators to the skin via the lateral intermuscular septum between brachioradialis and triceps, while the radial collateral artery gives off cutaneous perforators via the intermuscular septum between brachialis and brachioradialis. These cutaneous vessels anastomose with those from the medial side.

Cutaneous innervation

The skin over the anterior shoulder region is mobile and accommodates the great mobility of the shoulder girdle. The skin over the posterior aspect of the shoulder girdle is thicker and less mobile, being held to the underlying scapula by coarse connective tissue fibres. The cutaneous innervation reflects this difference in skin quality; the supraclavicular nerves (see Fig. 29.1) are long, mobile nerves passing deep to platysma before piercing the muscle and the investing cervical fascia over or immediately caudal to the clavicle. The medial clavicular skin, including the skin over the sternoclavicular joint, is innervated by the small, often multifilamentous, medial supraclavicular nerve. The skin over the clavicle and over a variable extent of the anterior chest skin, including a variable extent of the anterior axillary fold, is innervated by the large,

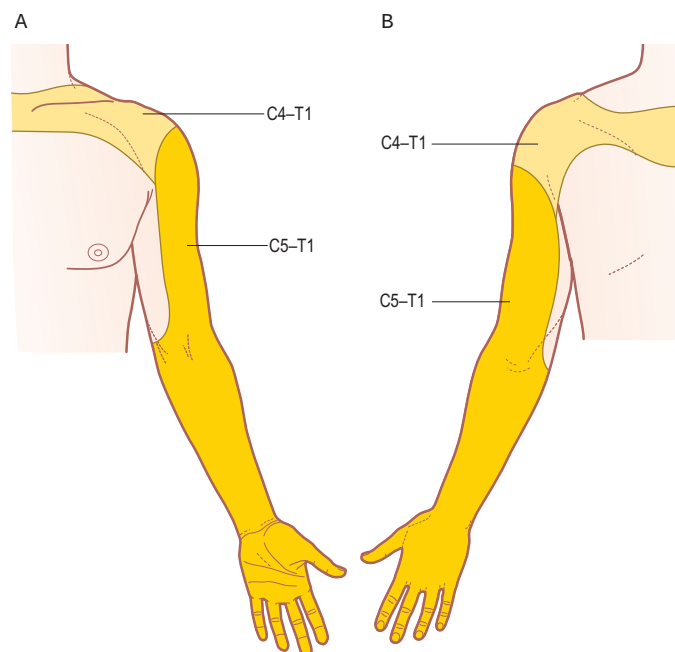


Fig. 48.1 The approximate area of sensory loss or disturbance after proven preganglionic (avulsion) lesion of C4–T1 and C5–T1. **A**, Anterior view. **B**, Posterior view.

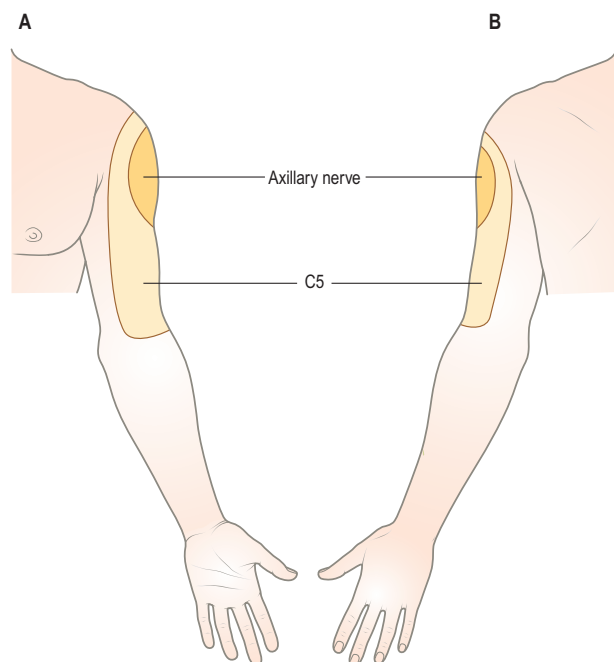


Fig. 48.2 The approximate area of sensory loss or disturbance after complete interruption of C5 and the axillary nerve. **A**, Anterior view. **B**, Posterior view. Sensory loss is greater in C5 lesions than in axillary lesions.

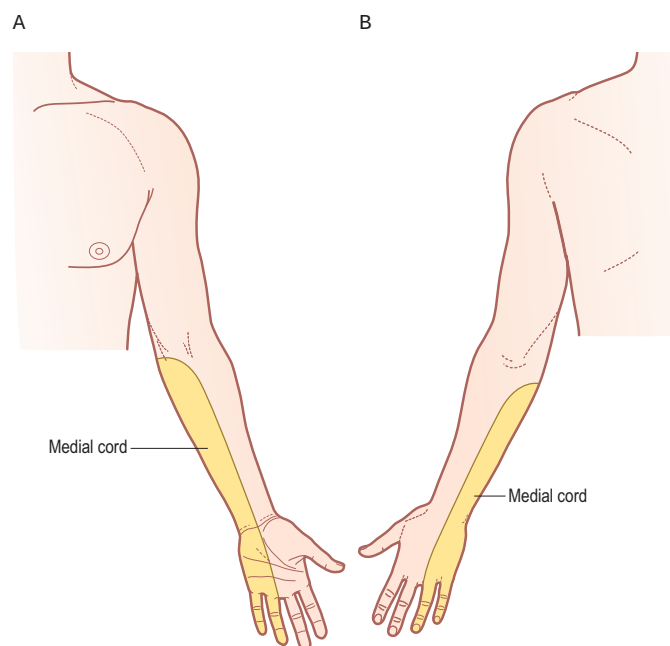


Fig. 48.4 The approximate area of sensory loss or disturbance after complete interruption of the medial cord. **A**, Anterior view. **B**, Posterior view.

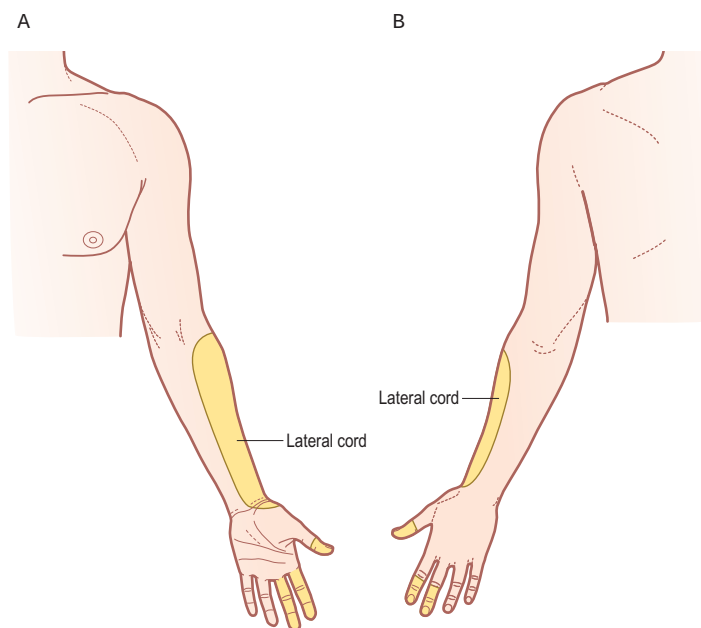


Fig. 48.3 The approximate area of sensory loss or disturbance after complete interruption of the lateral cord. **A**, Anterior view. **B**, Posterior view. Sometimes, loss of sensation is not complete; this is also true of the medial cord.

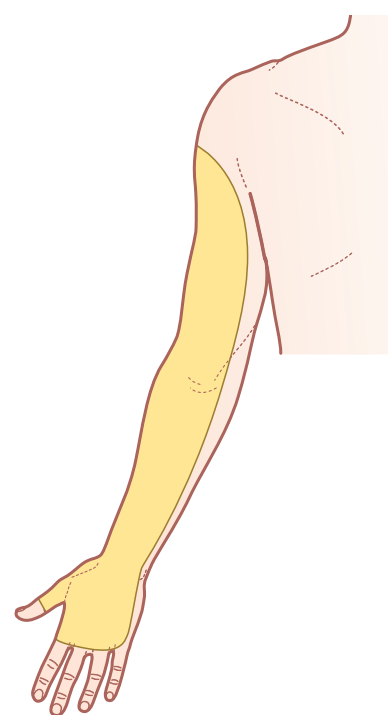


Fig. 48.5 The approximate area of sensory loss or disturbance after complete interruption of the posterior cord. Posterior view.

middle supraclavicular nerve. This nerve is frequently injured by injudicious incisions for the surgical treatment of fractures of the clavicle; a painful neuroma is a common outcome. The skin over the lateral clavicle, acromion and deltoid region is innervated by the lateral supraclavicular nerve or nerves. The floor of the axilla, together with part of the upper medial aspect of the arm, is supplied by the intercostobrachial nerve (lateral branch of the second intercostal nerve). Occasionally, the lateral branch of the third intercostal nerve contributes to the supply of the skin in the floor of the axilla. The lower aspect of the medial side of the upper arm is supplied by the medial cutaneous nerve of the arm. The lateral aspect of the upper arm is supplied by the upper lateral cutaneous nerve (a branch of the axillary nerve) and the lower lateral cutaneous nerve (a branch of the radial nerve). The posterior aspect is supplied by the posterior cutaneous nerve of the arm (a branch of

the radial nerve). **Figures 48.1–48.5** illustrate the approximate areas of sensory loss or disturbance after pre- or post-ganglionic lesions involving C4–T1.

SOFT TISSUES

Deep fascia

In common with locomotor muscles in the lower limb, the upper limb muscles are attached, in varying degrees, to their surrounding deep fascial envelopes. This arrangement increases the area (footprint) of attachment of a muscle within a fascial compartment, so increasing the effectiveness and efficiency of that muscle. For example, the fasciae overlying the three muscles of the scapular fossae (subscapularis, infraspinatus and supraspinatus) are well attached and thicker medially; the fibro-osseous cuff so formed around the medial parts of the muscles affords a greater area of attachment than would be available if

the muscles were attached only to the scapular fossae. The blood supply to the muscles of the upper limb is defined by their fascial compartmentation.

Compartment syndrome of the arm The muscular compartments of the arm (and thigh) are not so well defined as those of the forearm (and leg). Compartment syndrome is not so common in trauma of the arm but remains a concern, particularly after extensive reconstruction procedures and high-energy or crushing injuries of the limb.

Fascia over deltoid

The deep fascia over deltoid is thickest anteriorly and posteriorly, and is adherent in a linear fashion to the external aspect of the superficial fibrous septa of the central (lateral) part of that muscle. It blends with the pectoral fascia in front, and with the thick and strong fascia overlying infraspinatus behind. Above, it blends with the periosteum of the lateral part of the clavicle, the thin periosteum of the acromion, the crest of the scapular spine and the cranial part of the medial border of the scapula, caudal to the spine of the scapula, where it blends with the infraspinatus fascia. Below, it is continuous with the deep brachial fascia, blending with the lateral intermuscular septum (Leijnse et al 2008).

Pectoral and axillary fascia

The pectoral fascia is thin over pectoralis major. It is attached medially to the sternum and is continuous with the fascia of the rectus sheath caudally. Cranially, it blends with the periosteum of the clavicle and the anterior aspect of the capsule of the sternoclavicular joint. Laterally, it is continuous with the fascia over deltoid; it forms the roof of the infraclavicular fossa between the muscular attachments of pectoralis major and deltoid to the clavicle. The fascia is loosely adherent to the septum between the sternal and clavicular parts of pectoralis major. Inferolaterally, between pectoralis major and latissimus dorsi, the fascia thickens to form the floor of the axilla as the axillary fascia. At the caudal edge of pectoralis major, a deep lamina of fascia ascends to envelop the caudal border of pectoralis minor; it becomes the clavipectoral fascia at the upper edge of pectoralis minor. The hollow of the armpit is produced mainly by the action of this fascia in tethering the axillary skin to the floor of the axilla; it is sometimes referred to as the suspensory ligament of the axilla. The axillary fascia is pierced by the tail of the breast (see Fig. 53.22). The pectoral fascia envelops the lateral margin of latissimus dorsi, and the deep and superficial layers then ensheath that muscle and are attached behind to the spines of the thoracic and lumbar vertebrae, blending with the thoracolumbar fascia medially and caudally.

Clavipectoral fascia

The clavipectoral fascia is the cranial continuation of the deep lamina of the pectoral fascia and the medial continuation of the parietal layer of the subscapular bursal fascia. Laterally, it is continuous with the coraco-acromial ligament above and lateral to the coracoid. It envelops the coracoid, short head of biceps brachii and coracobrachialis. It covers the interval between those two muscles and pectoralis minor, which it envelops, and then traverses the interval between pectoralis minor and subclavius. It splits around subclavius and is attached to the clavicle anteriorly and posteriorly to the groove for subclavius. The posterior layer is contiguous with the deep cervical fascia, a condensation of which forms a tether around the central tendinous part of omohyoid, so indirectly connecting it to the clavicle. The deep aspect of the posterior lamina of the clavipectoral fascia blends with the sheath of the axillary vessels. Medially, it blends with the fascia over the first two intercostal spaces and is attached to the first rib, medial to subclavius. Occasionally, the fascia thickens to form a band between the first rib and coracoid process, the costocoracoid ligament, under which the lateral cord of the brachial plexus is closely applied (Atasoy 2004). The cephalic vein, thoraco-acromial artery and associated veins and lymphatic vessels, and the lateral pectoral nerve all pass through the fascia, immediately cranial to the upper border of pectoralis minor.

Subscapular fascia

The subscapular fascia is thin and attached to the entire circumference of the subscapular fossa. Subscapularis is partly attached to its deep surface medially, an example of the extension of the attachment zone of a muscle for more effective action. The fascia extends laterally and blends with the deep layer of the subscapular bursa in front of the tendon of subscapularis and with the external layer of the capsule of the rotator interval of the glenohumeral joint. Inferiorly, it continues as the fascia enveloping teres major.

Infraspinous fascia

The infraspinous fascia covers infraspinatus and is attached to the margins of the infraspinous fossa, except cranially and medially, where there is a loose attachment to the lower aspect of the spine of the scapula. Cranially and laterally, the infraspinatus fascia is continuous with the supraspinatus fascia, creating a single muscular compartment for the two muscles. The deep fibres of deltoid are attached to the infraspinatus fascia over a strip of about 1 cm for the entire length of the spine of the scapula; the examining finger cannot palpate the spine of the scapula from within the space between deltoid and infraspinatus. The fascia is contiguous with the strong lower border of the deltoid fascia along the overlapping posterior border of deltoid.

Supraspinous fascia

The supraspinous fascia completes the osseofibrous compartment containing supraspinatus; it is attached to the scapula around the boundaries of the attachment of supraspinatus. It is thick medially, where supraspinatus attaches to its own fascia, but thinner laterally under the coraco-acromial ligament.

Brachial fascia

The deep fascia of the upper arm, the brachial fascia, is continuous with the fasciae covering deltoid and pectoralis major; it forms a thin, loose covering for the anterior muscles of the arm and a more robust covering for the posterior muscles. Medially, just below the middle of the upper arm, it is perforated by the basilic vein, lymphatic vessels and, at various levels, by branches of the brachial cutaneous nerves.

The fascia is thickest distally, where it contains the brachial muscles in distinct compartments, anteriorly and posteriorly, and defined medially and laterally by tough septa. The lateral intermuscular septum is continuous with the fascia overlying the lateral part of deltoid proximally, and has an upward, thinner extension to the lateral crest of the intertubercular sulcus (groove) contiguous with the fascia over the anterior border of deltoid. It is attached to the supracondylar ridge of the lateral epicondyle of the humerus, and is perforated at the level of the junction of the upper three-fifths and lower two-fifths of the humerus by the radial nerve and the radial collateral branch of the profunda brachii artery passing into the anterior compartment from behind. It provides extension for the zone of attachment of lateral head of triceps posteriorly, and of brachialis, brachioradialis and extensor carpi radialis longus anteriorly. The medial intermuscular septum extends from the medial lip of the intertubercular sulcus, where it is contiguous with the fascia of teres major, then blends with the aponeurosis of attachment of coracobrachialis to the medial aspect of the humerus, and passes as a thick septum along the medial aspect of the humerus to the medial supracondylar ridge of the medial epicondyle. It gives attachment to the medial head of triceps posteriorly, and brachialis anteriorly. It is perforated by the ulnar nerve at about the same level as the radial nerve laterally, together with the superior ulnar collateral artery and the posterior branch of the inferior ulnar collateral artery. At the elbow, the brachial fascia is attached to the epicondyles of the humerus, so completing the muscular compartments, and the olecranon of the ulna, and is continuous with the antebrachial fascia.

Axillary neurovascular sheath

The axillary neurovascular sheath is closely contiguous with the posterior aspect of the clavipectoral fascia. The second part of the axillary artery lies behind pectoralis minor; most intimal ruptures of the vessel caused by distraction trauma occur in this part of the vessel.

Spread of infection



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BONES

CLAVICLE

The clavicle lies almost horizontally at the root of the neck (see Figs 46.7, 29.13). It is a crank-shaped cantilever that carries the scapula, so enabling the limb to swing clear of the trunk. It transmits part of the weight of the limb to the axial skeleton. The lateral or acromial end of the bone is flattened and articulates with the medial side of the acromion, whereas the medial or sternal end is enlarged and articulates with the clavicular notch of the manubrium sterni and first costal cartilage. The shaft is gently curved and resembles the italic letter *f* in shape, being convex forwards (the antecurve) in its medial two-thirds and

During forceful flexion/adduction/protraction of the upper limb across the trunk (e.g. when the scapula is wrenched from the lateral end of the clavicle in thoracoscapular dissociation), the axillary sheath containing the axial vessels and cords of the brachial plexus is forced into pectoralis minor, which acts as a fulcrum or guillotine, around which the vessels angle sharply, so rupturing the arteria intima. Acute neural injuries occur as a result of direct injury (traction, compression, or both) to the nerve trunks, causing acute sensorimotor deficits, arterial adventitial haematoma or acute aneurysm expanding within the axillary sheath to cause the syndrome of causalgia (rapidly progressive, severe pain in the distribution of the affected nerve trunks with neural deficits). Arterial thrombosis following intimal rupture is associated with late nerve deficits as a result of inadequate neural perfusion, local scarring with distortion of the axillary sheath and its contents, and poor posture of the affected limb.

The neurovascular bundle, enclosed within the sheath, is separated from the subscapular fascia by a virtual space traversed by the subscapular nerves; this permits full excursion of subscapularis without distortion of the neurovascular bundle during normal arm movements. The tip of the coracoid can be likened to the tip of the shaft of a steering oar attached to a vessel (the scapula) tethered within the thick clavipectoral fascia: the effect of pectoralis minor as an antagonist of trapezius in scapular position and motion control (see below) is optimized by the wide attachment of the clavipectoral fascia and coraco-acromial ligament to the clavicle and scapula (see [Fig. 48.18](#)).

When axillary suppuration occurs, the local fascial arrangement affects the spread of pus. Suppuration may be superficial or deep to the clavipectoral fascia. In the former, an abscess would appear at the edge of the anterior axillary fold or in the groove between deltoid and pectoralis major; in the latter, pus would tend to track upwards in the axillary neurovascular sheath and appear at the root of the neck, taking the direction of least resistance. Lymphangitis in the medial aspect of the arm suggests infection deep to the clavipectoral fascia with lymphatic obstruction. When an axillary abscess is incised, the knife should enter the axillary 'base', midway between the anterior and posterior margins and near the thoracic side, to avoid the lateral thoracic, subscapular and axillary vessels on the anterior, posterior and lateral walls, respectively.



Fig. 48.6 The left clavicle, showing sites of muscular and ligamentous attachment and articular surfaces. **A**, Superior view. Key: 1, sternocleidomastoid (clavicular head); 2, sternal end; 3, pectoralis major; 4, trapezius; 5, acromial end; 6, deltoid. **B**, Inferior view. Key: 1, pectoralis major; 2, for costoclavicular ligament; 3, for first costal cartilage; 4, for sternum; 5, sternohyoid; 6, subclavius; 7, deltoid; 8, for acromion; 9, trapezoid line; 10, trapezius; 11, conoid tubercle.

concave forwards (the retrocurve) in its lateral third (Fig. 48.6). The inferior aspect of the intermediate two-fifths is grooved in its long axis for the attachment of subclavius. Laterally, the internal architecture of the clavicle is trabecular; there is a medullary cavity in its medial two-thirds. The cortical bone is thickest at the transition zone between the antecurve and retrocurve. The female clavicle is typically shorter, thinner, less curved and smoother. Mid-shaft circumference is the most reliable single indicator of sex; a combination of this measurement with weight and length yields more reliable and consistent results. The clavicle is subcutaneous throughout its whole length and accordingly can be palpated for its entire length; the medial (sternal) end forms an expanded, blunt-faced margin of the jugular notch. The anterior border of the lateral (acromial) end is not readily distinguished from the acromion; the attachment of deltoid obscures the anterior aspect of the acromioclavicular joint. By contrast, the posterior border of the lateral end forms the anterior margin of a roughly parabolic hollow under the lateral fibres of trapezius, the posterior border being formed by the anterior border of the spine of the scapula. The hollow will admit at least one palpating finger. If the pulp of the finger is placed on the spine of the scapula as far laterally as the hollow will permit, the nail will inevitably point to the posterior capsule of the acromioclavicular joint; this landmark is always palpable, even in well-muscled individuals, when other shoulder landmarks can be obscured.

Lateral two-fifths

The lateral two-fifths of the clavicle are flattened and have a superior and an inferior surface, limited by an anterior and a posterior border. The anterior border is concave, thin and roughened, and may be marked by a small deltoid tubercle. The posterior border, also roughened by muscular attachments, is convex backwards. The superior surface is roughened near its margins but is smooth centrally, where it can be felt through the skin. The inferior surface presents two obvious markings. Close to the posterior border, at the junction of the lateral fourth with the rest of the bone, a prominent conoid tubercle gives attachment to the conoid part of the coracoclavicular ligament. A narrow, roughened strip, the trapezoid line, runs forwards and laterally from the lateral side of this tubercle, almost as far as the acromial end (see Fig. 48.6B). The trapezoid part of the coracoclavicular ligament is attached to it. A small, oval articular facet, for articulation with the medial aspect of the acromion, faces laterally and slightly downwards at the lateral end of the shaft.

Subclavius is attached to the groove on the inferior surface (see Fig. 48.6B). The clavipectoral fascia is attached to the edges of the groove. The posterior edge of the groove runs to the conoid tubercle, where fascia and conoid ligament merge. There is a laterally inclined nutrient foramen lateral to the groove. Deltoid (anterior) and trapezius (posterior) are attached to the lateral two-fifths of the shaft; both muscles are directly inserted into the clavicle but are also indirectly inserted into the clavicular periosteum. The coracoclavicular ligaments, attached to

the conoid tubercle and trapezoid line (see Fig. 48.6B), transmit the weight of the upper limb to the clavicle, and are counteracted by the cervical part of trapezius, which supports its lateral part (see Fig. 48.18).

Medial three-fifths

The medial three-fifths of the shaft of the clavicle are cylindrical or prismoid in form and have four surfaces, although the inferior surface is often reduced to a mere ridge. The anterior surface is roughened over most of its extent but is smooth and rounded laterally, where it forms the upper boundary of the infraclavicular fossa. The upper surface is roughened medially and smooth laterally. The posterior surface is smooth and featureless medially; its lateral half bears a groove in the long axis of the bone. The inferior surface is marked near its sternal end by a roughened oval impression, which is often depressed below the surface. Its margins give attachment to the costoclavicular ligament, which connects the clavicle to the upper surface of the first rib and its cartilage. Rarely, this area is smooth or raised to constitute an eminence that may form a synovial pseudarthrosis with the upper surface of the first rib; in older subjects, an extra-articular synostosis may form.

The medial three-fifths provide attachment, anteriorly, for the clavicular head of pectoralis major. The clavicular head of sternocleidomastoid is attached to the medial half of the superior surface, but the marking on the bone is not conspicuous. The smooth, posterior surface is devoid of muscular attachments, except at its lower part immediately adjoining the sternal end, where the lateral fibres of sternohyoid are attached. Medially, this surface is related to the lower end of the internal jugular vein (from which it is separated by sternohyoid), the termination of the subclavian vein, and the start of the brachiocephalic vein. More laterally, the clavicle arches in front of the trunks and divisions of the brachial plexus and the third part of the subclavian artery. The thyrocervical trunk and its branches, the suprascapular and transverse cervical vessels, are immediately behind and above the upper aspect of this surface. Subclavius is inserted in the subclavian groove on the inferior surface; the clavipectoral fascia is attached to the edges of the groove. The posterior lip of the groove is continuous with the conoid tubercle laterally and brings the clavipectoral fascia into continuity with the conoid ligament. A nutrient foramen is found in the lateral end of the groove, running in a lateral direction; the nutrient artery is derived from the suprascapular artery.

Sternal end

The sternal end of the clavicle is directed medially, downwards and forwards, and articulates with the clavicular notch of the manubrium sterni. The sternal surface, usually irregular and pitted, is quadrangular (sometimes triangular). Its uppermost part is slightly roughened for attachment of the interclavicular ligament, sternoclavicular capsule and articular disc. Elsewhere, the surface is smooth and articular, and it extends on to the inferior surface for a short distance, where it articulates with the first costal cartilage. The sternal end of the clavicle projects

upwards beyond the manubrium sterni; it can be felt and seen easily, forming the lateral wall of the jugular notch, behind which are the cricoid cartilage, cricothyroid membrane, the lower part of the thyroid cartilage of the larynx and the brachiocephalic vein. The sternal ends of each clavicle thus form a guide to the jugular notch: if the sternal end of the clavicle is displaced backwards (by traumatic dislocation), the landmarks of the notch are lost, the trachea is displaced and there will be difficulty breathing. If the trachea is displaced to one side (for instance, by pneumothorax), the features of the larynx cannot be palpated readily.

Ossification

The clavicle begins to ossify before any other bone in the body. The shaft of the bone is ossified in condensed mesenchyme from two primary centres, medial and lateral, that appear between the fifth and sixth weeks of intrauterine life, and fuse about the fourth–fifth day. Cartilage then develops at both ends of the clavicle. The medial cartilaginous mass contributes more to growth in length than does the lateral mass; the two centres of ossification meet between the medial three-fifths and lateral two-fifths of the clavicle. A secondary centre for the sternal end appears in the late teens, or even early twenties, usually 2 years earlier in females (Fig. 48.7). Fusion is probably rapid but reliable data are lacking. An acromial secondary centre sometimes develops

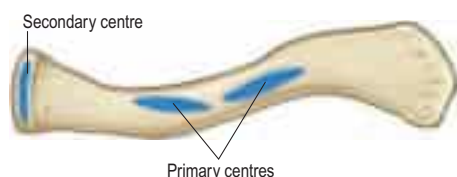


Fig. 48.7 The three constant centres of ossification of the clavicle.

at around 18–20 years, but this epiphysis is always small and rudimentary, and rapidly joins the shaft. Growth is usually complete by 23 years in both sexes.

SCAPULA

The scapula is a large, triangular bone that lies over the posterolateral chest wall, covering parts of the second to seventh ribs (see Fig. 46.7), with a vertical long (craniocaudal) axis. It has costal and dorsal surfaces; superior, lateral and medial borders; inferior, superior and lateral angles; and three processes: the spine, its continuation – the acromion, and the coracoid process (Figs 48.8, 48.9).

The superior and lateral borders and the supraspinous and infraspinous fossae converge laterally at the lateral angle of the scapula. This region comprises the glenoid fossa, the coracoid process and the neck of the scapula. Thus, three robust columns of bone converge at the neck region: the lateral border of the spine of the scapula, the coracoid, and the lateral border of the scapula. Load applied through the glenoid fossa or through the coracoid is transmitted into the scapula through these columns, which also provide a robust framework for the scapular body. The main processes, and thicker parts of the scapula, contain trabecular bone for load-bearing; the rest consists of a thin layer of compact bone for muscular attachments.

Costal surface

The costal surface, which is directed medially and forwards when the arm is by the side, is gently concave, especially in its upper part, matching the contour of the chest wall (see Fig. 48.8B).

Near the lateral border, there is a longitudinal rounded ridge, prominent near the neck but less so below, which is separated from the lateral border by a narrow, grooved area. Subscapularis arises from the subscapular fossa that comprises nearly the whole of the costal surface,

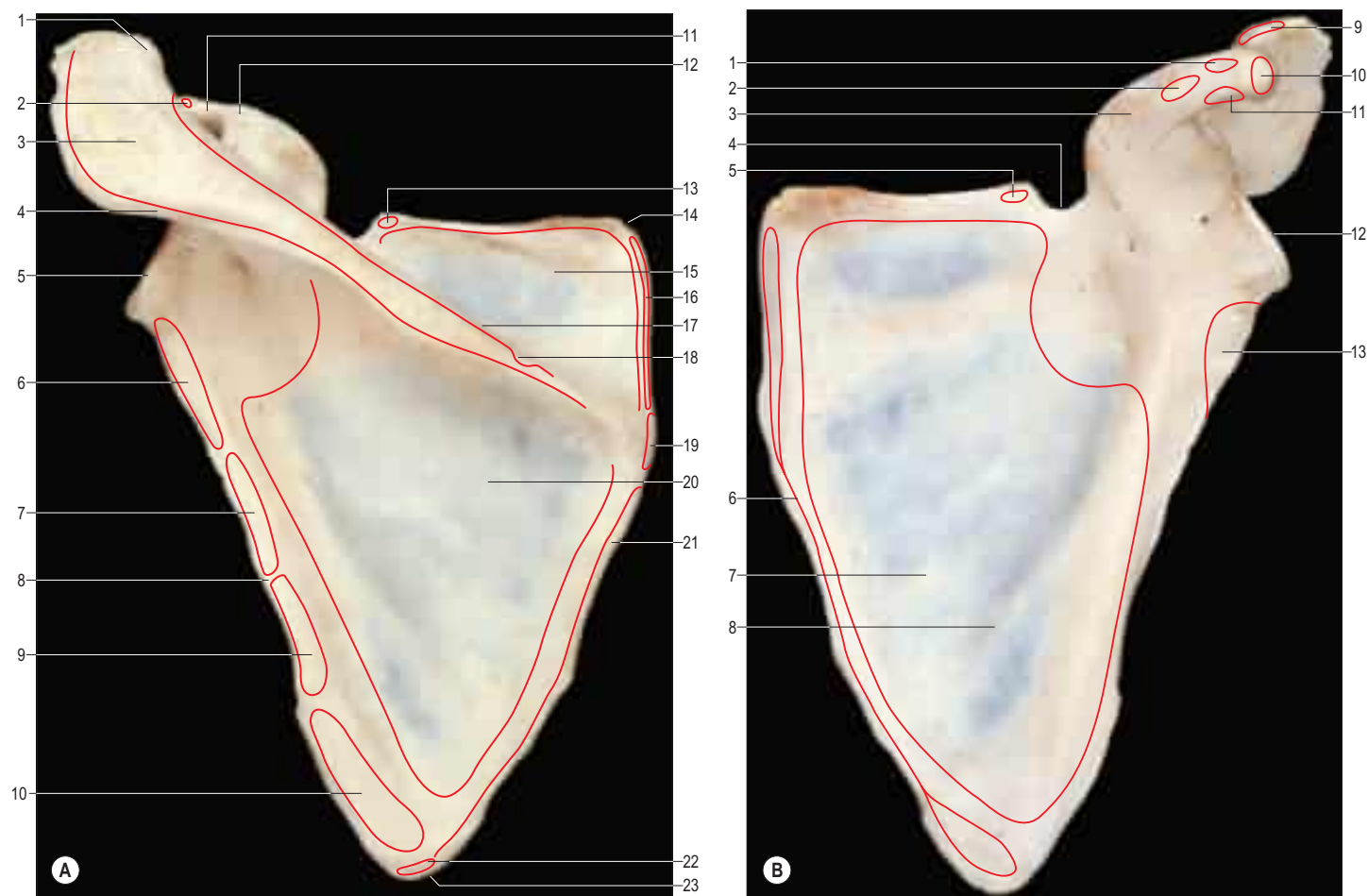


Fig. 48.8 The left scapula, showing sites of muscular and ligamentous attachment and articular surfaces. **A**, Posterior aspect. Key: 1, clavicular facet; 2, biceps brachii (short head); 3, acromion; 4, deltoid; 5, glenoid fossa; 6, triceps brachii (long head); 7 and 9, teres minor; 8, groove for circumflex scapular artery; 10, teres major; 11, conoid tubercle; 12, coracoid process; 13, omohyoid (inferior belly); 14, superior angle; 15, supraspinatus; 16, levator scapulae; 17, spine; 18, trapezius; 19, rhomboid minor; 20, infraspinatus; 21, rhomboid major; 22, latissimus dorsi; 23, inferior angle. **B**, Anterior aspect. Key: 1, attachment of trapezoid ligament; 2, attachment of conoid ligament; 3, acromion process; 4, suprascapular notch; 5, omohyoid (inferior belly); 6, serratus anterior; 7, subscapularis; 8, ridge for intermuscular tendon of subscapularis; 9, deltoid; 10, biceps brachii (short head) and coracobrachialis; 11, pectoralis minor; 12, glenoid fossa; 13, triceps brachii (long head).

The shaft of the clavicle is usually fractured as a result of a violent impact to the side of the shoulder. The fracture typically occurs at the junction of the lateral two-fifths and medial three-fifths, where the transition from antecurve to retrocurve occurs. The weight of the arm causes displacement of the lateral fragment downwards, inwards and into forward (ventral) rotation. If the fracture heals in this position, the dimensions and shape of the retroclavicular space are altered, particularly during elevation of the arm above shoulder level, and the brachial plexus (and its perfusion) and the subclavian vessels can be distorted (one of the many causes of the syndrome associated with thoracic outlet obstruction).

The clavicle does not ossify exclusively by intramembranous ossification. In 14 mm embryos, the clavicle is a band of condensed mesenchyme between the acromion and apex of the first rib, and is continuous with the sternal rudiment. Medial and lateral zones of early cartilage transformation ('precartilage') occur within this band; intramembranous centres of ossification appear, and soon fuse, in the mesenchyme between them. Sternal and acromial zones become true cartilage into which ossification extends from the shaft. Length increases by interstitial growth of these terminal cartilages; the latter develop zones of hypertrophy, calcification and advancing endochondral ossification like other growth cartilages. Diameter increases by subperichondral deposition in the extremities and subperiosteal deposition in the shaft. Epiphyses are endochondral and probably fuse in the same way as they do in long bones. Clavicle growth and development defects include those of ossification, of morphology (shape) and of length. Defects of ossification in the clavicle and those cranial bones that ossify by intramembranous ossification occasionally coincide, e.g. in cleidocranial dysostosis.

Defects of morphology and length occur in the immature clavicle in birth lesions of the brachial plexus; the lateral two-fifths are affected. In a typical case, the acromial end of the clavicle is hypercurved, ventrally rotated by up to 90° and short. The coracoclavicular ligaments are normal. The coracoid is long and more vertical than usual; the scapula is often small, and the scapular neck region short with variable glenoid dysplasia. These characteristic defects are predominantly associated with the upper trunk lesion (C5 and C6) that results in a muscular imbalance across the acromioclavicular and glenohumeral joints; the flexor and medial (internal) rotator muscles are relatively short. The short, tight pectoralis minor and coracobrachialis produce a distorting force on the coracoid, which elongates and rotates dorsally. With retro-pulsion of the humeral head posteriorly, the scapula rotates ventrally, particularly if there is posterior glenoid dysplasia. Deltoid shortening creates a downward distraction on the developing acromion, which lengthens and curves distally. The spine of scapula is parallel to the body of the scapula rather than dorsally directed; these deformities cause the acromioclavicular joint to be orientated more horizontally. As a result, the lateral clavicle hypercurves ventrally. The concept here is that dysmorphology is driven by asymmetrical muscle paralysis and partial recovery, and that *form* follows the neural lesion. While poor movement (causing a lack of 'motivation' for growth in *length*) might explain the clavicular shortening, an equally valid explanation might involve a vascular insufficiency of the lateral ossification centre of the clavicle (with a similar insufficiency of the lateral ossification centres of the scapula for the glenoid); both have a contribution from the supra-scapular artery, which is 'at risk' during injuries that lead to brachial plexus palsy. Length is a function of perfusion and shortening follows vascular insufficiency ([Kambhampati et al 2006](#)).

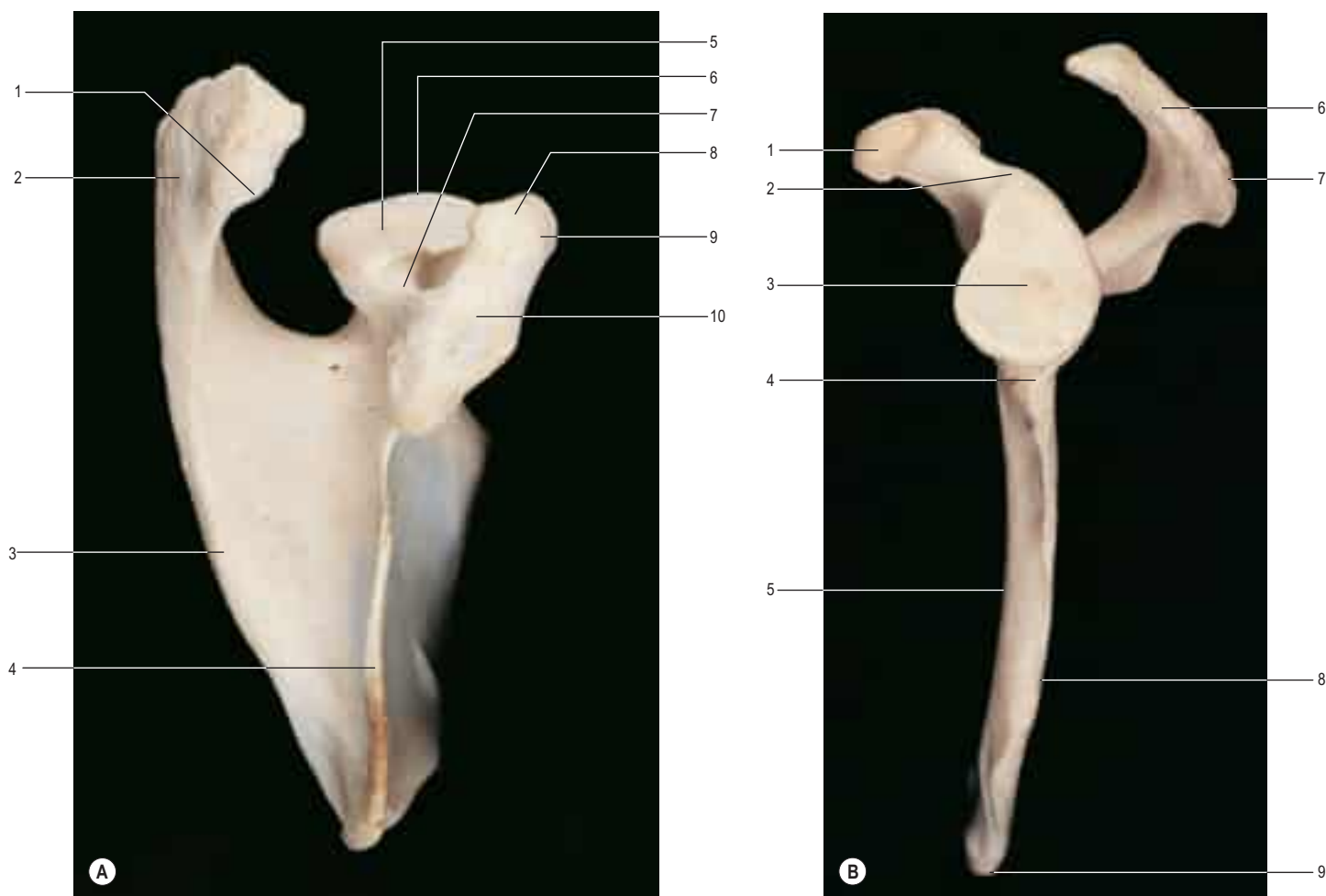


Fig. 48.9 The left scapula, showing sites of muscular and ligamentous attachment and articular surfaces. **A**, Superior aspect. Key: 1, facet for clavicle; 2, acromion process; 3, spine; 4, superior border; 5, head; 6, glenoid fossa; 7, neck; 8, conoid tubercle (for conoid ligament); 9, coracoid process; 10, attachment of trapezoid ligament. **B**, Lateral aspect. Key: 1, coracoid process; 2, supraglenoid tubercle; 3, glenoid fossa; 4, infraglenoid tubercle for long head of triceps brachii; 5, ventral surface; 6, acromion; 7, acromial angle; 8, lateral border; 9, inferior angle.

including the grooved area immediately adjoining the lateral border, but excluding the area next to the neck of the bone. Small, fibrous intramuscular septa are attached to four or five roughened ridges that subdivide this surface incompletely into a number of smooth areas. The anterior aspect of the neck is separated from subscapularis by a bursal protrusion of the synovial membrane of the shoulder joint (subscapular 'bursa'). The lower five or six digitations of serratus anterior are attached to an oval area near the inferior angle. The remainder of the muscle is inserted into a narrow strip along the ventral aspect of the medial border, which is wider above, where it receives the large first digitation. The longitudinal thickening of the bone near the lateral border provides a column of the necessary strength to withstand the pull of serratus anterior on the inferior angle during lateral scapular rotation, when the glenoid cavity is turned to face more directly upwards as the arm is raised from the side and carried above the head against gravity.

Dorsal surface

The dorsal surface is divided by the transverse, shelf-like spine of the scapula into two unequal parts: a smaller supraspinous fossa above and an infraspinous fossa below. The fossae are confluent at the spino-glenoid notch between the lateral border of the spine and the dorsal aspect of the neck; their central parts may be very thin (see Fig. 48.8A). The supraspinous fossa is bounded superiorly and anteriorly by the superior border, from the suprascapular notch and dorsal aspect of the root of the coracoid laterally, to the superior angle medially; and medially, by the upper part of the medial border. The infraspinous fossa is bounded by the medial and lateral borders and the inferior angle.

Supraspinatus is attached to the medial two-thirds of the supraspinous fossa on the dorsal surface; the fascia that covers the muscle is attached to the margins of the fossa. A fat pad lies between the supraspinatus fascia and the under-surface of trapezius. Teres minor is attached to the upper two-thirds of a flattened strip that adjoins the lateral border. The strip is grooved near its upper end by the circumflex scapular vessels passing between teres minor and the bone as they enter

the infraspinous fossa. The lower limit of the attachment of teres minor is indicated by an oblique ridge, which runs from the lateral border to the neighbourhood of the inferior angle and cuts off a somewhat oval area where teres major is attached. The dorsal aspect of the inferior angle may give origin to a small slip that joins the deep surface of latissimus dorsi. The infraspinous fossa is hollowed out laterally but is convex medially. Infraspinatus is attached to the infraspinous fossa, with the exception of an area near the neck of the bone. The strong infraspinatus fascia passes on to teres minor and teres major, and sends fascial partitions between them that reach the bone along the ridges marking the limits of their attachments.

Superior border

The superior border, thin and sharp, is the shortest. At its anterolateral end, it is separated from the root of the coracoid process by the suprascapular notch (see Fig. 48.8B), which can vary in shape and size. Medial to the suprascapular notch, the superior border gives origin to the inferior belly of omohyoid. The notch is bridged by the superior transverse ligament (or suprascapular ligament), which is attached laterally to the root of the coracoid process and medially to the limit of the notch. The ligament is sometimes ossified. The foramen, thus completed, transmits the suprascapular nerve to the supraspinous fossa, whereas the suprascapular vessels pass backwards above the ligament.

Lateral border

The lateral border has a triangular or rhomboidal cross-section. It forms a clearly defined, sharp, roughened ridge that runs sinuously from the inferior angle to the glenoid cavity. At its upper end, it widens into a rough, somewhat triangular, area: the infraglenoid tubercle (see Fig. 48.9B). The lateral border separates the attachments of subscapularis and teres minor and major. These muscles project beyond the bone and, with latissimus dorsi below, cover it so completely that it cannot be felt through the skin. The long head of triceps is attached to the infraglenoid tubercle.

The grooved part of the costal surface, the narrow, flat lateral strip of the dorsal surface and the adjacent thickened ridge (see Fig. 48.9B) are often included in the 'lateral column' in surgical practice. By definition, the remainder of the scapular body (costal and dorsal surfaces) and the medial border are called the 'medial column'.

Medial border

The medial border extends from the inferior to the superior angle. It is slightly thickened and flat, and is further thickened at the medial end of the spine. In its lower two-thirds, this border can easily be felt through the skin, but its upper third is more deeply placed and more difficult to palpate. It is thin and often angled opposite the root of the spine. Levator scapulae is attached to a narrow strip, extending from the superior angle to the root of the spine, and rhomboid minor is attached below this, opposite the root of the spine. Rhomboid major is attached to the remainder of the border.

Scapular angles

The inferior angle lies over the seventh rib, or over the seventh intercostal space. It can be felt through the skin and the muscles that cover it, and, when the arm is raised above the head, it can be seen to rotate forwards (protract) around the chest wall; when the arm is lowered, the scapula should restitute (retract) to its original position without a disturbance of rhythm. It is covered on its dorsal aspect by the upper border of latissimus dorsi, a small slip from which is frequently attached to the inferior angle. The superior angle, at the junction of the superior and medial borders, is obscured by the upper part of trapezius. It lies over the dorsal surface of the second rib and can be palpated deeply above and behind the clavicle.

The lateral angle, truncated and broad, comprises the glenoid fossa (cavity) and scapular neck, with the coracoid projecting forwards. When the arm is by the side, the fossa is directed forwards, laterally and slightly upwards. When the arm is raised above the head, it is directed almost straight upwards, i.e. the plane of the fossa is horizontal. The supraglenoid tubercle is a small, rough, sloping area at the cranial margin of the fossa; it often encroaches on the root of the coracoid process, and is the site of attachment for the tendon of the long head of biceps brachii. The infraglenoid tubercle is a larger, rough area at the caudal margin of the glenoid fossa, at the upper end of the lateral margin, and is the site of attachment for the tendon of the long head of the triceps brachii. The anatomical neck is the region of the scapula extending between the infraglenoid and supraglenoid tubercles anteriorly and posteriorly, lateral to the root of the coracoid process; anteriorly, there is no landmark to distinguish the anatomical neck from the costal surface of the body of the scapula. Posteriorly, the anatomical neck is limited by the spinoglenoid notch at the root of the spine of the scapula. The bone here has an asymmetric, triangular cross-sectional shape; the plane of the anterior surface of the neck makes an acute angle of about 60° with the plane of the glenoid fossa, but the dimension and shape of the posterior surface are more variable.

Spine of the scapula

The spine of the scapula forms a shelf-like projection on the upper part of the dorsal surface of the bone, and is triangular in shape (see Fig. 48.8A). Its lateral border is free, thick and rounded, and bounds the spinoglenoid notch, which lies between it and the dorsal surface of the neck of the bone. Its anterior aspect joins the dorsal surface of the scapula along a line that runs laterally and slightly upwards from the junction of the upper and middle thirds of the medial border. The plate-like body of the bone is bent along this line, which accounts for the concavity of the upper part of the costal surface. The dorsal border is the crest of the spine and is subcutaneous throughout nearly its whole extent. The crest expands into a smooth, triangular area at its medial end. Elsewhere, the upper and lower edges and the surface of the crest are roughened for muscular attachments. The upper surface of the spine widens as it is traced laterally and is slightly hollowed out. Together with the upper area of the dorsal surface of the bone, the upper surface of the spine forms the supraspinous fossa. The lower surface is overhung by the crest at its medial, narrow end, but is gently convex in its wider, lateral portion. Together with the lower area of the dorsal surface of the bone, the lower surface of the spine forms the infraspinous fossa, which communicates with the supraspinous fossa through the spinoglenoid notch.

Supraspinatus is attached to the upper surface of the spine of the scapula; the infraspinatus, by contrast, is not attached to the lower surface of the spine. The flattened triangular area at its root lies opposite the spine of the third thoracic vertebra and is covered by the tendon of trapezius; a bursa intervenes to enable the tendon to play over this part of the bone. The posterior fibres of deltoid are attached to the lower

border of the crest and, to a variable extent, to the medial border by attachment to the infraspinatus fascia. The middle fibres of trapezius are attached to the upper border of the crest. The lowest fibres of trapezius terminate in a flat, triangular tendon that glides over the smooth area at the base of the spine and inserts into a rough prominence, erroneously called the deltoid tubercle, on the dorsal or subcutaneous aspect of the spine near its medial end.

Acromion

The acromion projects forwards, almost at right angles, from the lateral end of the spine, with which it is continuous. The lower border of the crest of the spine becomes continuous with the lateral border of the acromion at the acromial angle, which forms a reliable, subcutaneous, bony landmark. The medial border of the acromion is short and is marked anteriorly by a small, oval facet, directed upwards and medially, for articulation with the lateral end of the clavicle. The lateral border, tip and upper surface of the acromion can all be felt through the skin without difficulty. There may be an accessory articular facet on the inferior surface of the acromion.

The acromion is subcutaneous over its dorsal surface, being covered only by the skin and superficial fascia. The lateral border, which is thick and irregular, and the tip of the process, as far round as the clavicular facet, give origin to the middle fibres of deltoid. The medial aspect of the tip gives attachment, below deltoid, to the lateral end of the coracoacromial ligament. The articular capsule of the acromioclavicular joint is attached around the margins of the clavicular facet. Behind the facet, the medial border of the acromion gives insertion to the horizontal (middle) fibres of trapezius. The inferior aspect of the acromion is relatively smooth and forms a protective arch over the shoulder joint, together with the coracoacromial ligament and the coracoid process. The tendon of supraspinatus passes below the overhanging acromion and is separated from it and from deltoid by the subacromial bursa.

Coracoid process

The coracoid process arises from the upper border of the neck of the scapula and its body is bent sharply so as to project forwards and slightly laterally (see Figs 48.8, 48.9). When the arm is by the side, the coracoid process points almost straight forwards. Its tip can be felt through the skin and is covered by the anterior fibres of deltoid, about 2.5 cm below the clavicle at the junction of the lateral fifth with the rest of the bone, at the lateral border of the infraclavicular fossa. It forms a short, curved cantilever fixed to the scapula at its base and is displaced by the actions of the muscles attached to its body and tip. Pectoralis minor pulls the coracoid (and, therefore, the scapula) forwards and medially around the chest wall, balanced by trapezius, its antagonist; this action forms the foundation on which coracobrachialis can help support the long lever arm of the humerus during humeral motion, together with its antagonist, the deltoid.

The infraclavicular brachial plexus and the axillary vessels lie below and medial to the coracoid. The acromial branch of the thoracoacromial artery passes above it, and the sensory branch of the lateral pectoral nerve to the rotator interval capsule, together with accompanying vessels, lies immediately below.

On the dorsal aspect of the coracoid process, at the point where it changes direction, a rough impression forms part of the region of attachment of the conoid portion of the coracoclavicular ligament. The trapezoid portion of the coracoclavicular ligament is attached to the upper aspect of the horizontal part of the process, anterior to the conoid part. Pectoralis minor is attached to the superior and medial aspects of the coracoid process. The wider, medial, end of the coracoacromial ligament is attached to the lateral border and is continuous inferiorly with the lateral aponeurotic part of the tendon of the short head of biceps brachii. The coracohumeral ligament is attached to the root of the coracoid at its lateral border. The interval between the anterior aspect of the scapular neck immediately medial to the glenoid fossa and the deep surface of the coracoid is often bridged by the glenocoracoid ligament. The inferior aspect of the coracoid process is otherwise smooth and saddle-shaped. When the arm is elevated, the upper border of subscapularis is apposed to, and runs under, this surface, which forms a pulley for the muscle, increasing the power generated during forceful medial (internal) rotation of the shoulder in elevation, such as during a serve in tennis. Coracobrachialis is attached to the deep aspect of the lower part of the medial side and the deep surface of the tip of the process, and the short head of biceps is attached to the lateral side and superficial aspect of the tip.

Scapular movements

Scapular movement is a product of interconnected suspension, motion and articulation mechanisms. The suspension mechanism consists of

the scapular articulation with the clavicle and the suspension muscles of the scapula and clavicle, notably trapezius, attached to the lateral clavicle, the acromion and the spine of scapula. The articulation with the clavicle comprises two linked systems: 'clavicle – acromioclavicular joint – acromion – spine of scapula – lateral scapular angle' and 'clavicle – coracoclavicular ligament – coracoid – lateral scapular angle – spine of scapula'. This mechanism therefore subserves the function of scapular rotation or tilt limited by the ellipsoid range of motion of the lateral clavicle. The motion mechanism consists of the agonist–antagonist force couple of serratus anterior passing to the chest wall anterolaterally, and levator scapulae, rhomboid major, rhomboid minor and serratus posterior (when present), which all pass to the vertebral column medially. This mechanism therefore subserves the function of scapular protraction and retraction around the elliptic paraboloid of the chest wall. The articulation mechanism consists of the scapular neck region and the glenoid fossa, which articulates with the humeral head. The muscles that subservise the function of holding the humeral head on the glenoid through the wide range of motion of the glenohumeral joint are the rotator cuff muscles, which take their attachment from the body of the scapula. This mechanism therefore concerns the relationship between the position and orientation of the glenoid fossa and the scapular body.

Ossification

The cartilaginous scapula is ossified from eight or more centres: one in the body, two each in the coracoid process and the acromion, and one each in the medial border, inferior angle and lower part of the rim of the glenoid cavity (Fig. 48.10). The centre for the body appears in the eighth intrauterine week. Ossification begins in the middle of the coracoid process in the first year or, in a small proportion of individuals, before birth; the process joins the rest of the bone about the fifteenth year. At or soon after puberty, centres of ossification occur in the rest of the coracoid process (subcoracoid centre), in the rim of the lower part of the glenoid cavity, frequently at the tip of the coracoid process, in the acromion, in the inferior angle and contiguous part of the medial border and in the medial border. A variable area of the upper part of the glenoid cavity, usually the upper third, is ossified from the subcoracoid centre; it unites with the rest of the bone in the fourteenth year in the female and the seventeenth year in the male. A horseshoe-shaped epiphysis appears for the rim of the lower part of the glenoid cavity; thicker at its peripheral than at its central margin, it converts the flat glenoid cavity of the child into the gently concave fossa of the adult. The base of the acromion is formed by an extension from the spine; the rest of the acromion is ossified from two centres that unite and then join the extension from the spine. The various epiphyses of the scapula have all joined the bone by about the twentieth year.

The normal fetal scapula differentiates in somitic ectoderm at about 5 weeks *in utero* juxtaposed to the fourth, fifth and sixth cervical vertebral anlagen. As the neural axis and cardiac primordia develop between the ninth and twelfth weeks, the primordial scapula migrates caudally to its usual position at a level between the second and seventh ribs. With further development, the scapula alters shape.

Ligaments

The intrinsic scapular ligaments are strengthening ligaments (coraco-acromial ligament) and bounding ligaments (superior transverse scapular or suprascapular ligament, inferior transverse scapular or spinoglenoid ligament, and the variable glenocoracoid ligament).

Coraco-acromial ligament The coraco-acromial ligament is a strong triangular band between the coracoid process and acromion. It is attached apically to the acromion anterior to its clavicular articular surface, and by its base along the whole lateral border of the coracoid. Together with the coracoid process and acromion, it completes an arch above the humeral head. It may be composed of two strong marginal bands with a thinner centre; when pectoralis minor is inserted into the humeral capsule instead of the coracoid process, which happens occasionally, its tendon passes between the bands. The subacromial bursa facilitates movement between the coraco-acromial arch and the subjacent supraspinatus and shoulder joint, functioning as a secondary synovial articulation. When the coraco-acromial ligament is cut, it is impossible to appose the two cut surfaces afterwards: this suggests there is resting tension in the ligament, which forms a 'tension band' between the acromial and coracoid processes of the scapula, so 'pre-tensioning' both processes against the distracting forces of deltoid and trapezius, and pectoralis minor and the conjoined short head of biceps brachii and coracobrachialis, respectively.

Superior transverse scapular (suprascapular) ligament The superior transverse scapular (suprascapular) ligament converts the scapular

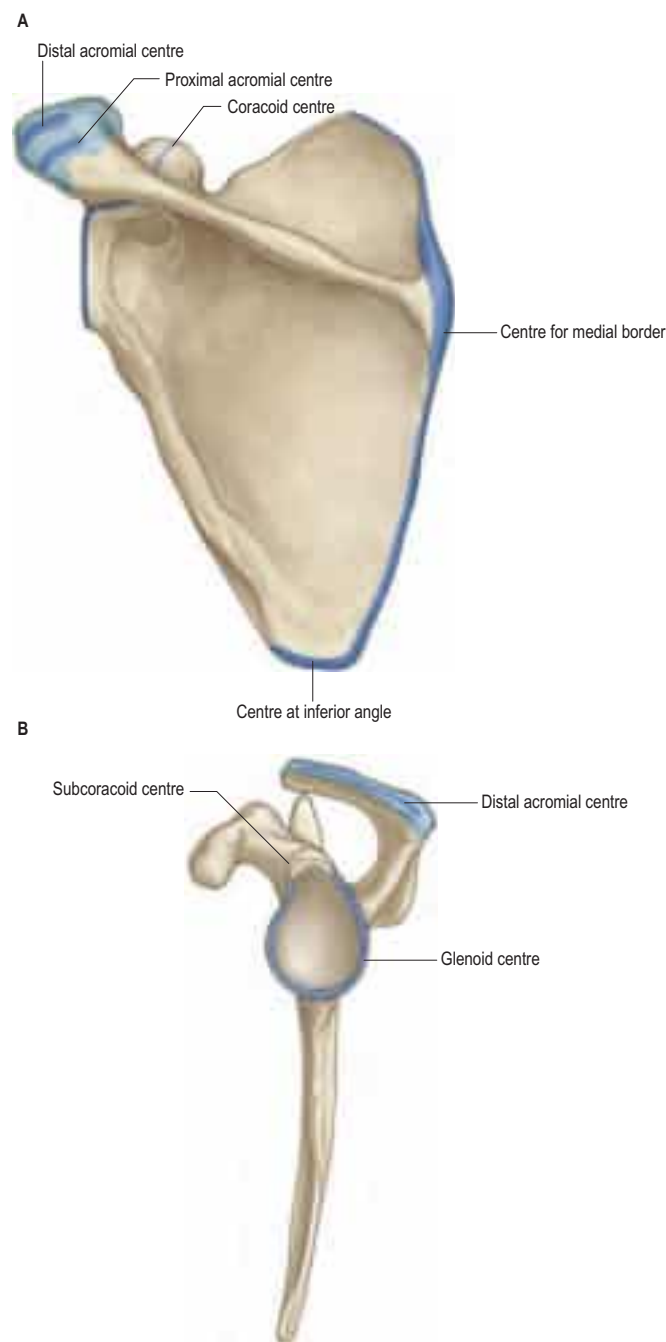


Fig. 48.10 Ossification of the scapula. **A**, Dorsal aspect. **B**, Lateral aspect. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

notch into a foramen; it is sometimes ossified. A flat fasciculus, it narrows towards its attachments to the base of the coracoid process, blending with the lower fibres of the conoid ligament, and to the medial side of the scapular notch. The suprascapular nerve traverses the foramen, and the suprascapular vessels cross above the ligament.

Inferior transverse scapular (spinoglenoid) ligament When present, the inferior transverse scapular ligament is a membranous ligament that may stretch from the lateral border of the spine of the scapula to the glenoid margin (see Fig. 48.33). It forms an arch over the branches of the suprascapular nerve and vessels entering the infraspinous fossa to supply infraspinatus.

HUMERUS

The humerus, the longest and largest bone in the upper limb, has expanded ends and a shaft (Figs 48.11–48.12). The spheroidal humeral head forms an enarthrodial articulation with the glenoid fossa of the scapula. The lesser tubercle projects from the front of the shaft, close

Initially, the horizontal diameter exceeds the vertical but the ratio gradually decreases until mature dimensions are reached, an adaptation thought to provide upright hominids with increased range, and therefore freedom of use, of the upper limb. Avian gene-deletion studies have shed light on the development of the scapula. Scapular body development appears to be controlled by the *Emx2* gene, also expressed in the developing mesonephros. The development of the acromion and the spine of the scapula are controlled by the *Pax1* gene, and glenoid and coracoid development is controlled by the *Hoxc6* gene (Huang et al 2000), one of the homeobox genes responsible for somite differentiation and development; development of the glenoid and coracoid appears likely to be more intimately related to the development of the upper limb bud as a whole. The development of the scapular body appears to be independent of that of the processes and glenoid region. Adult scapulae may show a line of fusion across the body and spine of the scapula, suggesting that the primordia of the processes (including the glenoid) and the primordium of the body migrate towards each other; since the trapezius is derived from branchial arch mesoderm it would appear that the development of the spine of the scapula and acromion and trapezius are intimately related, but separate from the body. This hypothesis is supported by the observation that; in the condition of so-called Sprengel shoulder (more correctly termed congenital undescended scapular syndrome), the development of the scapular body is altered, and there is intramuscular ossification of variable parts of the medial scapulospinal muscles, but the development of the processes (including the glenoid) is unaffected (Dhir and Lambert 2014).

The suprascapular nerve may be entrapped by the ligament if this is thickened or ossified and this may cause the syndrome of neurostenalgia, a typical pain of unremitting burning or aching nature, due to stenosis or distortion of the nerve trunk by extrinsic compression. Weakness of supraspinatus and infraspinatus may subsequently occur. Surgical release of the ligament is often helpful; this may be undertaken by open or arthroscopic surgical exposure and division of the ligament.

A ganglion (an outpouching of the synovial membrane of the glenohumeral joint) may occur immediately behind and above the posterosuperior corner of the glenoid fossa; this can compress the branch of the suprascapular nerve that supplies infraspinatus as it courses around the lateral aspect of the spine of the scapula under the inferior scapular ligament. Since the nerve cannot move aside (it is 'held' under the ligament), it can be readily distorted; there is characteristic severe pain, followed by specific atrophy of infraspinatus. Decompression of the ganglion and division of the ligament are curative in the early stage of this syndrome. An intraneural ganglion may arise from the articular branch of the suprascapular nerve; muscular pain and atrophy would be expected to be more rapid than in the case of the synovial ganglion (Spinner et al 2006).

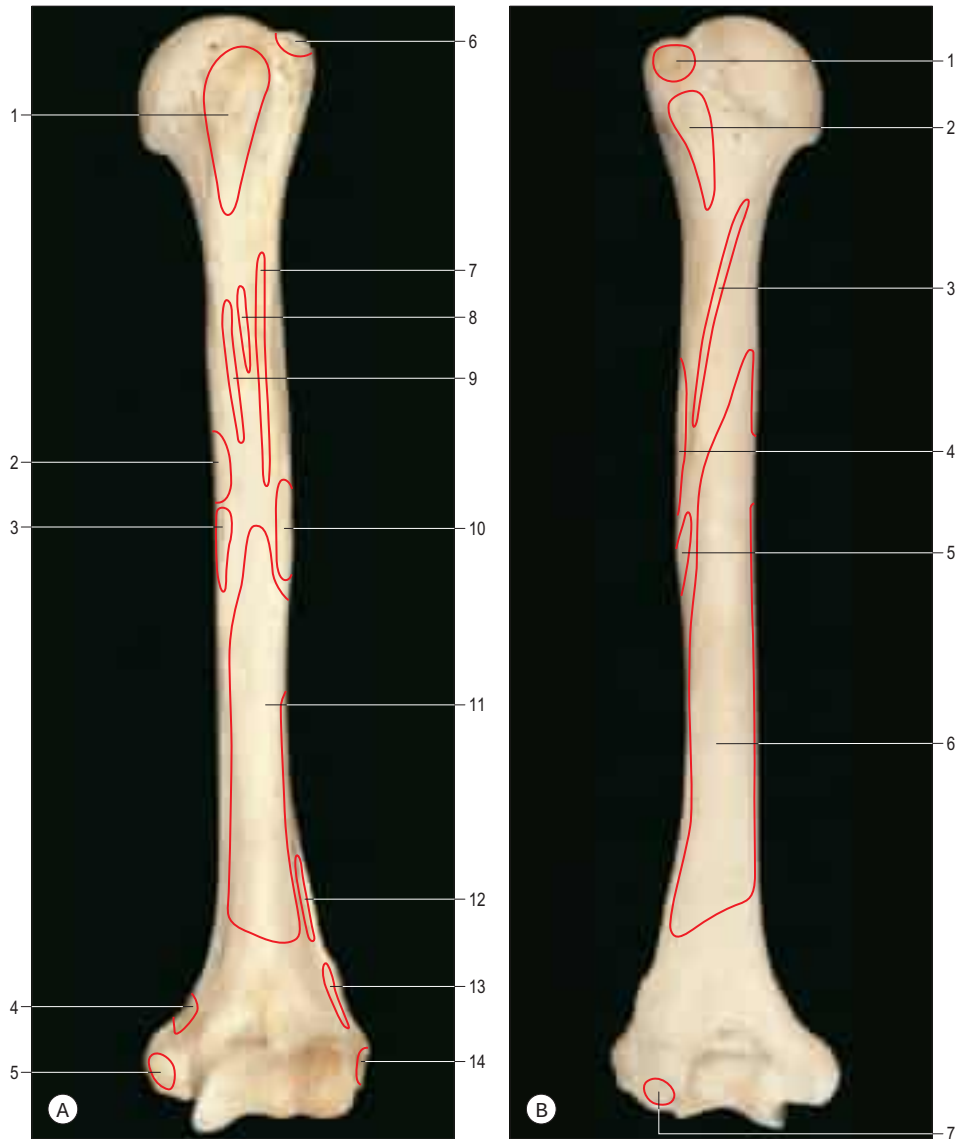


Fig. 48.11 The left humerus showing sites of muscular attachments. **A**, Anterior aspect. Key: 1, subscapularis; 2, triceps brachii (medial head); 3, coracobrachialis; 4, pronator teres (humeral head); 5, common flexor origin; 6, supraspinatus; 7, pectoralis major; 8, latissimus dorsi; 9, teres major; 10, deltoid; 11, brachialis; 12, brachioradialis; 13, extensor carpi radialis longus; 14, common extensor origin. **B**, Posterior aspect. Key: 1, infraspinatus; 2, teres minor; 3, triceps brachii (lateral head); 4, deltoid; 5, brachialis; 6, triceps brachii (medial head); 7, anconeus.



Fig. 48.12 The proximal end of the left humerus. **A**, Anterior aspect. Key: 1, head; 2, anatomical neck; 3, surgical neck; 4, greater tubercle; 5, lesser tubercle; 6, intertubercular sulcus; 7, shaft. **B**, Posterior aspect. Key: 1, greater tubercle; 2, surgical neck; 3, shaft; 4, radial groove; 5, head; 6, anatomical neck.

to the head, and is limited on its lateral side by the well-marked groove, the intertubercular sulcus, by which it is separated from the greater tubercle. The distal end is adapted to the forearm bones at the elbow joint and carries the medial and lateral epicondyles with the articular surfaces for the radius and ulna between them. The axis between the epicondyles is the inter-epicondylar axis.

The shaft of the human (and primate) humerus is relatively *medially* rotated with respect to the humeral head, compared with quadrupedal ancestors; this gives the characteristically greater range of *external* rotation at the glenohumeral joint than that obtaining in other species. This is reflected in the spiroidal architecture of the adult humeral medullary cavity, the arrangement and relationship of the posterior compartment of brachial muscles and the radial nerve, and the disposition to long spiroidal fractures caused by external twisting forces.

With the arm by the side in the anatomical position, and with the medial and lateral epicondyles in the same (frontal) plane (i.e. the inter-epicondylar axis is in the frontal plane), the humeral *head* is rotated posteriorly at an angle averaging 15–20° in European cadaveric specimens (Boileau and Walch 1997). Since the glenoid fossa of the scapula faces anterolaterally, the humerus is not rotated posteriorly relative to the scapula in the anatomical position. It is important to remember this position of the bone when movements of the arm and forearm are considered; movements are recorded relative to the trunk (starting in the anatomical position) or relative to the scapula, and it is important to define which method is in use.

Proximal end

The proximal end of the humerus consists of the head, anatomical neck, and the greater and lesser tubercles. It joins the shaft at an ill-defined 'surgical neck', which is closely related on its medial side to the axillary nerve and posterior humeral circumflex artery (see Figs 48.22, 48.31, 48.33). The proximal end of the humerus makes up the upper fifth of the length of the bone.

Head

The head of the humerus forms rather less than half a spheroid; in sectional profile, it is spheroidal (strictly, ovoidal) (see Fig. 48.12). Its smooth articular surface is covered with hyaline cartilage that is thicker centrally. When the arm is at rest by the side, it is directed medially, backwards and upwards to articulate with the glenoid cavity of the scapula. The humeral articular surface is much more extensive than the glenoid cavity, which means that only a portion of it is in contact with the cavity in any one position of the arm.

Anatomical neck

The anatomical neck of the humerus immediately adjoins the margin of the head, forming a slight constriction that is least obvious next to the greater tubercle. A slight roughness indicates the line of capsular attachment of the shoulder joint, other than at the intertubercular sulcus, where the long tendon of biceps brachii emerges. Medially, the capsular attachment diverges from the anatomical neck and descends 1 cm or more on to the shaft.

Lesser tubercle

The lesser tubercle is anterior to and just distal to the anatomical neck. It is palpable through the thickness of deltoid about 3 cm below the anterior edge of the acromion. It can be distinguished from the coracoid by rotation of the humerus under the palpating finger; the lesser tubercle slips away from the examining finger, the tip of the coracoid does not. The lateral edge of the lesser tubercle is sharp and forms the medial border of the intertubercular sulcus. Subscapularis is attached to the lesser tubercle (see Fig. 48.11A). The transverse ligament of the shoulder (transverse humeral ligament) is attached to the upper aspect of the lateral margin of the tubercle.

Greater tubercle

The greater tubercle is the most lateral part of the proximal end of the humerus and projects beyond the lateral border of the acromion. Its posterolateral aspect, near the anatomical neck, bears three smooth, flattened impressions for the attachment of supraspinatus (uppermost), infraspinatus (middle) and teres minor (lowest and placed on the posterior surface of the tubercle) (see Fig. 48.11). The attachments of subscapularis and teres minor are not confined to their respective tubercles, but extend for varying distances on to the adjacent metaphysis. The projecting lateral surface of the tubercle presents numerous vascular foraminae and is covered by deltoid, producing the normal rounded contour of the shoulder. A part of the subacromial bursa may cover the

upper part of this area and separate it from deltoid. The intertubercular sulcus (bicipital groove) lies between the tubercles. It contains the long tendon of biceps, its synovial sheath, and an ascending branch from the anterior circumflex humeral artery. The rough lateral lip of the groove is marked by the bilaminar tendon of pectoralis major, and its medial lip by the tendon and muscular insertion of teres major. The floor of the groove provides attachment for a frequent upward extension of the tendon of pectoralis major, and for the tendon of latissimus dorsi more caudally (Dancker 2013).

Shaft

The shaft of the humerus is almost cylindrical in its proximal half. Distal to the deltoid tuberosity, it is equilaterally triangular in cross-section, flaring out into a broad isosceles triangular cross-section just above the elbow joint. It can be readily palpated laterally and medially, but the muscles of the anterior and posterior compartments obscure the bone to palpation anteriorly and posteriorly. It has three surfaces and three borders, which are only distinct towards the elbow joint. The shaft comprises the middle three-fifths of the humerus.

Surfaces

The posterior surface, between the medial and lateral borders, is broad, flat and convex distally. The medial head of triceps is attached to the posterior surface over an elongated triangular area, the apex of which is placed on the medial part of the bone above the level of the lower limit of insertion of teres major. The area widens below and covers the whole surface almost down to the lower end of the bone. The lateral head of triceps is attached to a ridge, sometimes rough, that descends obliquely and laterally above the attachment for the medial head. Above triceps, the axillary nerve and the posterior circumflex humeral vessels wind around the bone on the deep surface of deltoid. Below the attachment of the lateral head of triceps, but above the attachment of the medial head, a shallow groove runs from the upper medial aspect of the surface downwards and laterally towards the lateral border; it contains the radial nerve and its branches, and the profunda brachii vessels.

The anterolateral surface is bounded by the anterior and lateral borders, and is smooth and featureless in its upper part, which is covered by deltoid. About, or a little above, the middle of this surface, deltoid is attached to the deltoid tubercle. Further distally, the surface gives origin to the lateral fibres of brachialis, which extend upwards into the floor of the lower end of the groove for the radial nerve (see Fig. 48.11). The nerve perforates the lateral intermuscular septum close to the bone at the level of the junction of the upper three-fifths and lower two-fifths of the humerus, and enters the anterior compartment of the arm close to the anterolateral surface, between brachialis medially and brachioradialis laterally.

The anteromedial surface is bounded by the anterior and medial borders. Below the intertubercular sulcus, the rest of the upper half of the surface is smooth and devoid of muscular attachment; its lower half is occupied by the medial part of brachialis (see Fig. 48.11A). Coracobrachialis is attached to a long rough strip in the middle third of the medial border, posterior to the brachial neurovascular sheath, which lies in close relation to the humerus in the middle third of this surface. The humeral head of pronator teres is attached to a narrow area close to the lowest part of the medial supracondylar ridge; the ridge itself gives attachment to the medial intermuscular septum of the arm.

A little below its midpoint, the nutrient foramen, which is directed downwards, opens close to the medial border. A hook-shaped process of bone, the supracondylar process, ranging from 2 to 20 mm in length, occasionally projects from the anteromedial surface of the shaft, approximately 5 cm proximal to the medial epicondyle. It is curved downwards and forwards, and its pointed apex is connected to the medial border, just above the epicondyle, by a fibrous band, to which part of pronator teres is attached. The foramen completed by this fibrous band usually transmits the median nerve and brachial artery, but sometimes encloses only the nerve, or the nerve plus the ulnar artery (in cases of high division of the brachial artery). A groove that lodges the artery and nerve usually exists behind the process.

Anterior border

The anterior border starts on the front of the greater tubercle and runs downwards almost to the lower end of the bone. Its proximal third forms the lateral lip of the intertubercular sulcus and is roughened for muscular attachments. The succeeding portion is also roughened and forms the anterior limit of the deltoid tubercle; the lower half of the border is smooth and rounded.

Lateral border

The lateral border is most conspicuous at the lower end of the bone, where it is thickened to form the lateral supracondylar ridge, and its sharp edge is roughened along its anterior aspect. In its middle and upper thirds, the border is barely discernible, but in a well-marked bone it can be traced upwards to the posterior surface of the greater tubercle. A little above its middle, it is marked by a V-shaped, roughened area: the deltoid tubercle. The limbs of the V are broad; the groove for the radial nerve runs downwards and laterally behind the posterior limb and fades away on the lower part of the anterolateral surface. The lateral intermuscular septum is attached to the lateral border, and is a condensation of the fascia over the lower part of deltoid and the neighbouring brachialis, forming a septum between the anterior and posterior muscular compartments. The septum is most obvious in the lower three-fifths of the arm, and is perforated by the radial nerve and accompanying vessels.

Medial border

The medial border, although rounded, can be identified without difficulty in the lower half of the shaft, where it becomes the medial supracondylar ridge. In its proximal third, the medial border is indistinct until it broadens out to form a triangular area. The lateral border of this area forms the medial lip of the intertubercular sulcus, and the medial border runs upwards as the calcar humerale to the anatomical neck, where it is roughened, with vascular apertures, and forms the area of attachment of the inferior part of the shoulder capsule. In its middle third, the medial border is interrupted by a wide, shallow groove, the radial (spiral) groove that crosses the bone obliquely, passing downwards and forwards from its posterior to its anterior surface.

Fractures of the humeral shaft

Humeral shaft fractures are common; the pattern of the fracture and the displacement of the fragments depend on the force of injury and on the level at which the bone is broken. If there is wide displacement at the time of fracture, closely associated nerves and vessels are at risk of direct injury. Nerves that are 'fixed' in relation to the bone and attached septa are at further risk of indirect injury by stretching or, later, by the callus of bone healing. (For a discussion of radial nerve injury and fractures of the humeral shaft, see Lambert (2005).)

Distal end

The distal end of the humerus is described in on [page 838](#).

Ossification

The proximal humerus and shaft are ossified from four centres: one each in the shaft, head, greater and lesser tubercles (Fig. 48.13). The centre for the shaft appears near its middle in the eighth week of intrauterine life, and gradually extends towards the ends. Most growth occurs at the distal end of the bone. Ossification begins in the head before birth (20%) or in the first 6 months afterwards; the greater tubercle starts to ossify during the first year in females and second year in males; the lesser tubercle begins to ossify at about the fifth year.

By the sixth year, the centres for the head and tubercles have joined to form a single large epiphysis, hollowed out on its inferior surface to adapt to the conical upper end of the metaphysis. This macroscopic topography provides for mechanical stability in the physis during

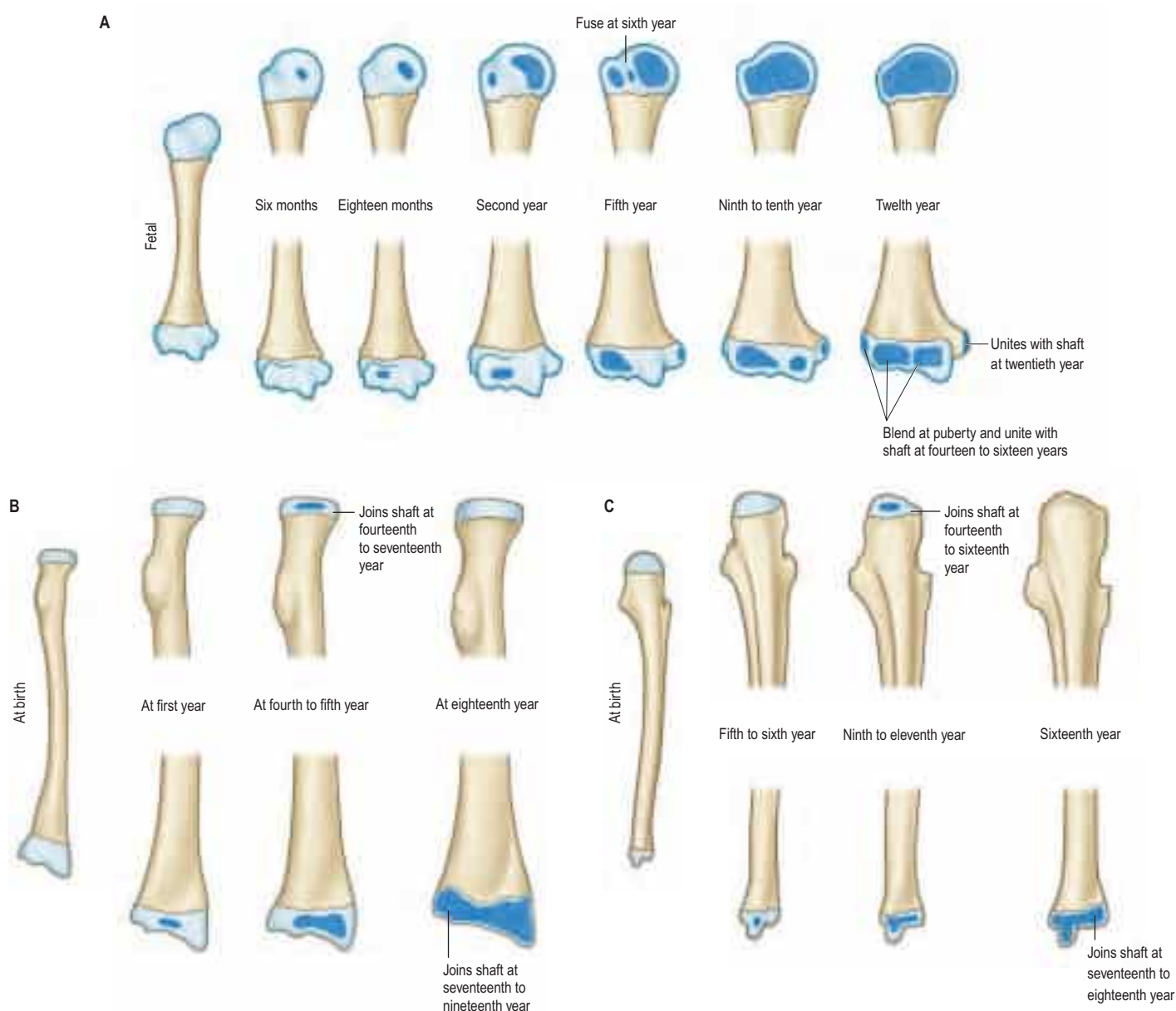


Fig. 48.13 The stages in ossification of (A) the humerus, (B) the radius and (C) the ulna (not to scale).

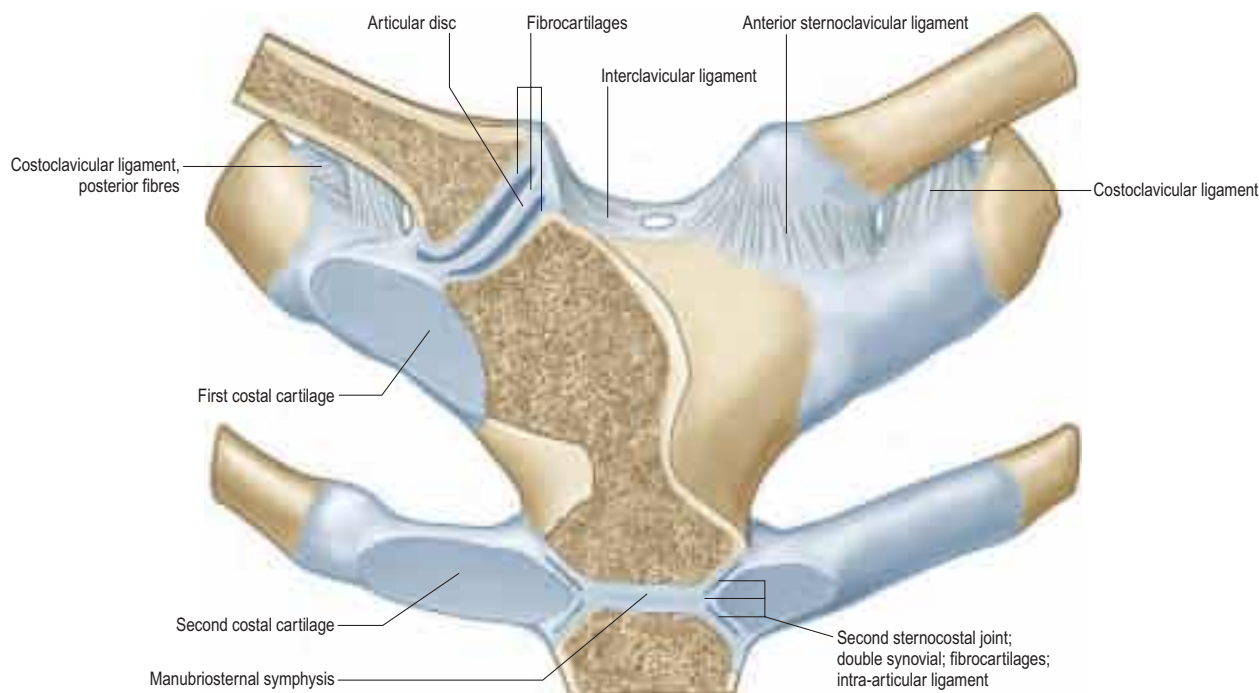


Fig. 48.14 The sternoclavicular joints, anterior aspect. The joint on the left side is intact and the joint on the right side is shown in coronal section.

growth, although rotational stability is still relatively poor; adolescent proximal humeral physal fracture-separations are commonly unstable in rotation. The proximal humeral epiphysis fuses with the shaft of the humerus at about the thirteenth or fourteenth year in females, beginning on the medial aspect of the physal line, and between the fourteenth to sixteenth year in males.

JOINTS

STERNOCLAVICULAR JOINT

The sternoclavicular joint is a synovial sellar (saddle, or hyperbolic paraboloid) joint with an intra-articular fibrocartilaginous disc. It is the only skeletal articulation between the upper limb and the axial skeleton (Sewell et al 2013).

Articulating surfaces The articulating surfaces are the sternal end of the clavicle and the clavicular notch of the sternum, together with the adjacent superior surface of the first costal cartilage (Fig. 48.14). The larger clavicular articular surface is covered by fibrocartilage, which is thicker than the fibrocartilaginous lamina on the sternum. The joint is convex vertically but slightly concave anteroposteriorly, and is therefore sellar; the clavicular notch of the sternum is reciprocally curved but the two surfaces are not fully congruent. An articular disc completely divides the joint.

Fibrous capsule The capsule is thickened anteriorly and posteriorly, but superiorly and especially inferiorly, it is little more than loose areolar tissue.

Ligaments There are two sets of associated ligaments. The intrinsic ligaments are the anterior and posterior sternoclavicular ligaments; the extrinsic ligaments are the midline interclavicular ligament and the costoclavicular ligaments on each side.

Anterior sternoclavicular ligament The anterior sternoclavicular ligament is broad and attached above to the anterosuperior aspect of the sternal end of the clavicle. It passes inferomedially to the upper anterior aspect of the manubrium, spreading on to the first costal cartilage.

Posterior sternoclavicular ligament The posterior sternoclavicular ligament is a weaker band posterior to the joint. It descends inferomedially from the posterior aspect of the sternal end of the clavicle to the posterior aspect of the upper manubrium.

Interclavicular ligament The interclavicular ligament is continuous above with the deep cervical fascia. It unites the superior aspect of the

sternal ends of both clavicles; some fibres are attached to the superior manubrial margin.

Costoclavicular ligament The costoclavicular ligament is like an inverted cone, but short and flattened. It has anterior and posterior laminae that are attached to the upper surface of the first rib and costal cartilage, and ascends to the margins of an impression on the inferior clavicular surface at its medial end. Fibres of the anterior lamina ascend laterally and those of the posterior lamina (which are shorter) ascend medially (see Fig. 48.14). They fuse laterally and are closely related to the attachments of subclavius, particularly the tendon of origin; it can be hard to distinguish the lateral border of the ligament from the tendon. The ligament merges medially with the capsule. It can be ossified in older adults, particularly females. A few deep fibres of pectoralis major attach to the external surface of the ligament, adjacent first rib, first costal cartilage and manubrium sterni.

Articular disc The fibrocartilaginous articular disc divides the cavity of the joint into two compartments between the sternal and clavicular surfaces. It is attached above to the posterosuperior border of the articular surface of the clavicle, below to the first costal cartilage near its sternal junction, and by the rest of its circumference to the capsule, and therefore adapts to the contour of the clavicular surface. It is thicker peripherally, especially superoposteriorly and inferomedially; the central part of the disc may be perforated in later life.

Vascular supply The sternoclavicular joint is supplied by branches from the internal thoracic and suprascapular arteries.

Innervation The sternoclavicular joint is innervated superficially by branches from the medial supraclavicular nerve and deeply by the nerve to subclavius.

Factors maintaining stability There is almost no bony articular congruence at the sternoclavicular joint. However, the strength of its associated ligaments and the articular disc produce durable stability. These factors make sternoclavicular joint dislocation rare; fracture of the clavicular shaft is far more common for the same force directed along the clavicle. (See Sewell et al (2013)) for a review of the mechanisms and treatment of sternoclavicular instability.)

Movements The capsule around the clavicular attachment is more lax. Movements between the clavicle and the disc are more extensive than those between the disc and sternum. The sellar shape of the articular surfaces permits translation or gliding in approximately anteroposterior and vertical planes, with rotation about the long axis of the clavicle. Close-packing coincides with maximum posterior rotation associated with full scapular rotation, i.e. in high elevation of the arm above shoulder height.

In this position, the tension developed in the anterior sternoclavicular ligament and anterior component of the costoclavicular ligament causes the clavicle to undergo an obligatory posterior translation. It also acts as a check-rein on further rotation and displacement, so protecting the relatively weaker posterior sternoclavicular ligament from overload. Although the function of subclavius is unknown, a consideration of the simple biomechanics of forceful elevation of the arm above shoulder height suggests that this muscle has an important role in decelerating the elevating arm, so protecting the medial 'clavicular–costal–sternal' ligamentous structures from recurrent stretching that is likely to accelerate degeneration and injury. The common innervation of the sternoclavicular joint (the deep afferent–mechanoreceptor system) and subclavius suggests an intimate functional relationship between these structures. In daily activities, in which the upper limb is used largely in front of the trunk, the sternal end of the clavicle glides on the sternal facet about the fulcrum provided by the costoclavicular ligament. All joints in which polyaxial gliding occurs, producing shear forces (rotation with translation), possess either intra-articular synovial bursae or intra-articular fibrocartilaginous discs; the latter degenerate over time, producing characteristic exophytic degenerative arthritis. In this context, it is interesting that degeneration of the sternoclavicular joint and ossification of the costoclavicular ligaments (claviculocostal synostosis) are almost exclusively found in females above the age of 50 years.

ACROMIOCLAVICULAR JOINT

The acromioclavicular joint is a synovial plane joint and has an intra-articular fibrocartilaginous disc.

Articulating surfaces The articulating surfaces are between the acromial end of the clavicle and the medial acromial margin (Fig. 48.15). The joint is approximately plane but either surface may be slightly convex, the other being reciprocally concave; both surfaces are covered by fibrocartilage. The clavicular surface is a narrow, oval area that faces laterally or inferolaterally and overlaps a corresponding facet on the medial acromial border. The long axis is anteroposterior (Colegate-Stone et al 2010).

Fibrous capsule The capsule completely surrounds the articular margins. It is lined by synovial membrane and is strengthened superi-

orly by the acromioclavicular ligament and the fibres of the attachment of trapezius, posteriorly by the fibres of attachment of trapezius, and anteriorly by the fibres of attachment of deltoid. The inferior capsule is often incomplete in later life.

Ligaments There are two sets of associated ligaments. The intrinsic ligaments are the acromioclavicular ligaments; the extrinsic ligaments are the coracoclavicular ligaments.

Acromioclavicular ligaments The superior acromioclavicular ligament is quadrilateral. It extends between the upper aspects of the lateral end of the clavicle and the adjoining acromion. Its parallel fibres interlace with the aponeuroses of trapezius and deltoid. The inferior acromioclavicular ligament is thin, and often perforated in later life; it extends between the inferior surface of the lateral end of the clavicle and the adjoining acromion. It provides attachment for the intra-articular disc when this is present and complete.

Coracoclavicular ligament The coracoclavicular ligament connects the clavicle and the coracoid process of the scapula (Fig. 48.16). Though separate from the acromioclavicular joint, it is an efficient and important accessory ligament because it helps to maintain the apposition of the acromion to the clavicle, and so contributes to the suspension mechanism of the scapula (see above). The trapezoid and conoid parts of the ligament, usually separated by fat or, frequently, by a bursa, connect the posterior (more horizontal) part of the coracoid process and the lateral end of the subclavian groove of the clavicle: these adjacent areas may even be covered by cartilage to form a coracoclavicular joint.

The trapezoid part is anterolateral and is broad, thin and quadrilateral, ascending slightly from the upper coracoid surface to the trapezoid line on the inferior clavicular surface. Its anterior border is free, and its posterior border is joined to the conoid part, forming an angle that projects backwards and upwards. Quadrilateral ligaments twist when the adjoining bones rotate with respect to each other, and as they twist, the apposed surfaces must approach each other; such ligaments, therefore, act to oppose excessive rotation of the adjoining bones.

The conoid part is posteromedial and is a dense, almost vertical, triangular or conical band. Its broader base is attached superiorly to the conoid tubercle of the clavicle; its inferior apex is attached posteromedially to the dorsal surface and root of the coracoid process immediately

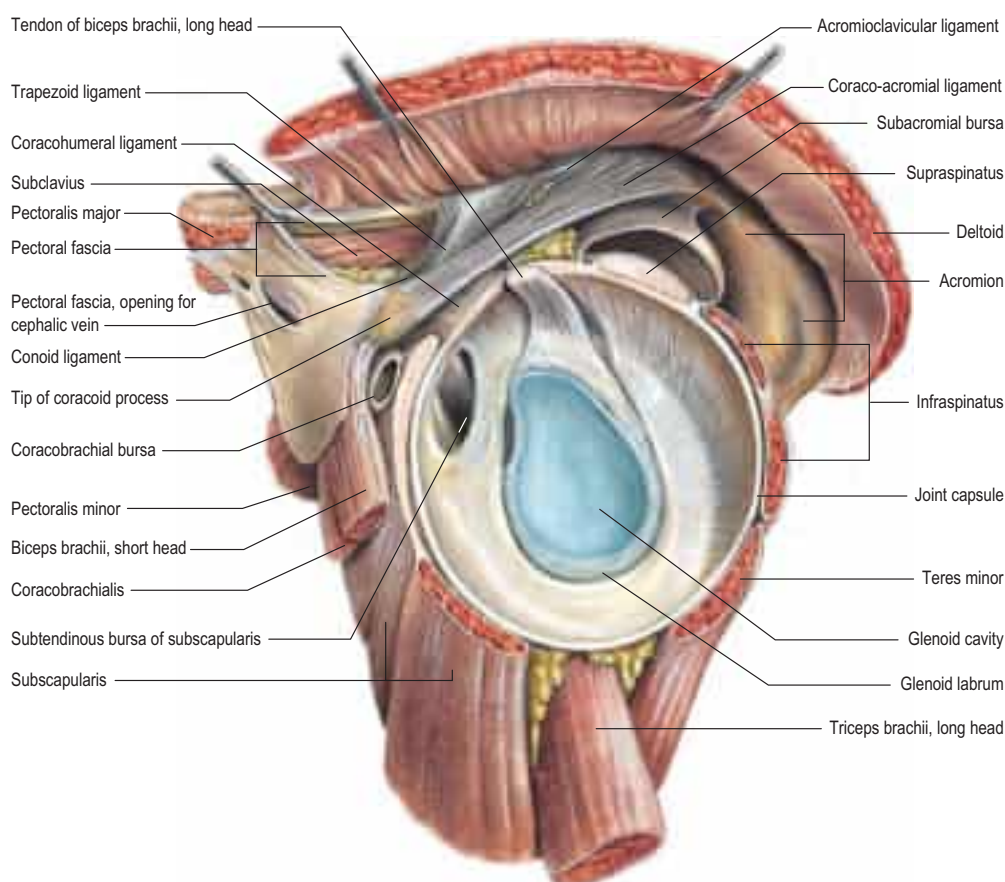


Fig. 48.15 The interior of the left shoulder joint, anterolateral aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

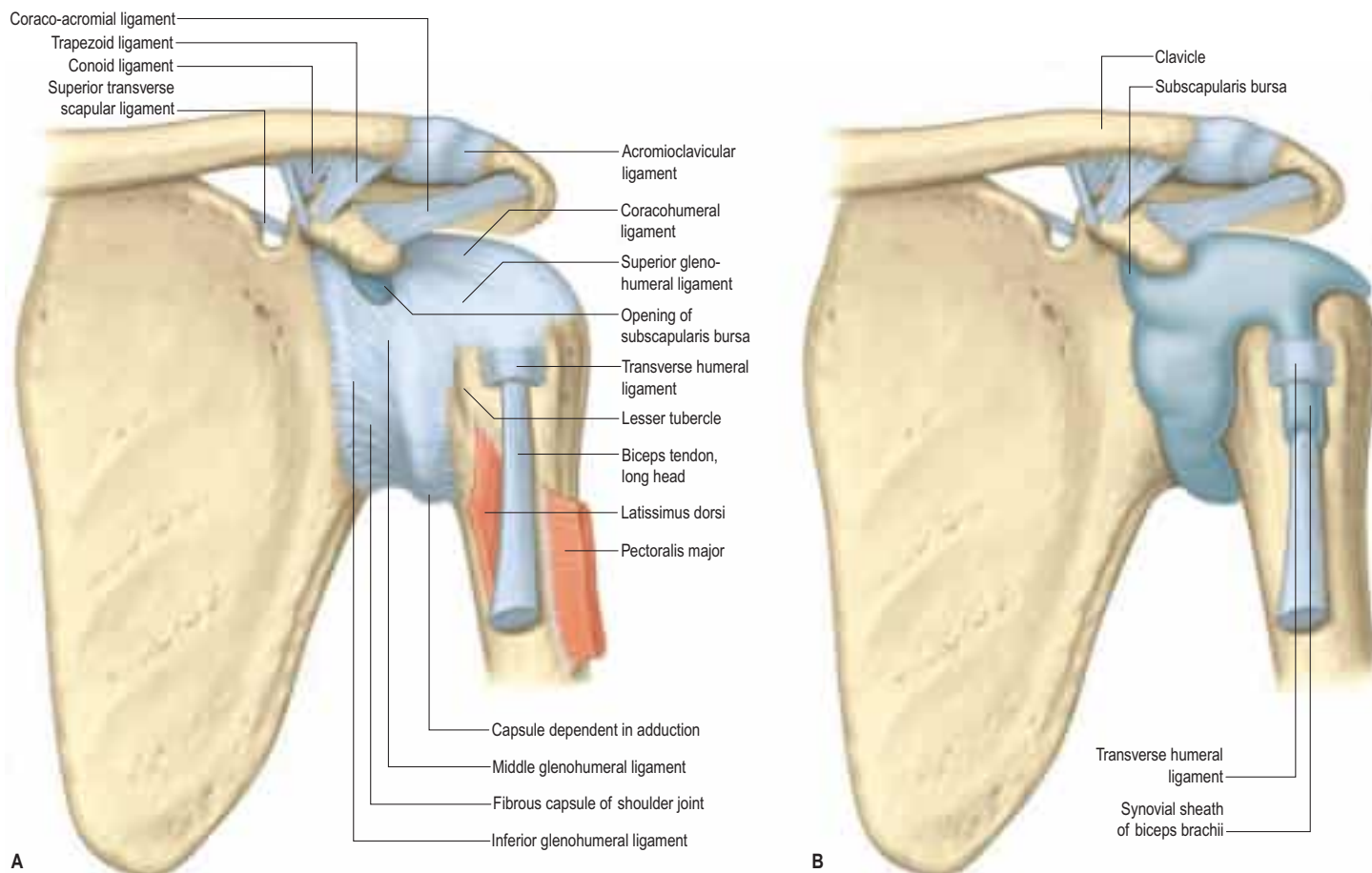


Fig. 48.16 **A**, The ligaments associated with the left shoulder, anterior aspect. **B**, A deeper view, showing the subscapularis bursa.

lateral to the scapular notch, and is contiguous with the superior transverse scapular ligament. Conical ligaments are constructed to oppose distraction; the conoid ligament, therefore, helps to keep the coracoid and the clavicle closely apposed. If the trapezoid ligament is disrupted, the conoid ligament forms a vertical fulcrum around which the coracoid can rotate under the clavicle, particularly if the acromioclavicular ligaments are also disrupted. This is the anatomical basis for the spectrum of acromioclavicular joint dislocation.

Articular disc The articular disc is a complete fibrocartilaginous partition in juvenile and adolescent joints, creating medial and lateral compartments; over time, the centre of the disc perforates, and in adults (more than 20 years old), it is often incomplete inferiorly.

Vascular supply The acromioclavicular joint receives its arterial supply from branches of the suprascapular and thoraco-acromial arteries.

Innervation The acromioclavicular joint is innervated by branches from the suprascapular and lateral pectoral nerves. It has been suggested that the density of nociceptors is greatest in the inferior acromioclavicular ligament and capsule.

Factors maintaining stability The acromioclavicular ligaments provide the greatest resistance to anteroposterior displacement of the acromioclavicular joint, while the coracoclavicular ligaments resist rotation and vertical translation of the joint.

Movements Movements at the joint complement those of the sternoclavicular joint (Figs 48.17–48.20). The sternal end of the clavicle is robustly supported by strong ligaments, and a similar, but less robust, arrangement of ligaments suspends the scapula from the lateral end of the bone. The motion of the acromioclavicular joint is limited by its small surface area and the articular capsule.

MOVEMENTS AT THE SCAPULOTHORACIC ‘JOINT’

The sternoclavicular and the acromioclavicular joints, in combination with the fascial space between the scapula and underlying chest wall,

are known collectively as the scapulothoracic articulation. Scapular movements on the thoracic wall are facilitated by loose areolar tissue between subscapularis and serratus anterior, and between serratus anterior and the chest wall; the latter gliding plane is the scapulothoracic joint. It is a virtual space. There may be a bursa between the superior angle of the scapula and the dorsal aspect of the second rib.

The following account should be read together with the description of movements of the glenohumeral joint. It is important to keep in mind the fact that the purpose of scapular motion is the appropriate positioning of the glenohumeral joint in space, and the purpose of scapular stability is to provide a foundation for glenohumeral motion.

Elevation and depression

Scapular elevation, as in ‘shrugging the shoulders’, is generated by trapezius acting on the lateral clavicle, acromion and spine of the scapula. If levator scapulae is less active, the scapula rotates laterally (i.e. the inferior angle protracts around the chest wall), with dorsal rotation at the acromioclavicular joint; this is resisted by the rhomboid muscles and controlled by pectoralis minor. If levator scapulae is more active, then the scapula ascends with less rotation. The sternal end of the clavicle, rotating about an anteroposterior axis through the bone above the medial attachment of the costoclavicular ligament, slides down over the articular disc. This is resisted by subclavius and by tension in the costoclavicular ligament and sternoclavicular joint capsule. In scapular depression, ventral rotation occurs at the acromioclavicular joint, and the clavicle slides up on the disc at the sternoclavicular joint. The movements are checked by the cervical fibres of trapezius, the interclavicular and superior sternoclavicular ligaments and the articular disc. Usually, gravity alone is sufficient; when necessary, the lowest part of serratus anterior and pectoralis minor are active depressors.

Protraction and retraction

Protraction (forward movement) round the thoracic wall occurs in pushing, thrusting and reaching movements, usually with some lateral rotation. The acromion advances over the clavicular facet to the limit, and the shoulder is simultaneously advanced by forward movement of the lateral end of the clavicle and posterior translation of its sternal end

The three long nerves innervating the scapulothoracic 'joint', i.e. accessory, long thoracic and dorsal scapular, innervate the three major muscle groups of scapulothoracic motion, namely: trapezius, serratus anterior and the medial scapular stabilizing muscles (levator scapulae and the rhomboids).

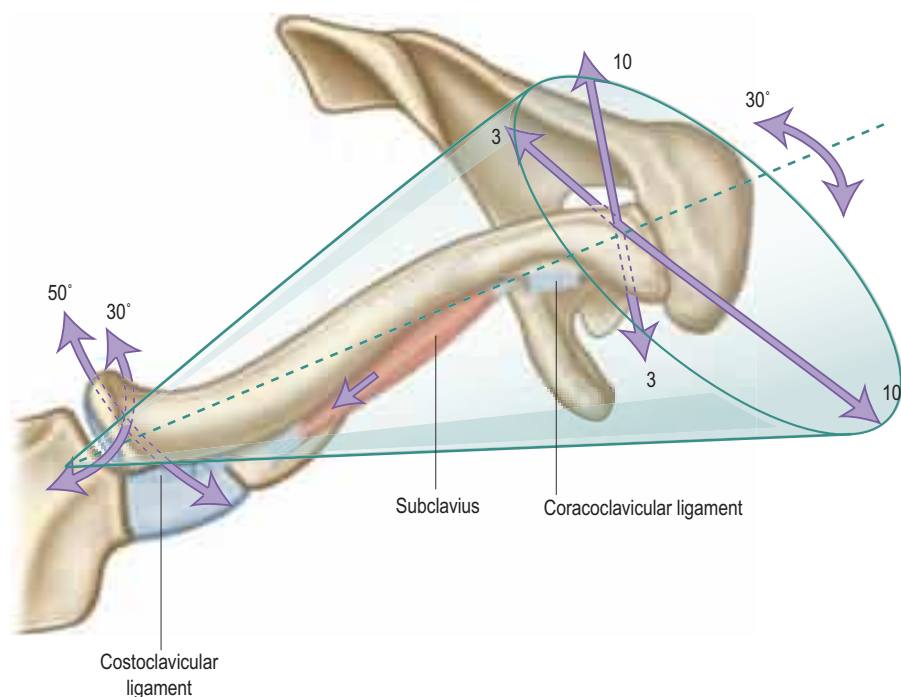


Fig. 48.17 Clavicular motion about the medial fulcrum of the sternoclavicular joint. The lateral (acromial) end of the clavicle describes an asymmetric conical path during motion of the upper extremity. The maximal range of rotation of the clavicle (subtended by the sternoclavicular joint) is 30°; the maximal range of rotation of the scapula (sternoclavicular and acromioclavicular joints combined) is about 60° with respect to the sternum. The scapula is suspended from the distal clavicle at the acromioclavicular joint, which can be considered as the true joint (the cavity bounded by the acromioclavicular capsule and ligaments); it is also part of an extended articulation that comprises the true joint and the coracoclavicular suspensory ligaments. Arguably, the ligaments might be better defined as the claviculocoracoid ligaments, to acknowledge their role as the spiroid fulcrum around which the body of the scapula, and therefore the glenoid fossa, rotate during protraction and retraction of the scapula. The action of subclavius is to resist upward displacement of the distal clavicle, or to decelerate the clavicle moving into elevation. Linear measurements are given in centimetres.

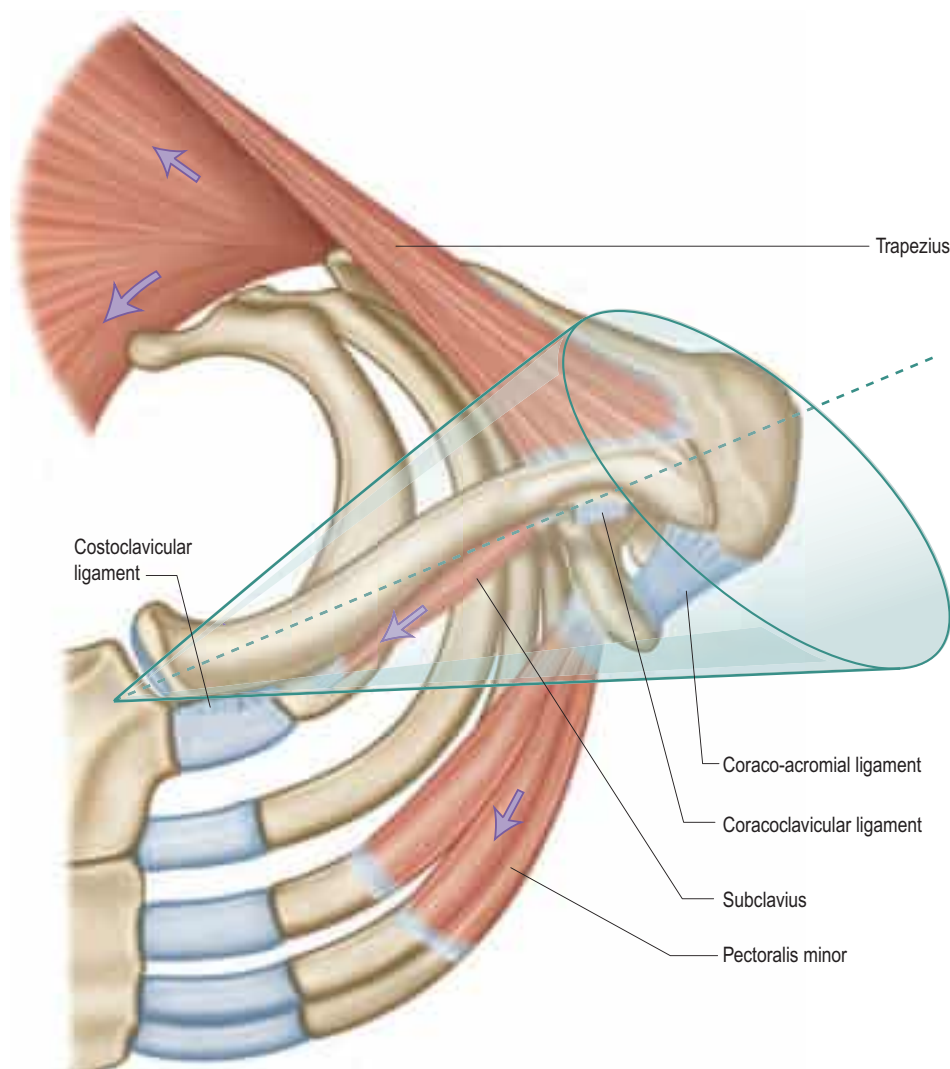


Fig. 48.18 The distal clavicle is moved by trapezius, which has an extensive attachment along the spine of the scapula and the medial aspect of the distal third of the clavicle. The antagonist of trapezial power is pectoralis minor, which, rarely, has an extension or slip of tendon that joins the coraco-acromial ligament. The latter acts as a tension band between the 'outriggers' of the acromion and coracoid. If surgically divided, the cut surfaces of the coraco-acromial ligament cannot be readily reapposed, suggesting a mechanical role in pre-tensioning the acromion and coracoid against the pull of trapezius. Shortening or tightness of pectoralis minor will create fixed protraction of the scapula, which then rotates ventrally around the chest wall. As a consequence, the acromion tilts and the space between it and the cranial surface of the rotator cuff below (the subacromial space) diminishes, creating the conditions in which forceful coaptation of the apposed surface of the rotator cuff and the coraco-acromial ligament may occur. This is the painful condition of 'impingement'.

over the sternal facet, carrying the disc with it. Pectoralis major, together with the anterior sternoclavicular ligament and posterior lamina of the costoclavicular ligament, check backward slide of the sternal end. Serratus anterior and pectoralis minor are prime movers and maintain continuous apposition of the scapula, especially its medial border, in smooth gliding on the thoracic wall, with the rhomboid muscles controlling the rate and range of motion. The upper part of latissimus dorsi

also acts like a strap across the inferior scapular angle in protraction and lateral rotation.

In scapular retraction, i.e. bracing back the shoulders, these movements are reversed and checked at the sternoclavicular joint by the posterior sternoclavicular ligament and anterior lamina of the costoclavicular ligament. Trapezius (horizontal and lower fibres) and the rhomboids are prime movers.

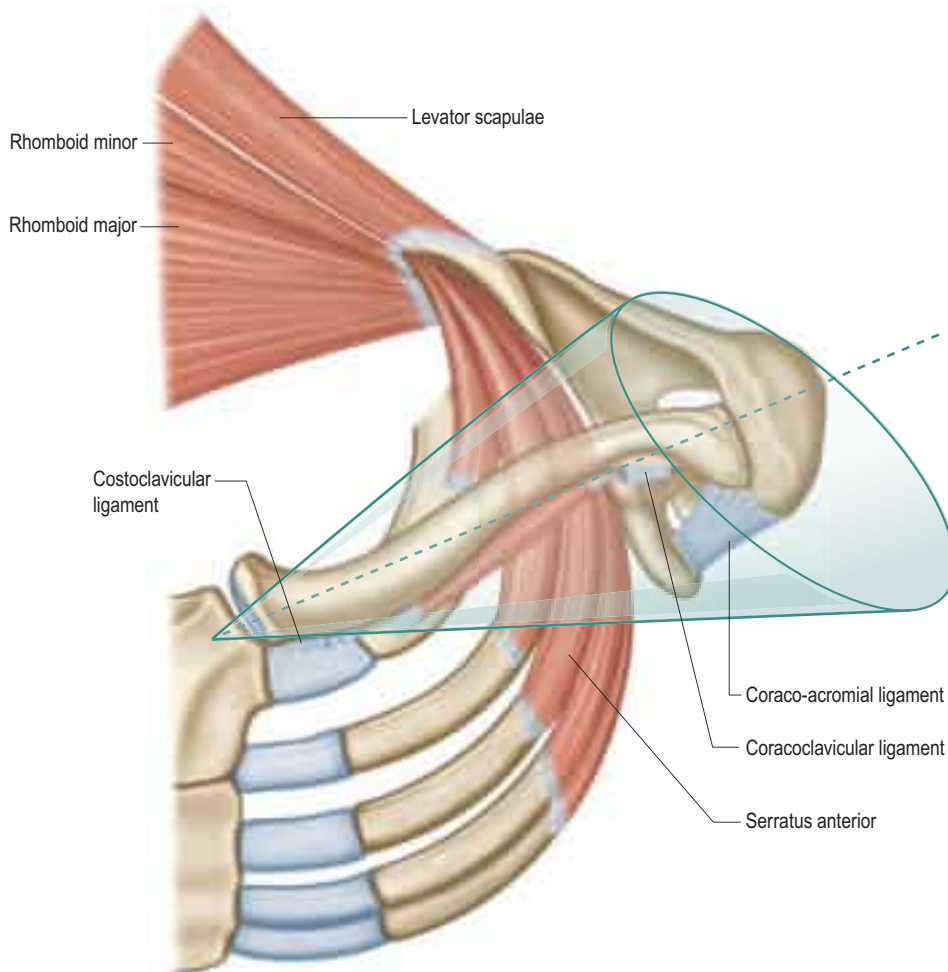


Fig. 48.19 Motion of the medial border of the scapula is governed by the opposing actions of the medial parascapular group (levator scapulae and rhomboids minor and major) and the scapulothoracic group (serratus anterior), causing retraction and protraction, respectively.

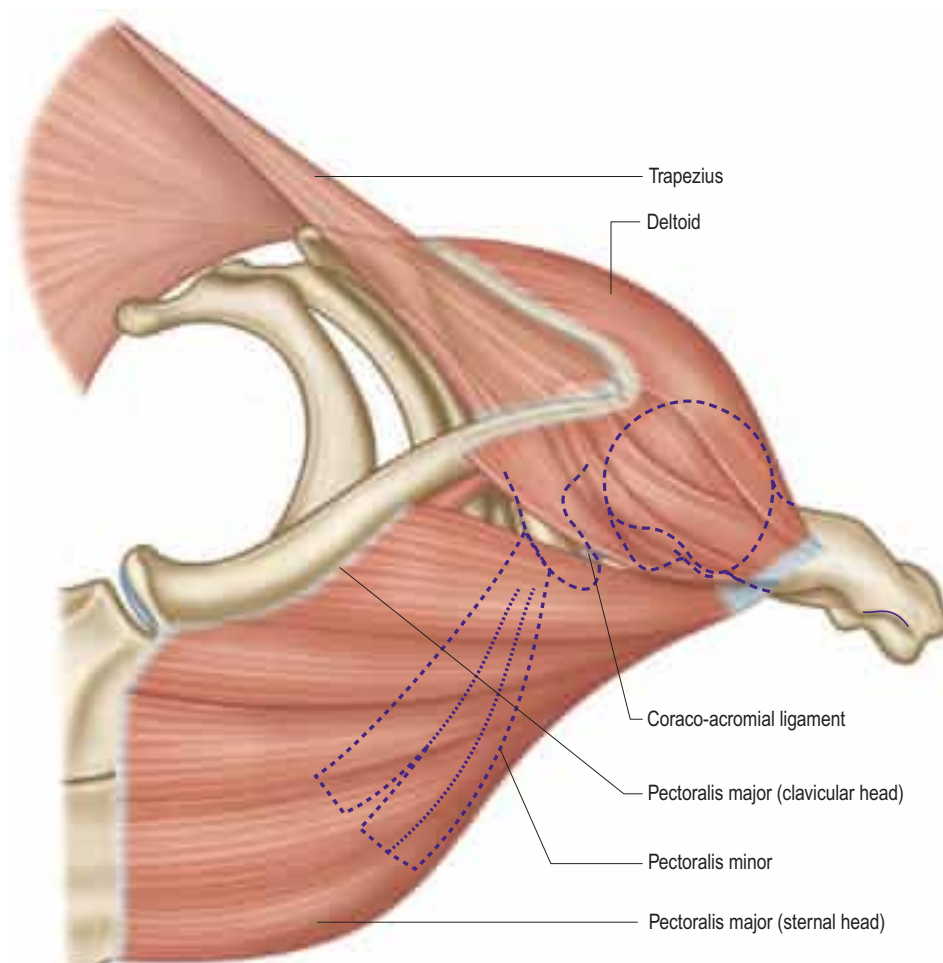


Fig. 48.20 The closely related humeral attachments of pectoralis major and deltoid (between one-third and one-half of the way down the shaft of the bone) enable the pectoralis major–deltoid muscular envelope to move the elbow, forearm and hand in space, using the shoulder as a fulcrum for effective delivery of power through a wide range of motion from in front of the trunk to behind. The relative coarseness of these muscles underlies their ability to produce great torque in upper-extremity motion; motion generated by these muscles is ‘fine-tuned’ by the rotator cuff working concordantly with the outer envelope (each segment or part of the rotator cuff can be said to have an equivalent segment or part in the outer envelope of muscles; this pattern is considered as a force-couple generating stable motion of the humerus on the scapula at the glenoid).

Lateral and medial rotation

Lateral (upward) rotation of the scapula increases the range of humeral elevation by turning the glenoid cavity to face almost directly up, e.g. when raising the arm above the head. This movement is always associated with humeral elevation and rotation at the glenohumeral joint, and with protraction of the scapula. Scapular rotation requires movement at both sternoclavicular and acromioclavicular joints; the sternoclavicular joint permits elevation of the lateral end of the clavicle, a movement that is almost complete when the arm is abducted to 90°. The acromioclavicular joint moves in the first 30° of abduction, when the conoid ligament becomes taut, and is subsequently accompanied by clavicular rotation at the sternoclavicular joint around the longitudinal axis of the bone. The medial end is depressed further as the lateral end continues to rise. Some acromioclavicular movement also occurs in the final stages of humeral abduction. Trapezius (cervical fibres) and serratus anterior (the lower part) are prime movers, with the rhomboids providing resistance to motion.

Medial (downward) rotation is usually effected by gravity; gradual active lengthening of trapezius and serratus anterior is sufficient to control it. When more force is needed, levator scapulae, the rhomboids and, in the initial stages, pectoralis minor are prime movers in returning the scapula to a position of rest.

GLENOHUMERAL (SHOULDER) JOINT

The glenohumeral joint is a synovial multiaxial spheroidal joint between the roughly hemispherical head of the humerus and the shallow glenoid fossa of the scapula (Fig. 48.21). It is the most mobile joint in the body and the most frequently dislocated. Its anatomy is a compromise between the requirements for motion and stability; both depend on the surrounding muscular and soft tissue envelope more than on the shape and surface area of the articulating surfaces.

The humeral head is held to the concave glenoid fossa by the compressive action of the rotator cuff muscles; the stabilizing mechanism of the glenohumeral joint is one of concavity compression. The rotator cuff muscles are attached to the proximal humerus through a musculo-tendinous envelope (see below), the deep surface of which fuses with the lateral part of the articular fibrous capsule. The humeral head is then contained in a spherical space bounded medially by the glenoid fossa and elsewhere by the deep surface of the fibrous capsule, a form of osseofibrous acetabulum. The walls of this 'acetabulum' (i.e. the rotator cuff muscles and tendons) are contained within a roughly spherical space bounded by the coracoid anteriorly; the coraco-acromial ligament anterosuperiorly; the acromion and spine of the scapula posteriorly, posterosuperiorly and posterolaterally; and the deep surface of deltoid anterolaterally and laterally. This arrangement creates two gliding planes, one internal (the glenohumeral articular gliding plane) and one external (the subcoracoid-subacromial-subdeltoid plane); the latter is continuous with the plane under the scapula (the scapulothoracic gliding plane). Gliding is facilitated by the synovial lined cavities of the glenohumeral joint and the subcoracoid, subacromial and subdeltoid bursae, which are nearly always contiguous. Infection, inflammation and injury commonly cause adhesions and fibrosis (with subsequent contracture) in these planes; restriction of motion follows, specific to the site and extent of adhesion. The surfaces of both gliding planes are extensive and fibrosis is consequently very restricting.

Articulating surfaces The articular surfaces are reciprocally curved and are correctly termed ovoids. The surface area of the humeral convexity is approximately four times that of the glenoid concavity, which means that only a small portion of the head opposes the glenoid in any position (Fig. 48.22). The radius of curvature of the glenoid fossa in the coronal plane is greater than that of the humeral head, and is deepened by a fibrocartilaginous rim, the glenoid labrum (see Fig. 48.15; Fig. 48.23). Both articular surfaces are covered by hyaline cartilage, thickest centrally and thinner peripherally over the humerus, and *vice versa* over the glenoid cavity. In most positions, their curvatures are not fully congruent and the joint is loose-packed; close-packing (full congruence) is probably only reached with the humerus abducted and laterally rotated.

Glenoid labrum The glenoid labrum is a fibrocartilaginous rim around the glenoid fossa. Anteriorly, it is triangular in section; posteriorly, it is more ovoid; while inferiorly, it may be flattened and almost deficient. It varies in size and thickness. The attachment to the edge of the glenoid fossa is secure superiorly where it blends with the anterior and posterior fasciculi of the long tendon of biceps (Vangsness et al 1994), less secure posteriorly, but relatively more secure anteroinferiorly (Bankart 1923). There may be an aperture between the free deep edge of the labrum and the anterior glenoid rim above the midpoint of the fossa. In slightly less than one-fifth of normal shoulders, there may be no anterior labrum, in which case its place taken by a cord-like middle glenohumeral ligament. The labrum may be hypoplastic or absent in patients with collagen deficiency but with normal shoulder function (hypermobility syndrome); these shoulders are not necessarily unstable. The labrum deepens the glenoid cavity and so may assist in stabilizing the humeral head on the fossa, although the mechanism of this function remains uncertain. It may protect the bone and probably assists lubrication.

Fibrous capsule A fibrous capsule envelops the joint (see Figs 48.16, 48.22). It is attached medially to the glenoid neck at a variable distance from the glenoid labrum, so forming recesses anterior and posterior to the scapular neck; it includes the supraglenoid tubercle, to which the long head of biceps attaches, and which is therefore intracapsular. Laterally, it is attached to the anatomical neck of the humerus, i.e. near the articular margin, except inferomedially, where it descends more than 1 cm on the calcar humerale. The fibres of the capsule are orientated in a spiral fashion, so that, in elevation of the arm, the capsule tightens, so bringing the articular surfaces into closer apposition and contributing to the concavity compression. In some individuals, a deeper band of fibres orientated in a reverse obliquity to the main capsular fibres passes from the region of the lesser tubercle to the lower part of the anterior rim of the glenoid fossa. This is the anterior oblique band; its contribution to stability is uncertain.

The lateral part of the external surface of the fibrous capsule is blended with the tendons of supraspinatus (superiorly), infraspinatus and teres minor (posteriorly), subscapularis (anteriorly) and by the tendon of the long head of triceps at the infraglenoid tubercle (inferiorly). The rotator interval is a medially based, triangular area of uncovered capsule between the superior edge of subscapularis and the anterior edge of supraspinatus, with the base of the coracoid as the medial boundary of the interval. It is reinforced by the coracohumeral ligament, the superior glenohumeral ligament, and fibres from the supraspinatus tendon laterally. The axillary nerve and posterior circumflex

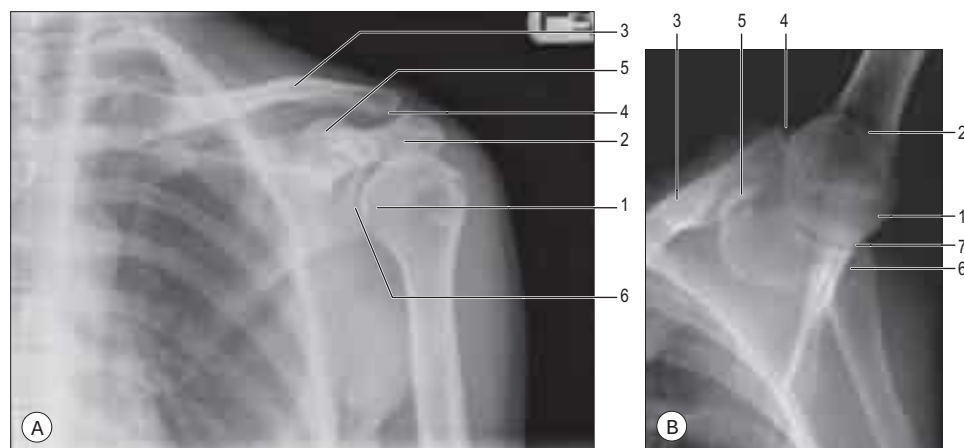


Fig. 48.21 Radiographs of the left shoulder of an 18-year-old female. **A**, Anteroposterior view. **B**, Axillary view with the arm abducted. Key: 1, head of humerus; 2, acromion; 3, clavicle; 4, acromioclavicular joint; 5, coracoid process; 6, glenoid fossa; 7, glenohumeral articulation.

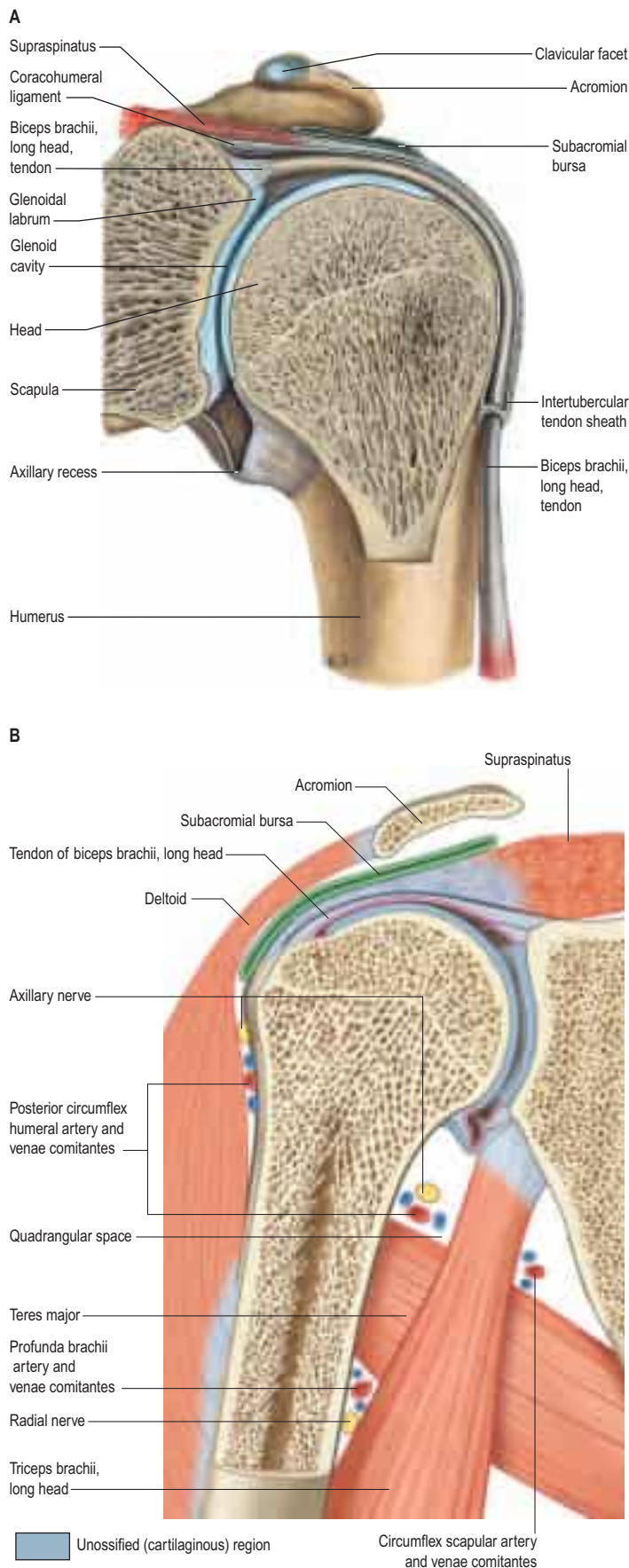


Fig. 48.22 Coronal sections through the left shoulder joint, posterior aspect. **A**, Anterior view. **B**, Posterior view. Note the subacromial bursa and contents of the quadrangular space. (A, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

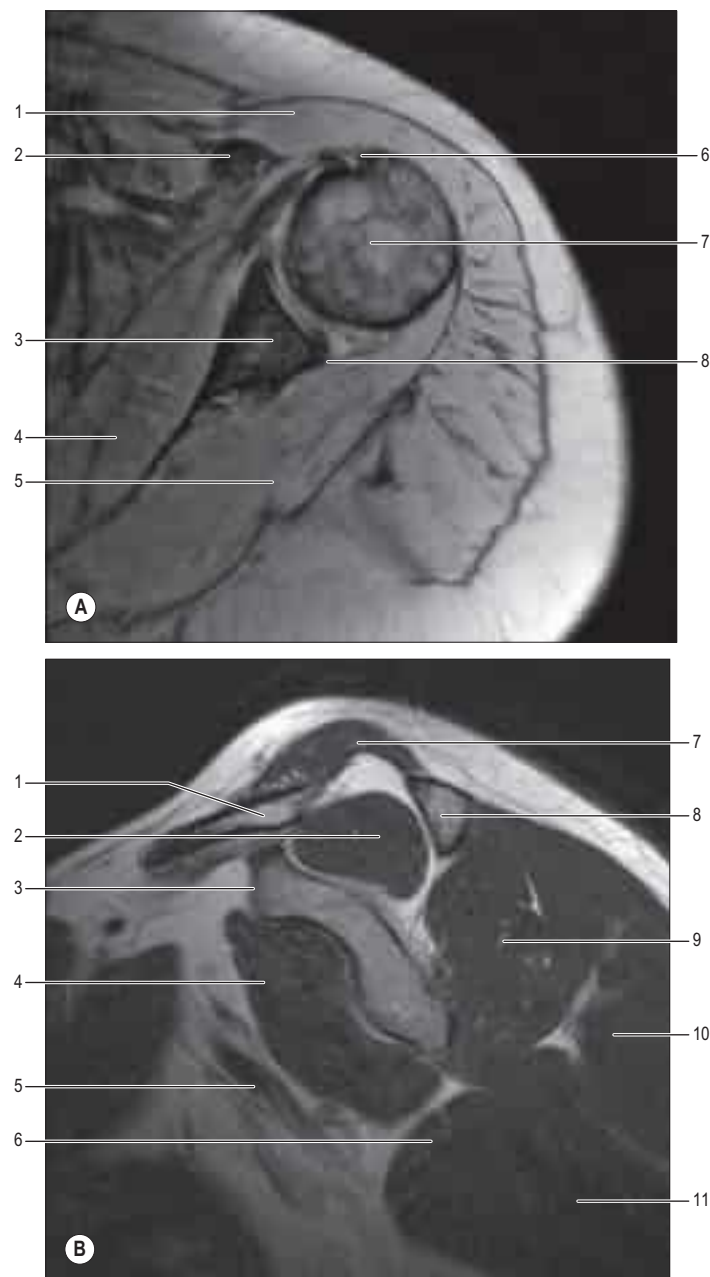


Fig. 48.23 MRI of the shoulder. **A**, Axial T1 weighted view. Key: 1, deltoid; 2, coracobrachialis; 3, glenoid; 4, subscapularis; 5, infraspinatus; 6, tendon of the long head of biceps; 7, head of humerus; 8, posterior labrum. **B**, Median sagittal oblique image at the base of the coracoid. Key: 1, clavicle; 2, supraspinatus; 3, coracoid; 4, subscapularis; 5, axillary artery; 6, latissimus dorsi; 7, trapezius; 8, scapular spine; 9, infraspinatus; 10, deltoid; 11, triceps. Compare A with Figure 48.46.

humeral vessels are separated from the inferior capsule by the inferior border of subscapularis. The capsule is least supported inferiorly, and subjected to the greatest strain in full abduction, when it is stretched tightly across the humeral head. It is supported in this position by the proximal part of the long head of triceps and the broad bulk of teres major.

There are two openings in the capsule: below the coracoid process, connecting the joint cavity to a bursa that may envelop the upper border of the tendon of subscapularis (anteriorly); and between the humeral tubercles, allowing passage of the long tendon of biceps and its synovial sheath.

Ligaments The ligaments associated with the glenohumeral joint are the glenohumeral (superior, middle and inferior), coracohumeral and transverse humeral. A ligament only restricts a specific motion of a joint when discrete parts of the ligament become tight; this only occurs at the limits of a specific motion of the joint when the ligament acts as a 'check-rein'. For movements that lie within the limit of motion, ligaments have no mechanical role. Mechanoreceptors and nociceptors are clustered at each end of the ligaments, more on the glenoid side than the humeral side (Halata and Baumann 2008). The capsule, ligaments

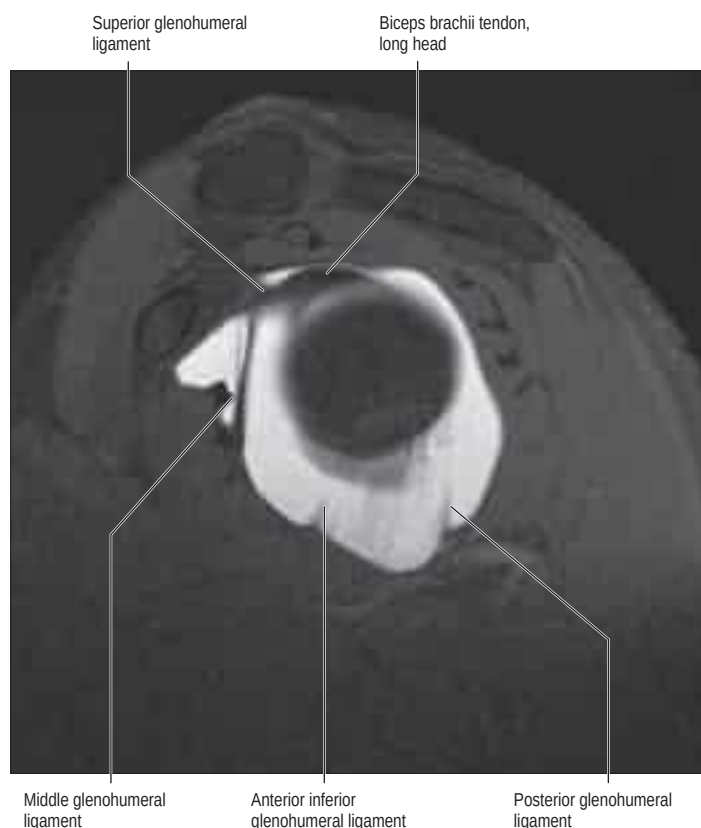


Fig. 48.24 A sagittal magnetic resonance arthrogram image following distension of the glenohumeral joint by intra-articular contrast injection in a 28-year-old male. Note the clearly outlined glenohumeral ligaments.

and labrum contain the deep afferent innervation of the glenohumeral joint and therefore contribute to stability of the joint through facilitation of the rotator cuff activity within the normal range of human shoulder motion.

Glenohumeral ligaments Three glenohumeral ligaments, only visible from within the joint, reinforce the capsule anteriorly and inferiorly (Fig. 48.24). The superior glenohumeral ligament passes from the supraglenoid tubercle, just anterior to the origin of the long head of biceps, to the humerus at the fovea capitis, near the proximal tip of the lesser tubercle on the medial ridge of the intertubercular sulcus. It forms an anterior cover around the long head of biceps and is part of the rotator interval. Together with the coracohumeral ligament, it is an important stabilizer in the inferior direction, helping to keep the humeral head suspended (the coracohumeral ligament is more robust than the superior glenohumeral ligament).

The middle glenohumeral ligament arises from a wide attachment below the superior glenohumeral ligament, along the anterior glenoid margin as far as the inferior third of the rim, and passes obliquely inferolaterally, enlarging as it does so, to attach to the lesser tubercle deep to the tendon of subscapularis, with which it blends. The width and thickness of this ligament may be as much as 2 cm and 4 mm, respectively. It provides anterior stability between 45° and 60° of abduction in the scapular plane. However, 30% of individuals may not have a middle glenohumeral ligament; these individuals are not more likely to have unstable shoulders. It may be thickened and cord-like in the Buford complex (a congenital glenoidal labrum variant), again with no apparent disadvantage for stability.

The thicker and longer inferior glenohumeral ligament complex is a hammock-like structure with anchor points on the anterior and posterior sides of the glenoid. It arises from the anterior, middle and posterior margins of the glenoid labrum, below the epiphysal line, and passes anteroinferiorly to the inferior and medial aspects of the neck of the humerus. The anterior, superior edge of the inferior ligament is thickened as the anterior band; there is an equivalent, but not so robust, thickening of the posterior capsule: the posterior band. Between the two bands is a region of thin capsule known as the axillary pouch. (For further details, consult [Burkart and Debski \(2002\)](#); [Di Giacomo et al \(2008\)](#).)

Coracohumeral ligament The coracohumeral ligament is attached to the dorsolateral base of the coracoid process and extends as two bands,

which blend with the capsule, to the greater and lesser tubercles (see Fig. 48.16). Portions of the coracohumeral ligament form a tunnel for the biceps tendon on the anterior side of the joint. The rotator interval is reinforced by the coracohumeral ligament. It also blends inferiorly with the superior glenohumeral ligament. The ligament tightens in external rotation of the glenohumeral joint.

Transverse humeral ligament The transverse humeral ligament is a broad band that passes between the humeral tubercles, and is attached superior to the epiphysal line (see Fig. 48.16). It converts the intertubercular sulcus into a canal, and acts as a retinaculum for the long tendon of biceps.

Synovial membrane The synovial membrane lines the capsule and covers parts of the anatomical neck. The long tendon of biceps brachii traverses the joint in a synovial sheath that continues into the intertubercular sulcus as far as the surgical neck of the humerus (see Figs 48.16, 48.22A). The intra-articular bicipital sheath may be connected to the articular surface of the superior capsule above by a mesotenon, which may be incomplete, presenting as one or several strands. This arrangement can only be the case if the biceps tendon was originally an extra-articular structure and invaginated, with its synovial sheath, into the glenohumeral joint as the upper limb rotated into external rotation relative to the trunk during development.

Bursae Many bursae adjoin the shoulder joint (see Fig. 48.16). They are usually found between the tendon of subscapularis and the capsule, communicating with the joint between the superior and middle glenohumeral ligaments; on the superior acromial aspect; between the coracoid process and capsule; between teres major and the long head of triceps; and anterior and posterior to the tendon of latissimus dorsi. The subacromial bursa, between deltoid and the capsule, does not communicate with the joint cavity but is prolonged under the acromion and coraco-acromial ligament, and between them and supraspinatus: it appears to be attached, together with the subdeltoid fascia, to the acromion. Bursae sometimes occur behind coracobrachialis and between the tendon of infraspinatus and the capsule, occasionally opening into the joint.

Vascular supply The glenohumeral joint is supplied by branches from the anterior and posterior circumflex humeral, suprascapular and circumflex scapular vessels.

Innervation The glenohumeral joint is innervated mainly from the posterior cord of the brachial plexus through the subscapular nerves. The capsule is supplied by the suprascapular nerve (posterior and superior parts), axillary nerve (anteroinferior) and the lateral pectoral nerve (anterosuperior, including the rotator interval capsule).

Factors maintaining stability The articulation between the relatively large humeral head and the shallow glenoid fossa allows a wide range of movement but is a challenge for the stability of the joint.

MOVEMENTS AT THE GLENOHUMERAL JOINT



The shoulder is capable of any combination of swing and spin over a range far wider than that of any other joint in the body. Flexion–extension, abduction–adduction, and medial (internal) and lateral (external) rotation all occur at the shoulder. A combination of all of these results in the motion of circumduction, which is not a useful movement to record in clinical practice. Although the majority of the movement of the shoulder occurs at the glenohumeral joint, the scapulothoracic articulation contributes to overall shoulder motion in all directions, including lateral rotation.

Muscles producing movements The muscles that produce movements at the glenohumeral joint are principally the scapulohumeral and thoracobrachial muscles: deltoid, pectoralis major and latissimus dorsi, assisted by coracobrachialis. The translating effect of these muscles on the shoulder joint is counteracted by the rotator cuff, a group of short muscles (subscapularis, supraspinatus, infraspinatus and teres minor), including teres major, that are attached closer to the joint, and that centre the head of the humerus in the glenoid fossa through the entire range of motion.

Flexion Flexion is carried out by pectoralis major (clavicular part), deltoid (anterior fibres) and coracobrachialis, assisted by biceps. The

In anatomical analysis of shoulder movements, it is usual to relate humeral movement to the scapula, rather than to conventional anatomical planes. However, in clinical practice, movements *are* related to the conventional anatomical planes. When the arm hangs at rest, the glenoid fossa faces almost equally forwards and laterally, and the humeral capitular and scapular (topographical) axes correspond, although the humerus, relative to the anatomical position, is medially rotated. With the humerus in the anatomical position, i.e. with the inter-epicondylar axis in the frontal plane, the humeral head axis is posteriorly directed, in line with the axis of the scapula.

Using the anatomical description, flexion carries the arm antero-medially on an axis through the humeral head orthogonal to the glenoid fossa at its centre. Abduction and adduction occur in a vertical plane orthogonal to that of flexion-extension and the axis is horizontal, through the humeral head and parallel with the glenoid plane. Pure abduction raises the arm anterolaterally in the plane of the scapula. However, when referred to the trunk, flexion and extension occur in the paramedian plane, and abduction and adduction occur in the coronal plane. In this sense, raising the arm vertically from flexion or raising it from abduction are both accompanied by humeral rotation in opposite directions. Whether 'scapular' or any other plane of abduction is described, these are selections from an infinite series. In scapular abduction, points on the humeral surface pursue vertical cords, but in rotation, they are horizontal. In 'pure' flexion-extension, in a plane orthogonal to the scapula, the axis of movement and the notional 'mechanical axis' are regarded as projected from the centre of the glenoid cavity.

Glenohumeral abduction is about 120° or more (see p. 831; see Fig. 48.21). Abduction may reach 180° after thoracoscapular arthrodesis for facioscapulohumeral dystrophy (Copeland and Howard 1978). Abduction is restricted to about 45° after glenohumeral arthrodesis. Some 60° of further abduction occurs at the sterno- and acromioclavicular articulations, and contralateral vertebral flexion also aids in bringing the arm to the vertical. During active elevation, movements at the glenohumeral and acromioclavicular joints are simultaneous, except in the initial few degrees, when most, often all, movement is glenohumeral. For every 15° of elevation, glenohumeral movement is

said to be 10° and scapular movement 5°, but this ratio is not linear throughout the range of motion. During the initial stages of abduction, the entire rotator cuff counteracts the strong upward component of the pull of deltoid, which would otherwise cause the humeral head to slide up the glenoid surface; the additive turning moments exerted by the combined deltoid and supraspinatus force-couple may then abduct the arm. Deltoid also acts with the other elements of the rotator cuff, according to the position of the arm in space, to maintain rotation, flexion and extension of the arm. It is important to consider the synergistic relation between the rotator cuff and deltoid as a continuum in which the rotator cuff muscles act to provide a stable base on which deltoid can work effectively, whatever the position of the arm.

In flexion, the humerus swings at right angles to the scapular plane, and scapular rotation cannot increase the elevation (120°) that is obtainable in full flexion. If the fully flexed humerus is also abducted, elevation increases *pro rata* until, when the humerus reaches the scapular plane, i.e. when true abduction is reached, 180° of elevation becomes possible. In medial or lateral rotation, the humerus revolves about one-quarter of a circle around a vertical axis; the range is greatest when the arm is pendant, and least when it is vertical. When the rotational range at the glenohumeral joint is assessed, the forearm should be flexed to a right angle at the elbow in order to prevent the effects of superadded pronation or supination in the pendant limb. In circumduction, which is a succession of the foregoing movements, the distal end of the humerus describes the base of a cone with its apex at the humeral head. This glenohumeral movement may be greatly increased by scapular movements, e.g. in acts of slinging objects with force.

The peculiar relation of the long head of biceps brachii to the shoulder joint may serve several purposes. By its connection with both the shoulder and elbow, the muscle harmonizes their actions as an elastic connection during all upper limb movements. It helps to prevent the humeral head impinging on the acromion when deltoid contracts and also to steady it in movements of the arm. In paralysis or absence (i.e. in rotator cuff tendon tears) of supraspinatus, the long head of biceps may also help initiate abduction of the arm, particularly when the humerus is laterally rotated.

sternocostal part of pectoralis major is a major force in flexion forwards to the coronal plane from full extension.

Extension Extension is carried out by deltoid (posterior fibres) and teres major, from the dependant position. When the fully flexed arm is extended against resistance, latissimus dorsi and the sternocostal part of pectoralis major act powerfully until the arm reaches the coronal plane.

Abduction Abduction is carried out by supraspinatus and deltoid. The effect of deltoid is mainly upward and, unless opposed, this would displace the humerus upwards. Subscapularis, supraspinatus, infraspinatus and teres minor exert a centralizing force and so apply a balancing force; together with deltoid, they constitute a 'couple' to produce abduction in the scapular plane.

Medial rotation Medial rotation is carried out by pectoralis major, deltoid (anterior fibres), latissimus dorsi, teres major and, with the arm pendant, subscapularis.

Lateral rotation Lateral rotation is carried out by infraspinatus, deltoid (posterior fibres) and teres minor. Lateral rotation is important for clearance of the greater tubercle and its associated tissues as it passes under the coraco-acromial arch during elevation of the arm.



Glenohumeral joint dislocation The glenohumeral joint is the most frequently dislocated joint in the body. The vast majority of dislocations are anterior, and occur when the forearm is forced backwards into greater lateral (external) rotation when it is in abduction, lateral rotation and extension. A dislocated shoulder loses its normal contour because the humeral head is displaced anterior to the glenoid rim and, sometimes, under the coracoid, which means that the acromion process, rather than the greater tubercle, becomes the most lateral bony structure, affording the shoulder a 'squared-off' appearance.

MUSCLES

The shoulder girdle and shoulder joint musculature are arranged in layered, functional groups (Table 48.1). The muscles of the shoulder girdle may suspend or move the scapula; these functions are interdependent. The muscles of the shoulder joint may stabilize the joint, move the arm on the stable shoulder joint, or coordinate shoulder and elbow movement; these functions are also interdependent. The closer a muscle is to the centre of motion of the joint on which it acts, the greater its effect on the stability of that joint at rest and during motion. The farther away from the centre of motion, the greater the effect of the resultant muscular vector on translation of the joint surfaces. As an

example, both subscapularis and pectoralis major generate medial rotation of the glenohumeral joint; subscapularis contributes to concavity compression at the glenohumeral joint (i.e. stability), while pectoralis major causes anterior displacement into flexion and adduction in the absence of subscapularis; when these muscles are working synergistically, the arm is both stabilized at the glenohumeral joint, and flexed and adducted.

The clinical examination of the shoulder girdle muscles is demonstrated in Video 46.1.



MUSCLES OF SCAPULAR SUSPENSION

Trapezius acts on the spine of the scapula, acromion and lateral clavicle to rotate the acromioclavicular joint upwards and dorsally; movement is guided and limited by the length and shape of the clavicle. Pectoralis minor acts to control the position of the coracoid as trapezius acts on the scapula, while subclavius acts to control clavicular elevation and rotation. These muscles act on the processes of the scapula, while the coraco-acromial ligament acts as a tension band between the processes. Disruption of the continuity of this arrangement (fracture of the clavicle or coracoid, or dislocation of the acromioclavicular joint) will have a profound effect on scapular suspension.

Trapezius

Trapezius is derived from head paraxial mesenchyme with neural crest connective tissue, rather than from the dermomyotomes of lower somites, and is therefore unique among shoulder girdle muscles. It is attached to the spine of the scapula and acromion, segments of the scapula for which development is encoded by a single gene (804.e1).

Attachments Trapezius is a flat, triangular muscle that extends over the back of the neck and upper thorax (see Fig. 53.16). The paired trapezius muscles form a diamond shape, from which the name is derived. On either side, the muscle is attached to the medial third of the superior nuchal line, external occipital protuberance, ligamentum nuchae, and apices of the spinous processes and their supraspinous ligaments from C7 to T12. The occipital attachment is by a fibrous lamina, which is also adherent to the skin. The spinal attachment is by a broad triangular aponeurosis from the sixth cervical to the third thoracic vertebrae, and by short tendinous fibres below this.

Superior fibres descend, inferior fibres ascend, and the fibres between them are directed horizontally; all converge laterally on the scapula. The superior fibres are attached to the posterior border of the lateral third of the clavicle; the middle fibres to the medial acromial margin and superior lip of the crest of the scapular spine; and the inferior fibres pass into an aponeurosis that glides over a smooth triangular surface at the medial end of the scapular spine and is attached to a tubercle at its lateral apex.

Variants The clavicular attachment of trapezius varies in extent, sometimes reaching mid-clavicle, and occasionally blending with sternocleidomastoid. The vertebral attachment sometimes stops at the eighth thoracic spine. The occipital attachment may be absent. Cervical and dorsal parts are occasionally separate. The inferior part may be hypoplastic or absent.

Vascular supply The upper third of trapezius is supplied by a transverse muscular branch that arises from the occipital artery at the level of the mastoid process. It enters the muscle on its deep surface and gives off several musculocutaneous perforators to the overlying skin. The middle portion of trapezius, together with an area of overlying skin, is supplied by the superficial cervical artery or by a superficial branch of the transverse cervical artery, via musculocutaneous perforators. The lower third of trapezius is supplied by a muscular branch from the dorsal scapular artery, passing medial to the medial border of the scapula. It reaches the deep surface of the muscle either by piercing the rhomboids or by passing between rhomboids major and minor at the level of the base of the spine of the scapula. It anastomoses with the medial and lateral perforating branches of the posterior intercostal arteries.

Innervation Trapezius is innervated by the accessory nerve (see Chs 29 and 46 for more details). Sensory (proprioceptive) branches are derived from the ventral rami of C3 and C4.

Actions Trapezius cooperates with other muscles in steadying the scapula, controlling it during movements of the arm and maintaining the level and poise of the shoulder. Electromyographic activity is

Table 48.1 The relationship between functional muscle grouping and actions of the pectoral girdle*

Functional group	Specific muscles	Action
Occipitospinoscapular	Trapezius	Scapular suspension
Thoracoscapular	Pectoralis minor	Position and motion of the acromioclavicular joint in relation to the thorax and (neur)axis
Spinoclavicular	Trapezius	Deceleration of the lateral clavicle during motion
Thoracoclavicular	Subclavius	
Thoracoscapular (medial border)	Levator scapulae	Scapular motion
	Rhomboid minor	Position and motion of the glenohumeral joint in relation to the thorax and (neur)axis
	Rhomboid major	
	Serratus anterior	
	Serratus posterior superior	
Thoracohumeral	Serratus posterior inferior	
	Pectoralis major	Thoracobrachial motion
Scapulohumeral	Latissimus dorsi	Position of the arm
	Deltoid	
	Coracobrachialis	
	Biceps brachii (short head)	
	Teres minor	Concavity compression of the glenohumeral joint
	Infraspinatus	
	Supraspinatus	
	Subscapularis	
	Teres major	
	Triceps brachii (long head)	Coordination of shoulder and elbow motion
	Biceps brachii (long head)	

*As a principle, a muscle will act on a motion segment to alter the position of the distal extent of that segment in space. As an example, deltoid acts on the shoulder joint but its effect is to move the distal extent of the humerus, i.e. the elbow.

The axillary nerve and vessels may be injured during dislocation or, more commonly, during inexperienced attempts at relocation, particularly in older patients. This can lead to an inability to maintain abduction of the shoulder as a result of paralysis of deltoid. If, after relocation of the joint, the patient cannot *initiate* elevation (abduction) from the pendant position, then there may be an associated rotator cuff rupture and/or suprascapular nerve lesion. If the patient can initiate, but cannot maintain, abduction beyond 15° , then deltoid palsy should be suspected. Patients, particularly the elderly, can have a rotator cuff rupture and an axillary nerve palsy causing deltoid paralysis; it can be difficult to distinguish between the two. There may also be an area of hypoaesthesia or anaesthesia over the distal part of the muscle (sometimes referred to as the 'regimental badge patch' of skin; see [Fig. 48.2](#)), reflecting loss of function in the upper lateral cutaneous nerve of the arm (a branch of the axillary nerve), as well as venous occlusive changes (swelling) in the limb. After more violent dislocation, younger patients may have deltoid paralysis and an intact, but stretched, rotator cuff or partial tearing of the tendons, with a suprascapular nerve injury; these patients may have unexpected pain in the supraspinous fossa with weakness of infraspinatus (lateral rotation), as well as the anticipated anterior shoulder pain. Posterior dislocation is rare and typically occurs when violent movements produce marked medial rotation and adduction, e.g. in epileptic seizures or electric shock. The upper limb is held in fixed medial rotation, and external rotation is impossible to perform.

minimal in the unloaded arm, and heavy loads can be suspended with only a small contribution from the upper part; it is the lower part of the muscle that resists lateral rotation of the scapula under load. Acting with levator scapulae, the upper fibres elevate the scapula and, with it, the point of the shoulder; acting with serratus anterior, the lower fibres of trapezius protract and rotate the scapula medially (upwards) so that the arm can be raised above the head; acting with the rhomboids, it retracts the scapula, bracing the shoulder. With the shoulder fixed, trapezius may bend the head and neck backwards and laterally. Trapezius, pectoralis minor, levator scapulae, rhomboids and serratus anterior combine to produce a variety of scapular rotations see below.

If the muscle is denervated, the functions of trapezius cannot be substituted readily by muscle transfer; its unique suite of complex actions suggest that it should rarely, if ever, be used as a muscle for transfer elsewhere.



Testing The upper fibres of trapezius are palpated while the shoulder is shrugged against resistance (see Video 46.1). The lower fibres are palpated during scapular retraction. A better discriminative test is as follows: the hand is brought by medial rotation to the small of the back and the patient is asked to maintain it firmly in this position while the examiner tries to pull the hand off the back. The lower fibres of the trapezius are activated to hold the scapular spine horizontal. Failure to do so implies paresis or paralysis of trapezius, but this must be distinguished from congenital hypoplasia of the lower fibres of the muscle.

Trapezius palsy results in a characteristic posture: the shoulder is tilted laterally and forwards, the acromioclavicular joint is lowered, the superior pole of the scapula is rotated upwards and the inferior pole is displaced medially and, usually, away from the chest wall. This distinguishes trapezius palsy from that of serratus anterior, in which the entire medial border of the scapula tends to elevate away from the chest wall (so-called 'winging'). The scapula is displaced upwards and towards the spine (see p. 838). The altered posture and disordered motion of the scapula create distortion (narrowing and elongation) of the cervico-axillary sheath and its contents; symptoms of reduced venous return, lymphatic obstruction and disturbed neural perfusion (pain, paraesthesiae, dysaesthesiae) are almost universal.

Pectoralis minor

Attachments Pectoralis minor is a thin, triangular muscle lying posterior (deep) to pectoralis major (Fig. 48.25). It arises from the upper

margins and outer surfaces of the third to fifth ribs (frequently, second to fourth), near their cartilages, and from the fascia over the adjoining external intercostal muscles. Its fibres ascend laterally under cover of pectoralis major, converging in a flat tendon that is attached to the medial border and upper surface of the coracoid process of the scapula. Part or all of the tendon may appear to cross the coracoid process into the coraco-acromial ligament.

Variants Slips of the muscle are sometimes separated and vary in number and level. In rare cases, one passes from the first rib to the coracoid (pectoralis minimus). The costal attachments can be second to fifth ribs; third to fifth; second to fourth; or third to fourth. The muscle may be present or absent when pectoralis major is absent (in Poland's syndrome).

Relations Pectoralis major, the lateral pectoral nerve and pectoral branches of the thoraco-acromial artery are anterior. The ribs, external intercostals, serratus anterior, the axilla, axillary vessels, lymphatics and brachial plexus are all posterior. The upper border of pectoralis minor is separated from the clavicle by a triangular gap filled by the clavipectoral fascia, behind which are the axillary vessels, lymphatics and nerves. The lateral thoracic artery follows the lower border of the muscle. The medial pectoral nerves pierce and partly supply the muscle.

Vascular supply Pectoralis minor is supplied by pectoral and deltoid branches of the thoraco-acromial and superior and lateral thoracic arteries.

Innervation Pectoralis minor is innervated by branches of the medial and lateral pectoral nerves: C5, 6, 7, 8 and T1.

Actions Pectoralis minor assists serratus anterior in drawing the scapula forwards around the chest wall. With levator scapulae and the rhomboids, it rotates the scapula, depressing the point of the shoulder. It resists scapular rotation caused by trapezius. Both pectoral muscles are electromyographically quiescent in normal inspiration, but are active in forced inspiration.

Subclavius

Attachments Subclavius is a long, asymmetric triangular muscle between the clavicle and the first rib (see Fig. 48.25). It arises from

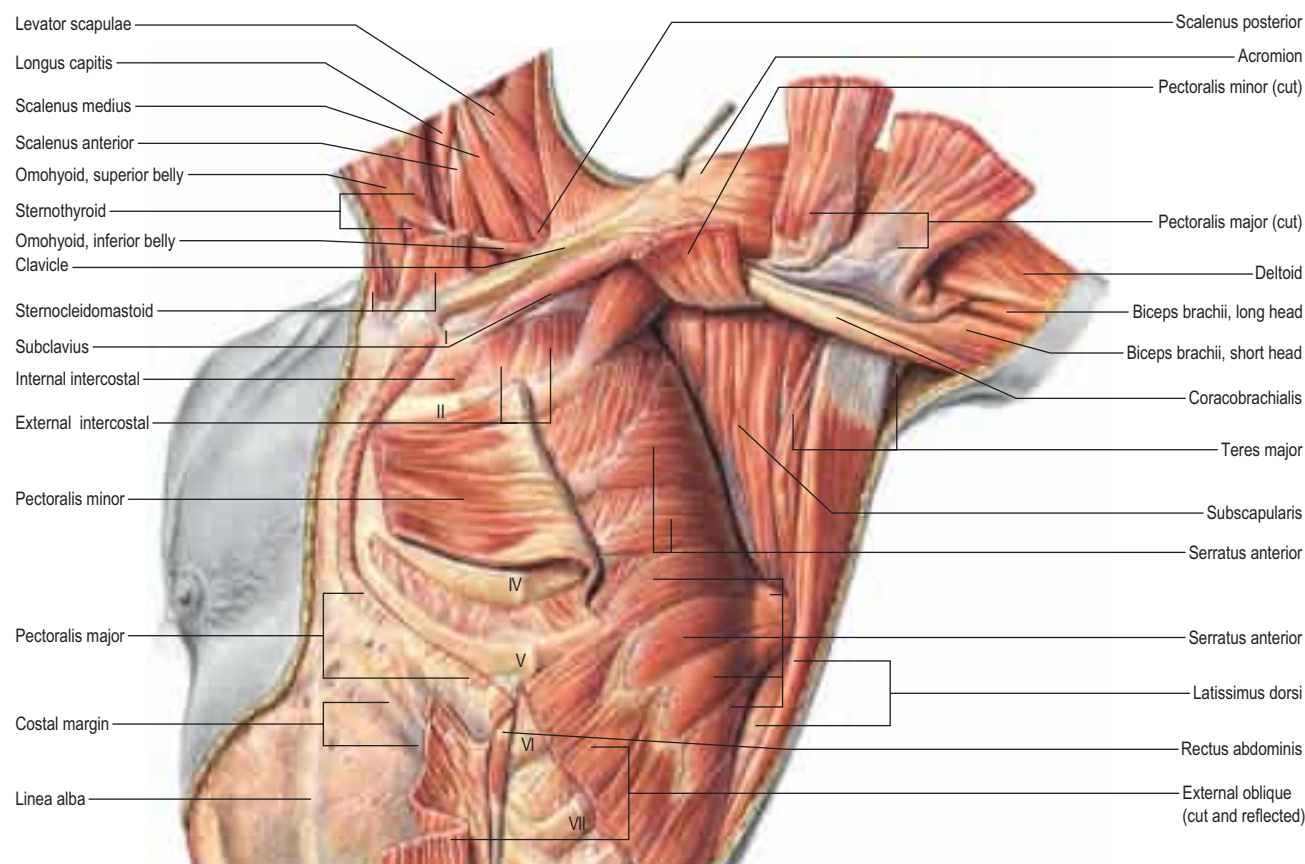


Fig. 48.25 Deep muscles of the chest and left shoulder girdle, anterior aspect. Roman numerals refer to costal cartilages. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

the junction of the first rib and its costal cartilage by a thick tendon, prolonged at its inferior margin and anterior to the costoclavicular ligament. It passes upwards and laterally to attach by direct muscular fibres into a groove on the undersurface of the middle third of the clavicle; the lower and most lateral fibres are the longest. Subclavius may be attached to the coracoid process or the upper border of the scapula as well as, or instead of, the clavicle.

Relations Posteriorly, subclavius is separated from the first rib by the subclavian vessels and brachial plexus; anteriorly, it is separated from pectoralis major by the anterior lamina of the clavipectoral fascia. The suprascapular artery, which has a variable pattern, descends behind the muscle, passing between it and omohyoid anteriorly and the subclavian vessels posteriorly; the artery, therefore, has a close relationship to the posterior surface of the clavicle, a feature that makes it susceptible to injury during clavicular fracture and surgical fixation.

Vascular supply Subclavius is supplied by the clavicular branch of the thoraco-acromial artery and the suprascapular artery.

Innervation Subclavius is supplied by the subclavian branch of the brachial plexus (the nerve to subclavius), C5 and 6. A muscle, however apparently insubstantial, that has a unique, solitary nerve supply must have an important function; subclavius is no exception (see above and below).

Actions Subclavius resists accelerated elevation and rotation of the clavicle during elevation of the shoulder girdle, and may help to close-pack the medial end of the clavicle against the articular disc of the sternoclavicular joint for greater stability under load. It is inaccessible to palpation and difficult to investigate by electromyography. Incidentally, it 'protects' the subclavian vessels and brachial plexus that lie behind it.

MUSCLES OF SCAPULAR MOTION

Levator scapulae

Attachments Levator scapulae is a slender muscle attached by tendinous slips to the transverse processes of the atlas and axis, and to the posterior tubercles of the transverse processes of the third and fourth cervical vertebrae (see Fig. 29.6; Fig. 48.26). It descends diagonally to attach to the medial scapular border between its superior angle and the triangular smooth surface at the medial end of the scapular spine.

Variants Levator scapulae varies considerably in its vertebral attachments and the extent to which it separates into slips. There may be accessory attachments to the mastoid process, occipital bone, first or second rib, the scaleni, trapezius and the serratus muscles.

Vascular supply Levator scapulae receives its arterial supply mainly from the transverse cervical and ascending cervical arteries. The vertebral extremity of the muscle is supplied by branches from the vertebral artery.

Innervation Levator scapulae is innervated directly by branches of the third and fourth cervical spinal nerves, and from the fifth cervical nerve via the dorsal scapular nerve.

Actions Levator scapulae elevates the superior angle of the scapula. If unopposed, this causes a lateral rotation of the scapula. With the cervical vertebral column fixed, levator scapulae acts with trapezius to elevate the scapula or to sustain a weight carried on the shoulder; with the shoulder fixed, the muscle inclines the neck to the same side.

Rhomboid minor

Attachments Rhomboid minor is a small, cylindrical muscle. It runs from the lower ligamentum nuchae and the spines of the seventh cervical and first thoracic vertebrae to the base of a smooth triangular surface at the medial end of the spine of the scapula, where dorsal and ventral layers enclose the inferior border of levator scapulae (see Fig. 48.26). The dorsal layer of rhomboid minor is attached to the rim of the triangular surface, dorsolateral to and below levator scapulae. The ventral layer is strong and wide, extending 2–3 cm medial to and below levator scapulae; here, the fasciae of rhomboid minor and serratus anterior are fused. Rhomboid minor is usually separate from rhomboid major, but the muscles overlap and are occasionally united.

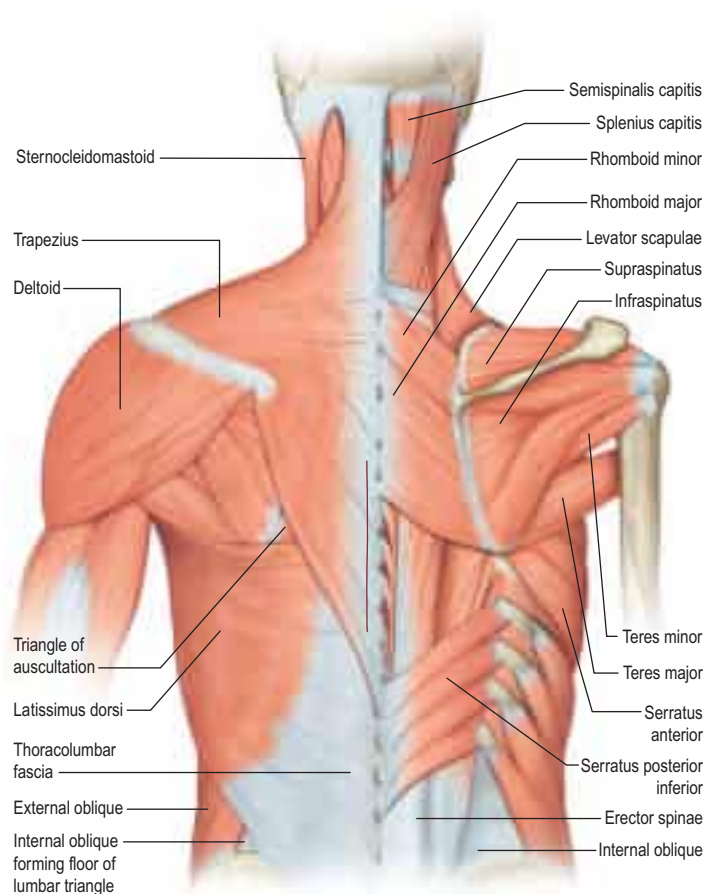


Fig. 48.26 Muscles of the back and shoulder girdles. Superficial muscles are shown on the left. Deeper muscles, revealed after excision of deltoid, latissimus dorsi and trapezius, are shown on the right.

Variants There is some variability in the vertebral and scapular attachments of rhomboids major and minor. A slip of muscle may extend from the upper border of rhomboid minor to reach the occipital bone (rhomboid occipitalis).

Rhomboid major

Attachments Rhomboid major is a quadrilateral sheet of muscle that arises by tendinous fibres from the spines and supraspinous ligaments of the second to fifth thoracic vertebrae, and descends laterally to the medial border of the scapula between the root of the spine and the inferior angle (see Fig. 48.26). Most of its fibres usually end in a tendinous band between these two points, joined to the medial border by a thin membrane. Occasionally, this is incomplete, in which case some muscular fibres are attached directly into the scapula. The attachments of rhomboid major (and also those of rhomboid minor, levator scapulae and serratus anterior) may be more extensive, with 'folds' or extensions passing to both dorsal and costal aspects of the scapula adjacent to its medial margin.

Vascular supply of the rhomboids Rhomboids major and minor are supplied by the dorsal scapular artery or the deep branch of the transverse cervical artery and by dorsal perforating branches from the upper five or six posterior intercostal arteries.

Innervation of the rhomboids Rhomboids major and minor are innervated by a branch of the dorsal scapular nerve, C4, 5.

Actions of the rhomboids Both rhomboid major and rhomboid minor retract the medial border of the scapula superiorly and medially, and are used in retraction of the shoulders.

Testing of the rhomboids Rhomboids major and minor can be palpated deep to trapezius on bracing the shoulder back against resistance with the arm in elevation at shoulder level.

Serratus anterior

Attachments Serratus anterior is a large muscular sheet that curves around the thorax. It arises from an extensive costal attachment and inserts on the scapula (see Fig. 48.25). Fleshy digitations arise anteriorly

from the outer surfaces and superior borders of the upper eight to ten ribs, and from fasciae that cover the intervening intercostal muscles. The line of origin of the muscle is long and slightly curved, and passes inferolaterally around the thorax. There are three parts: the upper fibres arise from the first and second ribs and intercostal fascia, the middle fibres arise from the third to sixth ribs, and the lower fibres interdigitate with the upper five slips of external oblique. The muscle follows the contour of the chest wall closely, reaching the medial border of the scapula anterior to subscapularis. The first (uppermost) digitation encloses, and is attached to, a triangular area of both the costal and the dorsal surfaces of the superior scapular angle. The next two or three digitations form a triangular sheet that is attached to the costal surface along almost its entire medial border. The lower four or five digitations converge to be attached by musculotendinous fibres to a triangular impression on the costal surface of the inferior angle; they enclose the inferior angle and are also attached to a smaller triangular part of its dorsal surface near its tip.

Variants Digitations may be absent, particularly the first and eighth, and sometimes also the intermediate part. Serratus anterior may be partly fused with levator scapulae, adjacent external intercostals or external oblique.

Vascular supply Serratus anterior is supplied by superior and lateral thoracic arteries, and by branches from the thoracodorsal artery before (occasionally, after) it divides in latissimus dorsi.

Innervation Serratus anterior is innervated by the long thoracic nerve, C5, 6 and 7, which descends on the external surface of the muscle.

Actions With pectoralis minor, serratus anterior protracts (draws forwards) the scapula, as a prime mover in all reaching and pushing movements. The upper part, with levator scapulae and the upper fibres of trapezius, suspends the scapula; only slight activity is sufficient to support the unloaded arm. The more extensive lower insertion pulls the inferior scapular angle forwards around the thorax, assisting trapezius in upward rotation of the bone, an action that is essential to raising the arm above the head. In the initial stages of abduction, serratus anterior helps other muscles to fix the scapula, so that deltoid acts effectively on the humerus and not the scapula. While deltoid is raising the arm to a right angle with the scapula, serratus anterior and trapezius are simultaneously rotating the scapula; the combination allows the arm to be raised to the vertical. To effect this upward rotation of the scapula, forward pull on the inferior angle by the lower digitations of serratus anterior is coupled with an upward and medial pull on the lateral end of the clavicle and acromion by the upper fibres of trapezius, and a downward pull on the base of the scapular spine by the lower fibres of trapezius. Conversely, slow downward scapular rotation, assisted by gravity, is achieved by controlled lengthening of these muscles. More powerful downward rotation requires balanced contraction of the upper fibres of serratus anterior, levator scapulae, rhomboids, pectoralis minor and the middle part of trapezius. When weights are carried in front of the body, serratus anterior prevents backward rotation of the scapula. Electromyography shows that serratus anterior is not active in normal human respiration. This may not apply to laboured respiration, e.g.

when asthmatics and athletes may be observed to fix the scapula by grasping a rail or other support.

Testing The muscular digitations of serratus anterior can be seen and felt when the outstretched hand pushes against resistance.

When serratus anterior is paralysed, the medial border of the scapula, and especially its lower angle, stand out prominently. The arm cannot be raised fully. Pushing is ineffective; indeed, attempts to do so produce further projection, known as 'winging' of the scapula (Streit et al 2012; see p. 831).

Serratus posterior inferior

Serratus posterior inferior is described on page 942 (see Fig. 48.26).

Serratus posterior superior

Serratus posterior superior is described on page 941.

MUSCLES OF THORACOBRACHIAL MOTION

The large muscles from the trunk to the arm, and from the scapula to the arm, act on the shoulder joint. Pectoralis major and latissimus dorsi are attached to the metaphysis of the humerus below the centre of rotation of the joint, while deltoid and coracobrachialis are attached to the midpoint of the diaphysis or shaft, some way from the centre of rotation of the shoulder joint. These four muscles move the arm in relation to the shoulder joint, so that, although they act through the shoulder joint, their role is to position the elbow joint (and, thus, the hand) in space. The continuous sheet of muscle that is pectoralis major and deltoid positions the elbow through a wide, asymmetrical, conical range of motion from adduction across the trunk with medial rotation, through flexion, abduction and external rotation, into extension at the shoulder. The cervical spinal segmental levels at which this range of movement is represented, C5 and C6, give rise to the anterior division innervation to pectoralis major (thus, pre-axial or predominantly flexor muscles) and the posterior division innervation (thus, post-axial or predominantly extensor muscles) to deltoid. Similarly, coracobrachialis (a flexor and adductor of the abducted arm) is innervated through the anterior division of C5, 6 and (predominantly) 7 segmental levels, while latissimus dorsi is innervated through the posterior division of C6, 7 and 8 (predominantly C7). These are examples of the *movements* of a joint, rather than individual muscles, being subserved in a biologically efficient manner by the minimum number of spinal segmental levels required to achieve the aim of positioning the elbow in space, while maintaining subtlety in effect.

Pectoralis major

Attachments Pectoralis major (Fig. 48.27) is a thick, fan-shaped muscle. It arises from the anterior surface of the sternal half of the clavicle (clavicular head); half the breadth of the anterior surface of the sternum down to the level of the sixth or seventh costal cartilage (sternal head); the first to the seventh costal cartilages (first and seventh often omitted); the sternal end of the sixth rib; and the aponeurosis of external oblique (rectus head). The clavicular fibres are usually separated from the sternal fibres by a slight cleft. The muscle converges to a flat tendon, approximately 5 cm wide, that is attached to the lateral

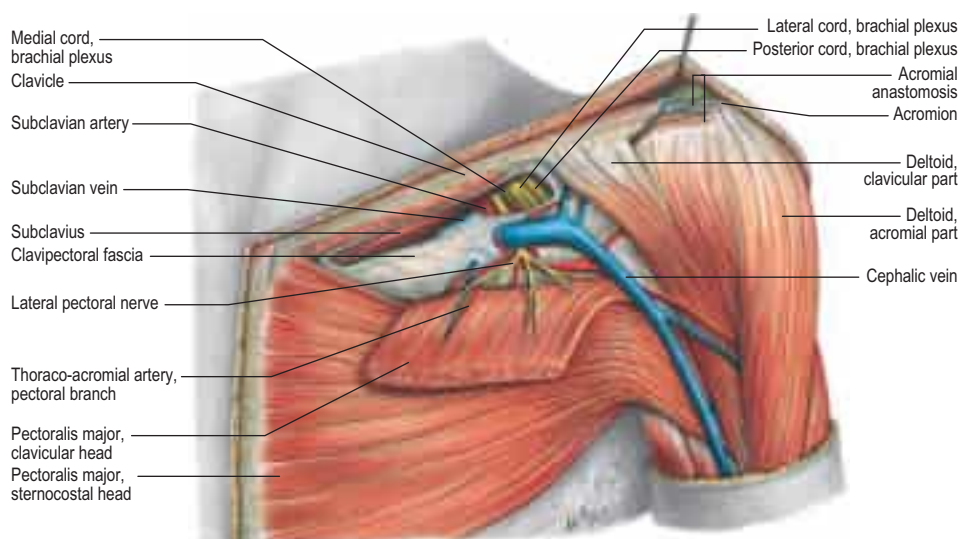


Fig. 48.27 The left deltopectoral groove, dissected to show the clavipectoral fascia and thoraco-acromial axis. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

lip of the intertubercular sulcus of the humerus. The tendon is bilaminar. The thicker anterior lamina is formed by fibres from the manubrium, joined superficially by clavicular fibres and deeply by fibres from the sternal margin and the second to fifth costal cartilages. Clavicular fibres may be prolonged into the tendon of deltoid. The posterior lamina receives fibres from the sixth (and, often, seventh) costal cartilages, sixth rib, sternum, and aponeurosis of external oblique. Costal fibres join the lamina without twisting. Fibres from the sternum and aponeurosis curve around the lower border, turning successively behind those above them, which means that this part of the muscle is twisted so that the fibres that are lowest at their medial origin are highest at their attachment on the humerus. The posterior lamina reaches higher on the humerus than the anterior, and gives off an expansion that covers the intertubercular sulcus and blends with the capsular ligament of the shoulder joint. An expansion from the deepest part of the lamina lines the intertubercular sulcus at its linear insertion; another expansion descends from its lower border into the deep fascia of the upper arm.

The rounded lower border of pectoralis major forms the anterior axillary fold and becomes conspicuous in abduction against resistance.

Variants The abdominal slip from the aponeurosis of external oblique is sometimes absent. The number of costal attachments and the extent to which the clavicular and costal parts are separated vary. Right and left muscles may decussate across the sternum. A superficial vertical slip, or slips, may ascend from the lower costal cartilages and rectus sheath to blend with sternocleidomastoid or to attach to the upper sternum or costal cartilages. This is sternalis (rectus sternalis).

Relations Skin, superficial fascia, platysma, medial and intermediate supraclavicular nerves, breast tissue and deep fascia are all anterior. The sternum, ribs and costal cartilages, clavipectoral fascia, subclavius, pectoralis minor, serratus anterior, external intercostal muscles and membranes are all posterior. Pectoralis major forms the superficial layer of the anterior axillary wall, and hence lies anterior to the axillary vessels and nerves and the upper parts of biceps brachii and coracobrachialis. Its upper border is separated from deltoid by the infraclavicular fossa, which contains the cephalic vein and deltoid branch of the thoraco-acromial artery. Pectoralis major is separated from latissimus dorsi on the medial axillary wall but the two muscles converge as they approach the lateral axillary wall; the floor of the intertubercular sulcus lies between their attachments.

Vascular supply Pectoralis major is supplied by one dominant vascular pedicle from the pectoral branch of the thoraco-acromial axis, supplemented by several smaller secondary segmental vessels from the deltoid and clavicular branches of the thoraco-acromial axis, and perforating branches of the internal thoracic arteries and superior and lateral thoracic arteries.

Innervation Pectoralis major is supplied through the medial and lateral pectoral nerves. Fibres for the clavicular part are from C5 and 6; those for the sternocostal part are from C7, 8 and T1.

Actions The two parts of pectoralis major can act separately or together. The whole muscle assists adduction and medial rotation of the humerus against resistance. It swings the extended arm forwards and medially, its clavicular part acting with the anterior fibres of deltoid and coracobrachialis; the sternocostal part is relaxed. The opposite movement is usually aided by gravity. When it is resisted, the sternocostal part acts together with latissimus dorsi, teres major and the posterior fibres of deltoid; the clavicular part is relaxed. With the raised arms fixed, e.g. gripping a branch, the same combination of muscles draws the trunk up and forwards. Pectoralis major is active in deep inspiration.

Testing To test the clavicular head, the abducted arm is flexed against resistance. To test the sternocostal head, the arm is adducted against resistance.

Poland's syndrome



Available with the *Gray's Anatomy* e-book

Chondro-epitrochlearis Chondro-epitrochlearis is a rare extra 'head' of pectoralis major, arising deep to the sternal and rectus

components of that muscle from the anterior aspect of the fifth, sixth and seventh ribs and their costal cartilages, and in the same plane as pectoralis minor.

Deltoid

Attachments Deltoid is a thick, curved triangle of muscle. It arises from the anterior border and superior surface of the lateral third of the clavicle, the lateral margin and superior surface of the acromion, and the lower edge of the crest of the scapular spine (other than its smooth, medial, triangular surface) (see Figs 48.26, 48.27). The fibres converge inferiorly to a short, substantial tendon that is attached to the deltoid tubercle on the lateral aspect of the midshaft of the humerus. Anterior and posterior fibres converge directly to this tendon. The intermediate part is multipennate; four intramuscular septa descend from the acromion to interdigitate with three septa ascending from the deltoid tubercle. The septa are connected by short muscle fibres that provide powerful traction (Leijnse et al 2008). The fasciculi are large, producing a coarse longitudinal striation. The muscle surrounds the glenohumeral articulation on all sides, except inferomedially, giving the shoulder its rounded profile. In contraction, its borders are easily seen and felt. The tendon gives off an expansion into the brachial deep fascia that may reach the forearm.

Variants Deltoid may fuse with pectoralis major or may receive additional slips from trapezius, the infraspinous fascia or the lateral scapular border. Teres minor shares a common innervation and could be considered as a fourth, posteroinferior, part of deltoid.

Relations The skin, superficial and deep fasciae, platysma, lateral supraclavicular and upper lateral brachial cutaneous nerves are all superficial. The coracoid process, coraco-acromial ligament, subacromial bursa, tendons of pectoralis minor, coracobrachialis, both heads of biceps brachii, pectoralis major, subscapularis, supraspinatus, infraspinatus, teres minor, long and lateral heads of triceps, circumflex humeral vessels, axillary nerve, and the surgical neck and upper shaft of the humerus, including both tubercles, are all deep. The anterior border of deltoid is separated from pectoralis major proximally by the infraclavicular fossa, which contains the cephalic vein and deltoid branches of the thoraco-acromial artery; distally, these muscles are in contact. The posterior border of deltoid overlies infraspinatus and triceps.

Vascular supply Deltoid is supplied by acromial and deltoid branches of the thoraco-acromial artery; the anterior and posterior circumflex humeral arteries; subscapular artery; and the deltoid branch of profunda brachii.

Innervation Deltoid is innervated by the axillary nerve, C5 and 6.

Actions Different parts of the muscle can act independently, as well as together. Anterior fibres assist pectoralis major in drawing the arm forwards and rotating it medially. Posterior fibres act as external rotators, and act with latissimus dorsi and teres major in drawing the arm backwards (into extension). The posterior fibres of deltoid provide up to 80% of the external rotation power of the arm when elevated into the plane of the scapula. The multipennate, acromial, part of deltoid is a strong abductor; aided by supraspinatus, it abducts the arm until the inferior joint capsule is tight. Movement takes place in the plane of the body of the scapula, which is the only way that scapular rotation can be fully effective in raising the arm above the head. In true abduction, acromial fibres contract strongly, while clavicular and posterior fibres prevent departure from the plane of motion. In the early stages of abduction, traction by deltoid is upward, but the humeral head is prevented from translating upward by the synergistic centralizing effect of the rotator cuff muscles (supraspinatus, subscapularis, infraspinatus and teres minor). Electromyography suggests that deltoid contributes little to medial (internal) or lateral (external) rotation but confirms that it takes part in most other shoulder movements.

Testing Deltoid can be seen and felt to contract when the arm is abducted against resistance in the scapular plane. Since this motion can also be achieved by supraspinatus, a more specific test for the activity of deltoid is to assess extension against resistance with the arm in 30° abduction in the scapular plane: this reduces the confounding effect of latissimus dorsi and triceps as extensors of the adducted arm. Subtle weakness can be determined using the deltoid lag sign (Hertel et al 1998).

Poland's syndrome is a rare congenital anomaly, in which there is hypoplasia of the thoracic chest wall muscles and ipsilateral hypoplasia of the arm and hand. It occurs in approximately 1:50,000 live births. The major clinical feature is absence of the sternocostal head of pectoralis major and all of pectoralis minor. In addition, there may be hypoplasia of latissimus dorsi, serratus anterior, external oblique, supraspinatus, infraspinatus, deltoid and the intercostal muscles, and hypoplasia of the hemithorax and ribs. Hypoplasia affecting the arm ranges from syndactyly to symbrachydactyly and ectrodactyly. The second, third and fourth fingers are the most affected; the wrist, forearm, upper arm and scapula are variably involved. There may be ipsilateral breast hypoplasia and absent nipple-areolar complex.

The aetiology is unclear. It has been suggested that the condition is caused by disruption of lateral plate mesenchyme 2–4 weeks after fertilization, or by disruption of the arterial blood supply to the subclavian vessels during the sixth and seventh weeks of embryonic life.

The muscle has an extensive attachment to the lateral crest of the intertubercular sulcus, the medial intermuscular septum and the medial epicondylar ridge. It is innervated by branches of the lateral pectoral nerve that perforate pectoralis minor or pass to the muscle in the plane between pectoralis minor and major. It is noted as a pterygium deep to the anterior axillary wall with a sharply defined inferior border in the axilla, made more noticeable by flexion and medial rotation against resistance with the arm elevated into the scapular plane. With the arm in the elevated position, contraction of the muscle causes a retropulsion of the glenohumeral joint, a rare cause of posterior glenohumeral instability.

Coracobrachialis

Coracobrachialis is the most medial of the three anterior compartment muscles of the arm. It has a common origin with the short head of biceps brachii. In turn, the short head of biceps shares a common distal attachment with the long head of biceps. The three anterior compartment muscles, therefore, form a single functional group that brings the supinated hand into flexion and medial rotation, towards the face.

Attachments Coracobrachialis arises from the deep surface of the apex of the coracoid process, deep to the tendon of the short head of biceps brachii, and by muscular fibres from the proximal 5–10 cm of this tendon. It ends on an impression, 3–5 cm in length, midway along the medial border of the humeral shaft between the attachments of

the medial head of triceps and brachialis (Fig. 48.28). Accessory slips may be attached to the lesser tubercle, medial epicondyle or medial intermuscular septum.

Relations Coracobrachialis forms an inconspicuous rounded ridge on the upper medial side of the arm; pulsation of the brachial artery can be felt and often seen in the depression behind it. The muscle is perforated and supplied by the musculocutaneous nerve. Anteriorly, it is related to pectoralis major above and, at its humeral insertion, to the brachial vessels and median nerve, which cross it. The tendons of subscapularis, latissimus dorsi, teres major, the medial head of triceps, the humerus and the circumflex humeral vessels are all posterior. The third part of the axillary artery and proximal parts of the median and

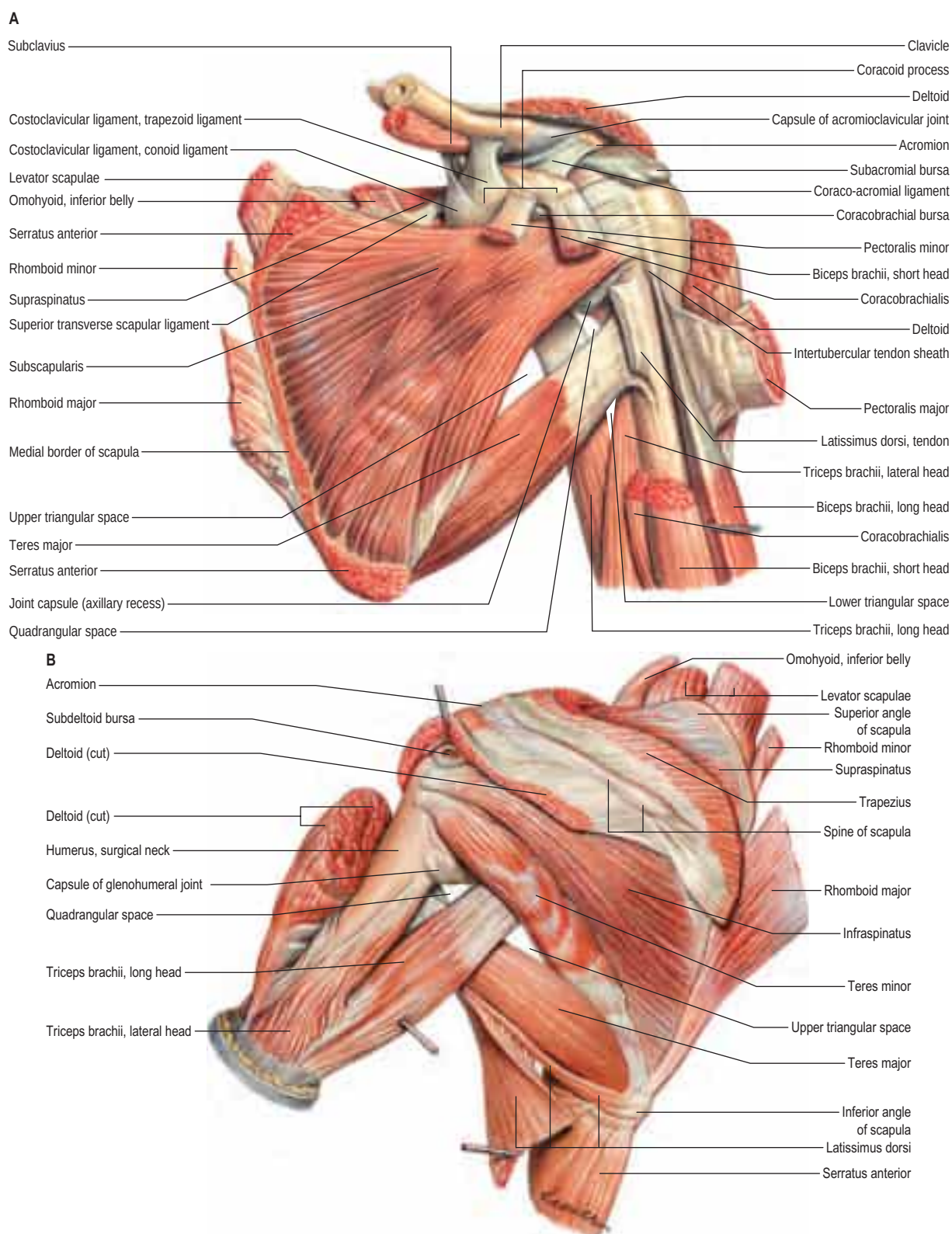


Fig. 48.28 Deep muscles of the left shoulder girdle. **A**, The anterior aspect after removal of the superficial muscles. **B**, Posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

musculocutaneous nerves are first medial and become posterior further distally; they may be partially obscured by the muscle if it is bulky. Biceps brachii and brachialis are lateral.

Vascular supply One or more branches from the axillary artery pass deep to the lateral root of the median nerve and the musculocutaneous nerve to reach the deep surface of coracobrachialis. Branches from the anterior circumflex humeral artery supply the deep surface of the muscle. The artery accompanying the musculocutaneous nerve sends a recurrent branch to the coracoid attachment and gives off a series of branches to coracobrachialis during its intramuscular course. Accessory branches from the thoraco-acromial artery supply the superficial part of coracobrachialis.

Innervation Coracobrachialis is innervated by the musculocutaneous nerve, C5, 6 and 7.

Actions Deltoid and coracobrachialis insert into the humerus approximately opposite each other. Coracobrachialis acts as an antagonist to the action of deltoid, particularly when the arm is in abduction and extension; it flexes the arm forwards and medially. In abduction, it acts with the anterior fibres of deltoid to resist departure from the plane of motion.

Testing Coracobrachialis can be tested by palpating its fibres during adduction and medial rotation of the flexed shoulder against resistance. Alternatively, the muscle can be palpated during forceful depression of the elbow against resistance or with the arm by the side.

Latissimus dorsi

Attachments Latissimus dorsi is a large, flat, triangular muscle that sweeps over the lumbar region and lower thorax, and converges to a narrow tendon (see Fig 48.26). It arises by tendinous fibres from the spines of the lower six thoracic vertebrae anterior to trapezius; from the posterior layer of thoracolumbar fascia, by which it is attached to the spines and supraspinous ligaments of the lumbar and sacral vertebrae; and from the posterior part of the iliac crest. It also springs by muscular fibres from the posterior part (outer lip) of the iliac crest lateral to erector spinae, and by fleshy slips from the three or four lower ribs, interdigitating with external oblique (see Fig 61.6). From this extensive attachment, fibres pass laterally with different degrees of obliquity (the upper fibres are nearly horizontal, the middle oblique, and the lower almost vertical), to form a sheet 12 or 13 mm thick that overlaps the inferior scapular angle, and may attach to it. The muscle curves around the inferolateral border of teres major to gain its anterior surface and ends as a flattened tendon, up to 7 cm long, anterior to the tendon of teres major. It is attached to the floor of the intertubercular sulcus of the humerus with an expansion to the deep fascia (Dancker 2013). The attachment extends higher on the humerus than that of teres major. As the muscle curves round teres major, the fasciculi rotate around each other, so that fibres that originate lowest at the midline insert highest on the humerus, and fibres that originate highest at the midline insert lowest on the humerus. A bursa sometimes occurs between the muscle and the inferior scapular angle. The tendons of latissimus dorsi and teres major are united at their lower borders but separated by a bursa near their humeral attachments.

Latissimus dorsi and teres major together form the posterior axillary fold. When the arm is adducted against resistance, this fold is accentuated, and the whole inferolateral border of latissimus dorsi can be traced to its attachment to the iliac crest. The radial nerve and the branch or branches it supplies to the medial head of triceps are close anterior relations of the tendon of the latissimus dorsi, 3–5 cm medial to the shaft of the humerus, before the radial nerve traverses the lower triangular space and passes into the posterior compartment of the arm (see quadrangular and triangular spaces below).

The lower, lateral margin of latissimus dorsi is usually separated from the posterior border of external oblique by the lumbar triangle. The base of this small triangle is the iliac crest and its floor the internal oblique; it should not be confused with the triangle of auscultation, medial to the scapula, which is bounded above by trapezius, below by latissimus dorsi and laterally by the medial border of the scapula. Part of rhomboid major is exposed in the triangle. If the scapulae are drawn forwards, by folding the arms across the chest, and the trunk is bent forwards, parts of the sixth and seventh ribs and the interspace between them (overlying the apex of the lower pulmonary lobe) become subcutaneous.

Variants Latissimus dorsi commonly receives some additional fibres from the scapula as it crosses the inferior scapular angle. A muscular

axillary arch, 7–10 cm in length and 5–15 mm in breadth, may sometimes be present; it crosses from the edge of latissimus dorsi, midway in the posterior fold, and passes anterior to the axillary vessels and nerves to join the tendons of pectoralis major, coracobrachialis or the fascia over biceps brachii. The vertebral and costal attachments of latissimus dorsi may be reduced or, in rare cases, increased. A fibrous slip usually passes from the tendon, near its humeral insertion, to the long head of triceps.

Vascular supply Latissimus dorsi is supplied by a single dominant vascular pedicle, the thoracodorsal artery, itself a continuation of the subscapular artery. The thoracodorsal artery and its accompanying venae comitantes enter the muscle at a single neurovascular hilum on its costal surface, 6–12 cm from the subscapular artery and 1–4 cm medial to the lateral border of the muscle. The artery gives off up to three large branches to serratus anterior before dividing at, or even before, the neurovascular hilum for latissimus dorsi. The basic pattern of branching is a bifurcation into lateral and medial branches. The larger lateral branch follows a course parallel to, and 1–4 cm from, the upper border of the muscle; the smaller branch diverges at an angle of 45° and travels medially. In a small number of cases, the artery trifurcates. This usually yields a small recurrent branch that returns to supply the proximal part of the muscle but, in some cases, it provides a third major branch that supplies the distal part of the muscle. Occasionally, the lateral branch gives off a further collateral to serratus anterior. The major branches give off 5–9 longitudinal branches that travel distally, parallel with the muscle fibres. Musculocutaneous perforators arising from these vessels supply the overlying skin.

In addition to the dominant vascular pedicle, latissimus dorsi is supplied inferiorly by several smaller, secondary, segmental vascular pedicles. These are dorsal perforating arteries derived from the ninth, tenth and eleventh posterior intercostal arteries and the first, second and third lumbar arteries. They all enter the muscle on its deep surface and anastomose within the muscle with branches of the thoracodorsal artery.

Innervation Latissimus dorsi is supplied by the thoracodorsal nerve, from the posterior cord of the brachial plexus, C6, 7 and 8 (predominantly C7). This nerve runs in the neurovascular pedicle and divides approximately 1.5 cm proximal to the point of bifurcation or trifurcation of the thoracodorsal artery. The neural and vascular branches travel together and so the pattern of branching of the nerve follows that of the artery closely.

Actions Latissimus dorsi is active in adduction, extension and, especially, in medial rotation of the humerus. Humeral adduction and extension are most powerful when the initial position of the arm is one of partial abduction or flexion, or a combination of the two. With the sternocostal part of pectoralis major and teres major, it adducts the raised arm against resistance. When the arms are raised above the head, as in climbing, it pulls the trunk upwards and forwards. It takes part in all violent expiratory efforts, such as coughing or sneezing; this is readily confirmed by palpation. Electromyography suggests that latissimus dorsi aids deep inspiration, but it is also active towards the end of forcible expiration, e.g. when blowing a sustained note on a musical instrument. When the arm is elevated, the stretched fibres of latissimus dorsi press on the inferior scapular angle (to which it may also be attached), keeping it in contact with the chest wall. Despite this range of actions, surgical transposition of the muscle does not appear to produce any serious restriction of normal activity.

Testing Latissimus dorsi is palpated when the abducted arm is adducted against resistance and can be felt to contract during coughing.

MUSCLES OF CONCAVITY COMPRESSION

The function of the rotator cuff muscles is to generate a concavity compressive force in all positions of the arm at the shoulder. The term 'rotator cuff' is conventionally taken to mean the tendinous parts of the four rotator cuff muscles: subscapularis, supraspinatus, infraspinatus and teres minor. Since teres minor, parts of infraspinatus and most of subscapularis do not have tendons, this use of the term excludes important parts of the rotator cuff apparatus. In this description, the term 'rotator cuff' will be used to define the entire musculotendinous apparatus that centralizes the humeral head on the glenoid. The tendons of the rotator cuff will be referred to specifically.

It is also maintained that the inferior capsule of the glenohumeral joint is exposed as a weak part of the capsule, and so predisposed to

injury (dislocation). However, in elevation of the arm above 90°, the bulk of teres major forms a muscular buttress to the lower part of the joint and, in this position, clearly draws the humeral head into the glenoid, i.e. it has a centralizing effect. Teres major is a part of the centralizing function of the rotator cuff, and should be considered along with the conventional definition of the rotator cuff.

The mechanism of the rotator cuff has been likened to that of the fibrous 'endoskeleton' of the heart, in which the muscular chambers are constructed around the fibrous architecture of the valvular system, and muscle contraction is centralized on this fibrous support. The muscles of the rotator cuff are attached to their tendons in a similar manner: the fibrous architecture of the tendons provides a framework for the action of the rotator cuff so that it always generates a centralizing force on the glenoid (Gagey et al 1993).

The muscles of the rotator cuff combine in three groups (subscapularis and teres major, supraspinatus and infraspinatus, and teres minor) by virtue of their innervation. Subscapularis and teres major are innervated through the subscapular nerves (C5, 6 and 7, posterior division, posterior cord); supraspinatus and infraspinatus by the suprascapular nerve (C5, 6, upper trunk); and teres minor by the axillary nerve (C5, 6, posterior division, posterior cord). The common origin of their innervation suggests a common final action, namely: the generation of the centralizing compression force. In functional terms, supraspinatus and infraspinatus can be considered as one muscle on the grounds that they have the same innervation and the same vascular supply; their attachment to the scapula is a continuum broken only by the spine of scapula for the attachment of trapezius; and their actions are complementary.

Supraspinatus

Attachments Supraspinatus arises from the medial two-thirds of the supraspinous fossa and from the supraspinous fascia (Fig. 48.28). The fibres converge, under the acromion, into a thick tendon that passes obliquely through the muscle, crosses above the humeral head, and is attached to the highest facet of the greater tubercle of the humerus. The tendon has a posterior, flatter, component closer to the facet of attachment. The anterior, obliquely directed, fibres of supraspinatus are inserted into the anterior aspect of the thick tendon, while the more parallel posterior fibres are inserted into both the thick tendon and the flatter component. The thick tendon is contiguous with the transverse humeral and coracohumeral ligaments; it forms an anterior pillar to the posterosuperior part of the rotator cuff. The deep surface of the tendon blends into the articular capsule over its distal 1 cm or so. Fibrocartilage has been described at the tendinous insertion, as in other tendons attached to epiphysal bone.

Vascular supply Supraspinatus is supplied by the suprascapular and dorsal scapular arteries.

Innervation Supraspinatus is innervated by the suprascapular nerve, C5 and 6.

Actions The conventional view is that supraspinatus initiates abduction of the shoulder and assists deltoid in abduction thereafter. However, there is evidence that both supraspinatus and deltoid are involved throughout the range of abduction, including initiation of the movement. In complete axillary nerve lesions (complete, irreversible deltoid palsy), supraspinatus, acting with other cuff muscles, can initiate, regulate, achieve and maintain full abduction of the arm (although endurance is limited). If supraspinatus is detached, torn or paralysed, and deltoid is also paralysed, investigation and surgical exploration and repair of damaged rotator cuff tendon, muscle or nerve supply is mandatory. With a functioning rotator cuff but poor deltoid, the arm is functionally adept and useful, whereas with a poorly functioning rotator cuff and normal deltoid, arm function is potentially poor because stability of the joint is adversely compromised and deltoid cannot function to overcome this disability. As part of the rotator cuff, supraspinatus helps to stabilize the head of the humerus in the glenoid fossa during movements of the glenohumeral joint. With the arm dependent, even when moderately loaded, supraspinatus assists deltoid in prevention of downward displacement of the humerus.

Testing It is extremely difficult to palpate supraspinatus unless trapezius is thin. The tendon of supraspinatus is separated from the coracoacromial ligament, acromion and deltoid by the large subacromial bursa; when this is inflamed, abduction of the shoulder joint is painful. The tendon is the most frequently torn element of the musculotendinous cuff around the shoulder joint.

Infraspinatus

Attachments Infraspinatus is a thick, triangular muscle that occupies most of the infraspinous fossa (see Fig. 48.28). It arises by muscular fibres from the medial two-thirds of the fossa, by tendinous fibres from ridges on its surface and from the deep surface of the infraspinous fascia, which separates it from teres major and minor. Its fibres converge to a tendon that glides under the lateral border of the spine of the scapula, and then passes across the posterior aspect of the capsule of the shoulder joint to be attached to the middle facet on the greater tubercle of the humerus. The tendon fuses with that of supraspinatus immediately lateral to the thick lateral margin of the spine of the scapula; the tendon fibres decussate so that fibres of infraspinatus can be traced to the facet for supraspinatus, and vice versa (Mochizuki et al 2008). The tendon is sometimes separated from the capsule by a bursa, which may communicate with the joint cavity. The tendon is flattened, but is thickened in several places, reinforcing its posterior aspect as it attaches to the middle facet. From this region, fibres passing orthogonally within the deeper substance of the tendon of infraspinatus pass forwards into supraspinatus, then pass across the rotator interval to subscapularis. This band of fibres is termed the 'cable' (Burkhart et al 1993); it provides the mechanical link between the posterosuperior and anterior parts of the rotator cuff, thus ensuring that there is intrinsic coordination of the rotator cuff mechanism. The muscular part of infraspinatus is attached directly to the middle facet on the greater tubercle and is sometimes fused with teres minor laterally, although their tendons are usually separable closer to the joint capsule.

Vascular supply Infraspinatus is supplied by the suprascapular and circumflex scapular arteries.

Innervation Infraspinatus is innervated by the suprascapular nerve, C5 and 6.

Actions Infraspinatus is a lateral rotator of the humerus. Together with supraspinatus, subscapularis and teres minor, it helps to stabilize the head of the humerus in the glenoid fossa during shoulder movements.

Teres minor

Attachments Teres minor is a narrow, elongated muscle that arises from the upper two-thirds of a flattened strip on the dorsal surface of the scapula adjoining its lateral border, and from two aponeurotic laminae that separate it from infraspinatus and teres major (see Fig. 48.28). It runs upwards and laterally. The upper fibres end in a short tendon (which may be absent), attached to the lowest facet on the greater tubercle of the humerus. The lower fibres are attached directly into the humerus distal to this facet and above the origin of the lateral head of triceps. The tendon passes across, and blends with, the lower posterior surface of the capsule of the shoulder joint. Teres minor may be fused with infraspinatus.

Vascular supply Teres minor is supplied by the circumflex scapular artery, which pierces the origin of the muscle as it turns upwards in the infraspinous fossa, and by the posterior circumflex humeral artery.

Innervation Teres minor is innervated by a branch of the axillary nerve that is given off the main trunk as it enters the quadrangular space, C5 and 6.

Actions Teres minor acts as a lateral rotator and weak adductor of the humerus with the arm pendant. Together with supraspinatus, infraspinatus, subscapularis and teres major, it helps to stabilize the head of the humerus in the glenoid fossa during shoulder movements.

Subscapularis

Attachments Subscapularis is a bulky, triangular muscle that fills the subscapular fossa (see Figs 48.25, 48.28A). In its medial two-thirds, the fibres are attached to the periosteum of the costal surface of the scapula. Other fibres arise from tendinous intramuscular septa, attached to ridges on the bone, and from the aponeurosis that covers the muscle and separates it from teres major and the long head of triceps. The fibres converge laterally into a broad tendon above, which has a thickened, rounded upper margin, and is attached to the lesser tubercle of the humerus and the anterior part of the articular capsule, with a wide direct muscular attachment to the lower part of the lesser tubercle. The tendon is separated from the neck of the scapula by the large subscapular bursa, which may communicate with the shoulder joint.

Variation is unusual. A separate inferior slip (subscapularis accessorius) may pass from the lateral border of the scapula to the glenohumeral capsule or to the periosteum medial to the intertubercular sulcus of the humerus. This should be distinguished from the true infraglenoid muscle (Staniek and Brenner 2012).

Relations Subscapularis forms much of the posterior axillary wall. Its anterior surface is apposed inferomedially to serratus anterior, the axillary vessels, brachial plexus and subscapular vessels and nerves, and superolaterally to coracobrachialis and biceps. Its posterior surface is attached to the scapula and glenohumeral capsule. Its lower border contacts teres major and latissimus dorsi. Its upper border is contiguous with the capsule of the rotator interval; at the most lateral extent of the interval, the long tendon of biceps passes lateral to the attachment of subscapularis, covered by the transverse humeral ligament.

Vascular supply Subscapularis is supplied by small branches from the suprascapular, axillary and subscapular arteries.

Innervation Subscapularis is innervated by the upper and lower subscapular nerves, C5, 6 (posterior cord).

Actions Subscapularis is a medial rotator of the humerus. Together with supraspinatus, infraspinatus, teres minor and teres major, it helps to stabilize the head of the humerus in the glenoid fossa during shoulder movements.

The attachment of teres major lies in direct continuity with the attachment of subscapularis on the lesser tubercle. The muscles lie in virtually the same plane as they approach the humerus, share a similar innervation from the posterior cord of the brachial plexus, and perform the same actions. Like the spinati, subscapularis and teres major might be considered as a functional unit.

Teres major

Attachments Teres major is a large, thick, ovoid muscle that arises from the oval area on the dorsal surface of the inferior scapular angle, and from the fibrous septa interposed between the muscle and teres minor and infraspinatus (see Fig. 48.28). Its fibres ascend laterally and end in a short, flat tendon, 2–5 cm long, attached to the medial lip of the intertubercular sulcus of the humerus. The tendon lies behind that of latissimus dorsi, from which it is separated by a bursa; the tendons are united along their lower borders for a short distance but are readily separated.

Variants Teres major may be fused with the scapular part of latissimus dorsi, and may send a slip to join the long head of triceps or the brachial fascia.

Vascular supply Teres major is supplied by the thoracodorsal branch of the subscapular artery on its way to latissimus dorsi and by the posterior circumflex humeral artery (Dancker 2013).

Innervation Teres major is innervated by the lower subscapular nerve, C5, 6 and 7.

Actions Despite its name, teres major has a nerve supply and action that is distinct from teres minor. Teres major draws the humerus backwards and rotates it medially. Electromyographic studies are equivocal about its major role in movement, but its involvement as a contributor to static posture and arm-swinging is not contested.

Testing Teres major can be palpated posterior to the posterior axillary fold during adduction of the humerus against resistance.

Rotator cuff disease

The subacromial space is defined inferiorly by the superior surface of the rotator cuff (supraspinatus and the anterior part of infraspinatus); superiorly by the anterior edge and inferior surface of the anterior third of the acromion, coraco-acromial ligament and acromioclavicular joint, forming the coraco-acromial arch; and posteriorly by the posterior wall of the subacromial bursa. It is occupied by the subacromial bursa, lined by synovial membrane, that extends anterolaterally under deltoid as the subdeltoid bursa, and which facilitates movement of the rotator cuff under the coraco-acromial arch. The upper surface of the rotator cuff normally impinges, without symptoms, under the coraco-acromial arch when the humerus is abducted, flexed and medially rotated. If the upper surface of the rotator cuff is damaged, the bursa is thickened or inflamed, or the undersurface of the coraco-acromial arch is rough, then the impingement can become abrasive: this is the painful condition called 'impingement syndrome'.

MUSCLES THAT COORDINATE SHOULDER AND ELBOW MOTION

The muscles of the upper arm are coracobrachialis, which acts only on the shoulder joint; biceps and triceps, which cross both shoulder and elbow joints; and brachialis, which acts only at the elbow joint. Coracobrachialis is described on page 821.

Anterior compartment of the arm

Biceps brachii

Attachments Biceps brachii derives its name from its two proximally attached parts or 'heads' (see Figs 48.15, 48.16, 48.25, 48.28A). The short head arises by a thick, flattened tendon from the coracoid apex, together with coracobrachialis. The long head starts within the capsule of the shoulder joint as a long narrow tendon, running from the supraglenoid tubercle of the scapula at the apex of the glenoidal cavity, where it is continuous with the glenoidal labrum. The tendon of the long head, enclosed in a double tubular synovial sheath, arches over the humeral head, emerges from the joint under the transverse humeral ligament, and descends in the intertubercular sulcus, where it is retained by the transverse humeral ligament and a fibrous expansion from the tendon of pectoralis major. The two tendons lead into elongated bellies that, although closely applied, can be separated to within 7 cm or so of the elbow joint. At this joint, they end in a flattened tendon, which is attached to the rough posterior area of the radial tuberosity; a bursa separates the tendon from the smooth anterior area of the tuberosity. As it approaches the radius, the tendon spirals, its anterior surface becoming lateral before being applied to the tuberosity. The tendon has a broad medial expansion, the bicipital aponeurosis, which descends medially across the brachial artery to fuse with deep fascia over the origins of the flexor muscles of the forearm (Fig. 48.29). The tendon can be split without difficulty as far as the tuberosity; it can, thus, be confirmed that its anterior and posterior layers receive fibres from the short and long heads, respectively.

Occasionally, a third head arises from the superomedial part of brachialis and is attached to the bicipital aponeurosis and medial side of the tendon of insertion. It usually lies behind the brachial artery but it may consist of two slips, which descend in front of and behind the artery. Less often, other slips may spring from the lateral aspect of the humerus or intertubercular sulcus.

Relations Biceps is overlapped proximally by pectoralis major and deltoid; distally, it is covered only by fasciae and skin, and it forms a conspicuous elevation on the front of the arm. Its long head passes through the shoulder joint; its short head is anterior to the joint. Distally, it lies anterior to brachialis, the musculocutaneous nerve and supinator. Its medial border touches coracobrachialis and overlaps the brachial vessels and median nerve; its lateral border is related to deltoid and brachioradialis.

Vascular supply Biceps brachii is typically supplied by up to eight vessels originating from the brachial artery in the middle third of the arm. These vessels pass laterally, posterior to the median nerve, and divide into ascending and descending branches just before reaching the deep surface of the muscle. Smaller branches arise from the anterior circumflex humeral artery and the deltoid branch of the acromial division of the thoraco-acromial axis.

There is great variation in the arterial supply to the muscle. The main arterial supply may originate from the superior or inferior ulnar collateral arteries, subscapular artery, axillary artery, ulnar or radial arteries in cases of proximal bifurcation of the brachial artery, or the profunda brachii artery. There is often a single large pedicle, attending the nerve to biceps; injury to the muscle and musculocutaneous nerve at this level in the arm commonly results in a fibrotic, atonic muscle with distal sensory dysfunction.

Innervation Biceps brachii is innervated by the musculocutaneous nerve, C5 and 6, with separate branches passing to each belly.

Actions Biceps brachii is a powerful supinator, especially in rapid or resisted movements. It flexes the elbow – most effectively, with the forearm supinated – and acts, to a slight extent, as a flexor of the shoulder joint. The long head helps to check upward translation of the humeral head during contraction of deltoid, in synergy with the action of the rotator cuff. It is attached, via the bicipital aponeurosis, to the posterior border of the ulna, the distal end of which is drawn medially in supination. When the elbow is flexed against

Pain is generated during abduction of the arm in the scapular plane; typically, an arc of pain is experienced (between 60° and 120° of elevation, the impingement arc). Rotator cuff disease is a painful condition with a multifactorial aetiology, in which severe or chronic impingement of the rotator cuff tendons on the undersurface of the coraco-acromial arch is often a significant factor. The supraspinatus tendon is anatomically affected most by the impingement, which coincides with an area of reduced vascularity in this tendon. When associated with a tendinopathy from age-related degenerative changes within the tendon, impingement may be associated with partial or complete tears of the cuff. Clinically, this condition causes tenderness over the anterior portion of the acromion, and pain that typically occurs on abducting the shoulder between 60° and 120° (the painful arc).

Lowering the hand under the influence of gravity by extension at the elbow calls for controlled lengthening of biceps brachii. This is an example of a habitual movement in which muscle tension increases despite increasing length. (This is 'eccentric' loading of a muscle; during such loading, the muscle-tendon junction is at greater risk of rupture. 'Concentric' loading occurs when the muscle shortens as tension increases, see [p. 114](#).) As the hand descends and the elbow extends, the vertical through the centre of gravity of the forearm is carried further from the fulcrum of movement; therefore, the turning moment exerted by the load increases and must be matched by an increase in the moment exerted by the muscle.

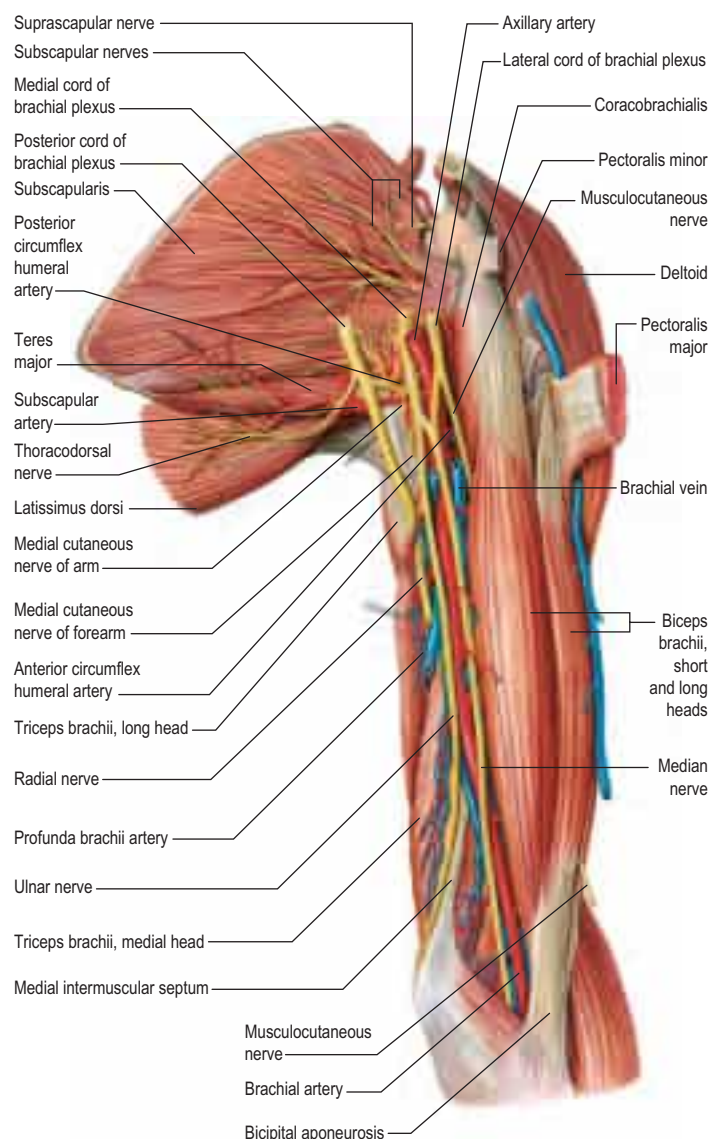


Fig. 48.29 Muscles, vessels and nerves of the left upper arm, anterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

resistance, the tendon of insertion and bicipital aponeurosis become conspicuous.

Testing With the forearm supinated, biceps brachii can be tested by palpating its fibres during elbow flexion against resistance.

Brachialis

Attachments Brachialis arises from the lower half of the anterior surface of the shaft of the humerus, starting on either side of the insertion of deltoid, and extending distally to within 2.5 cm of the cubital articular surface (Fig. 48.30). It also arises from the intermuscular septa, more from the medial than the lateral, since it is separated distally from the lateral intermuscular septum by brachioradialis and extensor carpi radialis longus. Its fibres converge to a thick, broad tendon that is attached to the ulnar tuberosity and to a rough impression on the anterior aspect of the coronoid process (see Fig. 49.5).

Brachialis may be divided into two or more parts. It may be fused with brachioradialis, pronator teres or biceps. In some cases, it sends a tendinous slip to the radius or to the bicipital aponeurosis.

Relations Biceps, the brachial vessels and the musculocutaneous and median nerves are anterior. The humerus and the capsule of the elbow joint are posterior. Pronator teres and the medial intermuscular septum, which separates it from triceps and the ulnar nerve, are medial. The radial nerve, radial recurrent and radial collateral arteries, brachioradialis and extensor carpi radialis longus are all lateral.

Vascular supply The blood supply to brachialis typically consists of two main arteries (superior and inferior), supplemented by a system of accessory arteries. The superior main artery originates from the brachial

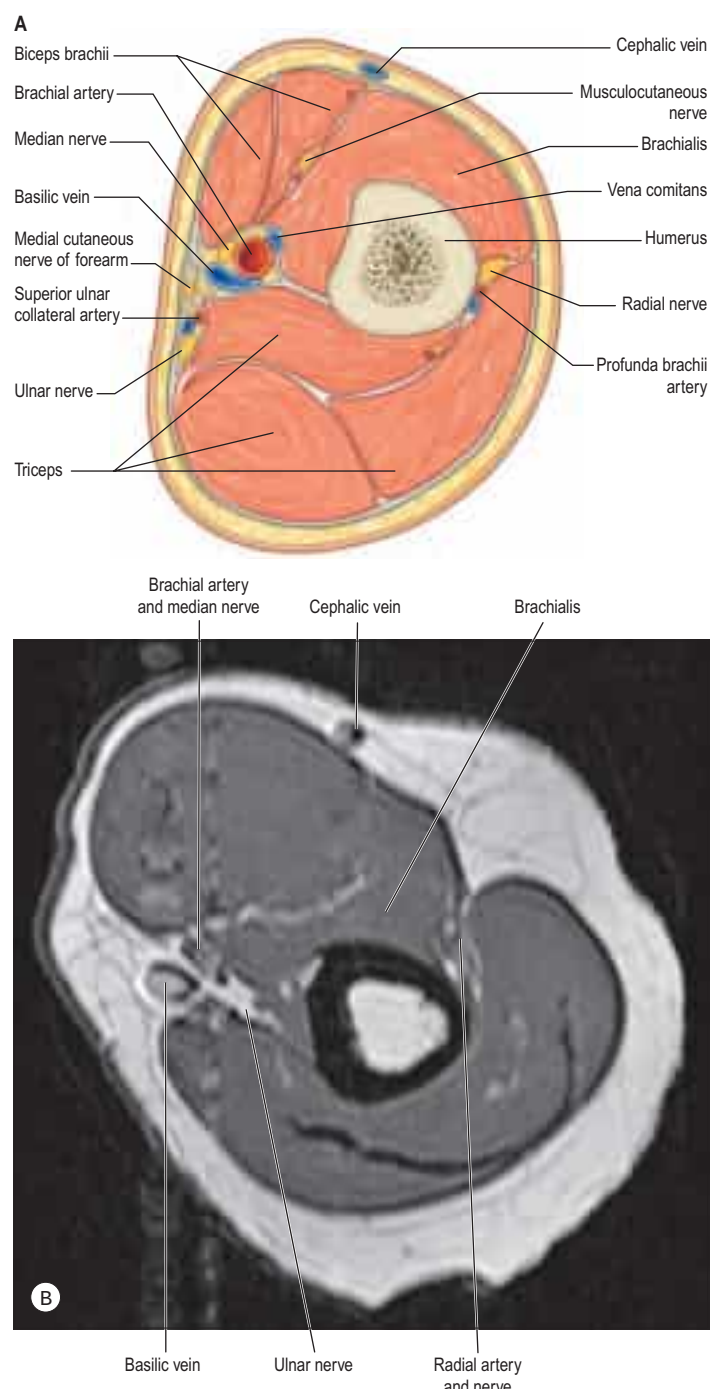


Fig. 48.30 A, A transverse section through the left arm, a little below the middle of the shaft of the humerus. **B**, A magnetic resonance imaging (MRI) scan at a similar level.

artery distal to the site of origin of the superior ulnar collateral artery and travels laterally to enter the anterior surface of the upper third of the muscle. The inferior main artery originates either from the superior ulnar collateral artery or directly from the brachial artery, and enters the mid-portion of the muscle. The accessory arteries are small and variable in number, origin and course. They can arise from the brachial artery, the superior and inferior ulnar collateral arteries, or the profunda brachii artery. The fact that brachialis has a greater vascular supply than biceps brachii suggests that its metabolic demands are greater; brachialis is the prime flexor of the forearm at the elbow, while biceps has more subtle actions on the shoulder and elbow, coordinating the two joints while positioning the hand in space.

Innervation Brachialis is innervated by at least two branches. The musculocutaneous nerve (C5 and 6) supplies a large medial part of the muscle, and the radial nerve (C7) supplies a small lateral part. This arrangement is consistent (Oberlin et al 1994).

Action Brachialis is a flexor of the elbow joint with the forearm either prone or supine, whether or not the movement is resisted.

Testing Brachialis can be tested by palpating its fibres during elbow flexion against resistance.

Posterior compartment of the arm

Triceps

Attachments Triceps fills most of the extensor compartment of the upper arm (see Figs 48.28–48.30). It has three heads (long, lateral and medial), from which it takes its name.

The long head arises by a flattened tendon from the infraglenoid tubercle of the scapula, blending above with the glenohumeral capsule. Its muscular fibres descend medial to the lateral head and superficial to the medial head, and join them to form a common tendon. The lateral head arises by a flattened tendon from a narrow, linear, oblique ridge on the posterior surface of the humeral shaft, and from the lateral intermuscular septum. The origin on the humerus ascends with varying obliquity from its lateral border above the radial groove and behind the deltoid tuberosity to the surgical neck medial to the insertion of teres minor. These fibres also converge to the common tendon. The medial head, which is overlapped posteriorly by the lateral and long heads, has a particularly extensive origin. It is attached to the entire posterior surface of the humeral shaft, below the radial groove from the insertion of teres major to within 2.5 cm of the trochlea; the medial border of the humerus; the medial intermuscular septum; and the lower part of the lateral intermuscular septum. Some muscular fibres reach the olecranon directly; the rest converge to the common tendon (Keener et al 2010). Some fibres from the medial head attach to the posterior capsule of the elbow joint, like those of articularis genus in the knee; these fibres are the articularis cubiti.

The tendon of triceps begins near the middle of the muscle. It has two laminae organized in a similar arrangement to the tendon of the gastrocnemii and soleus in the posterior compartment of the leg. The superficial lamina in the lower half of the superficial parts of the muscle forms an aponeurotic layer to which the deep fascia of the arm is adherent; the other lamina is more vertical. After receiving the muscle fibres, the two layers unite above the elbow and are attached, for the most part, to the upper surface of the olecranon. On the lateral side, a band of fibres continues down over anconeus to blend with antebrachial fascia.

Relations The long head of triceps descends between teres minor and major, dividing the wedge-shaped interval between them and the humerus into triangular and quadrangular spaces (Fig. 48.28). The lateral head of triceps forms an elevation, parallel and medial to the posterior border of deltoid, which stands out prominently when the elbow is actively extended. The mass that lies medial to it, and that disappears under deltoid, is the long head.

Vascular supply The blood supply to triceps is mainly from the profunda brachii artery and the superior ulnar collateral artery, with an accessory supply from the posterior circumflex humeral artery. The medial head of triceps is supplied on its anterior surface by two or three branches from the superior ulnar collateral artery. The posterior surface is supplied by a large proximal branch from the profunda brachii that passes medially anterior to the radial nerve. The long head of triceps is supplied on its anterior surface by two arteries, one arising from the axillary artery anterior to the tendon of latissimus dorsi and the other arising from either the brachial artery or the superior ulnar collateral artery. The posterior surface receives a recurrent branch from the posterior circumflex humeral artery immediately after it has traversed the quadrangular space. The lateral surface receives a number of small branches from the profunda brachii artery in its distal portion. The lateral head of triceps is primarily supplied by branches of the profunda brachii, with an additional supply from a branch of the posterior circumflex humeral artery.

Innervation Triceps is innervated by branches from the radial nerve, C6 (lateral head), 7 (long head) and 8 (medial head); there are separate branches for each head.

Actions Triceps is the major extensor of the forearm at the elbow joint. The medial head is active in all forms of extension, whereas the lateral and long heads are minimally active, except in extension against resistance, as in thrusting or pushing or supporting body weight on the hands with the elbows semi-flexed. When the flexed arm is extended at the shoulder joint, the long head may assist in drawing back and adducting the humerus to the thorax. The long head supports the lower part of the capsule of the shoulder joint, especially when the arm is raised.

Articularis cubiti probably draws up the posterior part of the capsule of the elbow joint during extension of the forearm. In forceful supination of the semi-flexed forearm, involving contraction of both supinator and biceps brachii, triceps contracts synergistically to maintain the semi-flexed position.

Testing Triceps can be tested by palpating its fibres during elbow extension against resistance.

Quadrangular and triangular spaces

The quadrangular space (quadrilateral space; foramen of Velpeau) is more a horizontal cleft or tunnel than a vertical space; it has defined boundaries. Anteriorly, the space is bounded by subscapularis, the capsule of the shoulder joint and teres minor superiorly, teres major inferiorly, the long head of triceps medially and the surgical neck of the humerus laterally. Posteriorly, the quadrangular space is bounded above by teres minor. The axillary nerve and the posterior circumflex humeral artery and vein pass obliquely through the space (see Figs 48.33, 48.47).

There are two triangular spaces (see Figs 48.28, 48.47). The upper triangular space is bounded above by subscapularis anteriorly, teres minor posteriorly, teres major inferiorly and the long head of triceps laterally. The circumflex scapular vessels pass through this space. The lower triangular space (the triangular interval) is bounded above by subscapularis anteriorly, teres major posteriorly, the long head of triceps medially and the humerus laterally. The radial nerve and the profunda brachii vessels pass through this space.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

Dorsal scapular artery

In the majority of cases, the dorsal scapular artery arises from the third, or less often the second, part of the subclavian artery. It passes laterally through the trunks of the brachial plexus in front of scalenus medius and then deep to levator scapulae to reach the superior scapular angle. Here, it descends with the dorsal scapular nerve under the rhomboids along the medial border of the scapula to the inferior angle. It supplies the rhomboids, latissimus dorsi and the inferior portion of trapezius and the skin over the inferomedial aspect of trapezius via musculo-cutaneous perforators. It anastomoses with the suprascapular and subscapular arteries and with posterior branches of some posterior intercostal arteries. It sends a small branch to scalenus anterior; sometimes, this arises directly from the subclavian artery.

Variants About one-third of the superficial cervical and dorsal scapular arteries arise as a common vessel – the transverse cervical artery – from the thyrocervical trunk, which then divides into a superficial and a deep branch (superficial cervical and dorsal scapular arteries, respectively) (see Figs 26.1A, 48.31). In this case, the dorsal scapular artery arises near the superior border of the scapula; it passes laterally, anterior to the brachial plexus and then posterior to levator scapulae.

Suprascapular artery

The suprascapular artery usually arises from the thyrocervical trunk of the subclavian artery, although it may arise from the third part of the subclavian artery (see Fig. 29.7A). It first descends laterally across scalenus anterior and the phrenic nerve, posterior to the internal jugular vein and sternocleidomastoid, then crosses anterior to the subclavian artery and brachial plexus, posterior and parallel with the clavicle, subclavius and the inferior belly of omohyoid, to reach the superior border of the scapula. Here, it passes above (sometimes, under) the superior transverse ligament, separating it from the suprascapular nerve, and enters the supraspinous fossa (Fig. 48.31, see Fig. 48.33), where it lies on the bone, and supplies supraspinatus. It descends behind the scapular neck, and passes through the spinoglenoid notch deep to the inferior transverse ligament to gain the deep surface of infraspinatus, where it anastomoses with the circumflex scapular and deep branch of the transverse cervical artery. Its muscular branches supply sternocleidomastoid, subclavius and infraspinatus. It also gives off a suprasternal branch that crosses the sternal end of the clavicle to supply the skin of the upper thorax, and an acromial branch that pierces trapezius to supply the skin over the shoulder. This last branch anastomoses with the thoracoacromial and posterior circumflex humeral arteries.

As the suprascapular artery passes over the superior transverse ligament, it gives off a branch that enters the subscapular fossa beneath

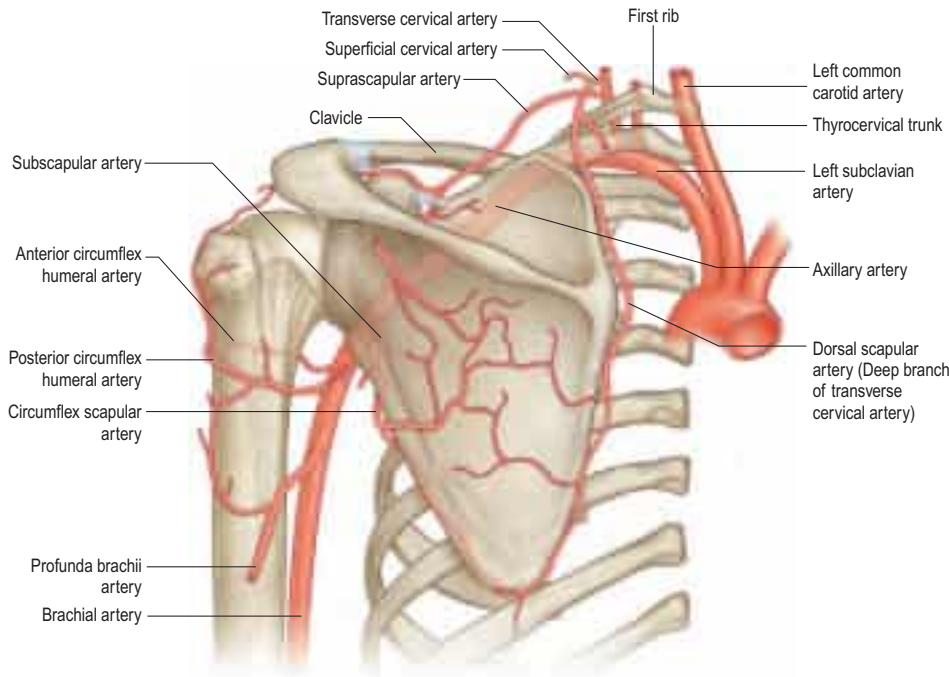


Fig. 48.31 The scapular anastomosis, left side. (With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)



Fig. 48.32 Axillary angiograms. **A**, View with the arm by the side. **B**, View with the arm elevated.

subscapularis and anastomoses with the subscapular artery and the deep branch of the transverse cervical artery. It also supplies the acromioclavicular and glenohumeral joints, the clavicle and the scapula.

Angiomyosomes

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Supraclavicular artery

This small vessel arises from either the transverse cervical or superficial cervical artery. It pierces the deep fascia just superior to the clavicle and anterior to trapezius, and supplies an area of skin over the lateral end of the clavicle.

Axillary artery

The axillary artery, a continuation of the subclavian artery, begins at the outer border of the first rib and ends nominally at the inferior border of teres major, where it becomes the brachial artery (see Fig. 48.31). Its direction varies with the position of the limb; it is almost straight when the arm is raised at right angles, concave upwards when the arm is

elevated above this, and convex upwards and laterally when the arm is by the side (Fig. 48.32). At first deep, it subsequently becomes superficial, when it is covered only by the skin and fasciae. Pectoralis minor crosses it and so divides it into three parts, which are proximal, posterior and distal to the muscle.

Relations of the first part

The skin, superficial fascia, platysma, supraclavicular nerves, deep fascia, clavicular fibres of pectoralis major and the clavipectoral fascia, lateral pectoral nerve and the loop of communication between the lateral and medial pectoral nerves, and the thoraco-acromial and cephalic veins are all anterior. The first intercostal space and external intercostal, first and second digitations of serratus anterior, long thoracic and medial pectoral nerves, and the medial cord of the brachial plexus are all posterior. The posterior cord of the brachial plexus is lateral and the axillary vein is anteromedial. The first part is enclosed with the axillary vein and brachial plexus in a fibrous axillary sheath, which is continuous with the prevertebral layer of the deep cervical fascia.

Relations of the second part

The skin, superficial and deep fascia, and pectoralis major and minor are all anterior. The posterior cord of the brachial plexus and the areolar tissue between it and subscapularis are posterior. The axillary vein is

Two branches of the subclavian artery perfuse the dorsal scapular suspension and medial motion muscles (the dorsal scapular artery), the posterosuperior rotator cuff muscles (the suprascapular artery) and the clavicle. These vessels correspond to the equivalent innervation for the same muscles. In this sense, innervation and angiomyotomes correspond. Pain afferents from the muscles, mostly small, unmyelinated axons, travel in the nerves; ischaemic ('fatigue-related') pain from these three groups of dorsal muscles is felt as a diffuse, burning sensation in the same region as the vascular supply, without radiation or referral, and without respite during rest for a characteristic period, somewhat equivalent to the period over which the symptoms of a chronic ischaemic muscular compartment syndrome may resolve. The suprascapular artery supplies a muscular compartment that is contained in a well-defined and relatively restricting fascia; it is, therefore, not a surprise that the ischaemic ('fatigue-related') pain generated in this compartment may present earlier and be more persistent than elsewhere in muscles less well contained by fascial sheaths.

medial, separated from the artery by the medial cord of the brachial plexus and the medial pectoral nerve. The lateral cord of the brachial plexus is lateral, separating the artery from coracobrachialis. The cords of the brachial plexus thus surround the second part on three sides, with the dispositions implied by their names, and separate it from the vein and adjacent muscles.

Relations of the third part

Pectoralis major and, distal to the muscle, skin and fasciae are anterior. The lower part of subscapularis and the tendons of latissimus dorsi and teres major are posterior. Coracobrachialis is lateral and the axillary vein is medial. Branches of the brachial plexus are arranged as follows: laterally, the lateral root and then trunk of the median nerve and, for a short distance, the musculocutaneous nerve; medially, the medial cutaneous nerve of the forearm between the axillary artery and vein anteriorly, and the ulnar nerve between these vessels posteriorly; anteriorly, the medial root of the median nerve; and posteriorly, the radial and axillary nerves, the latter only to the distal border of subscapularis. The dispositions of the branches of the brachial plexus around the axillary artery have, therefore, altered: there has been a rotation such that the nerves derived from the lateral cord are now more anterior, those from the medial cord more posterior, and those of the posterior cord come to lie more laterally.

Branches

The branches of the axillary artery are one from the first part (superior thoracic), two from the second part (thoraco-acromial, lateral thoracic) and three from the third part (subscapular, anterior and posterior circumflex humeral) (see Fig. 48.47).

Superior thoracic artery

The superior thoracic artery is a small vessel that arises from the first part of the axillary artery near the lower border of subclavius; it sometimes arises from the thoraco-acromial artery (see Fig. 48.47). It runs anteromedially above the medial border of pectoralis minor, then passes between it and pectoralis major to gain the thoracic wall. It supplies these muscles and the thoracic wall, and anastomoses with the internal thoracic and upper intercostal arteries.

Thoraco-acromial (acromiothoracic) artery

The thoraco-acromial artery is a short branch that arises from the second part of the axillary artery (see Figs 48.27, 48.47). It is, at first, overlapped by pectoralis minor, skirting its medial border; it next pierces the clavipectoral fascia and divides into pectoral, acromial, clavicular and deltoid branches, which supply pectoralis major and minor, an area of skin over the clavipectoral fascia, and the anterior portion of deltoid.

Pectoral branch The pectoral branch is the largest branch. It descends between the pectoral muscles, gives a branch to pectoralis minor, and then continues on the deep surface of pectoralis major. It enters the muscle and anastomoses with the intercostal branches of the internal thoracic and lateral thoracic arteries. It gives off perforating branches to the breast, and musculocutaneous perforators to the skin over pectoralis major.

Acromial branch The acromial branch crosses the coracoid process under deltoid, which it supplies, then pierces the muscle and ends on the acromion. It anastomoses with branches of the suprascapular artery, the deltoid branch of the thoraco-acromial artery and the posterior circumflex humeral arteries.

Clavicular branch The clavicular branch ascends medially between the clavicular part of pectoralis major and the clavipectoral fascia. It supplies the sternoclavicular joint and subclavius.

Deltoid branch The deltoid branch often arises with the acromial branch. It crosses pectoralis minor to accompany the cephalic vein between pectoralis major and deltoid, and supplies both muscles.

Lateral thoracic artery

The lateral thoracic artery arises from the second part of the axillary artery. Following the lateral border of pectoralis minor, it passes to the deep surface of pectoralis major as far distally as the fifth intercostal space (see Fig. 48.45). It supplies serratus anterior and the pectoral muscles, the axillary lymph nodes and subscapularis. It anastomoses with the internal thoracic, subscapular and intercostal arteries, and with the pectoral branch of the thoraco-acromial artery. In females, it is large and has lateral mammary branches, which curve round the lateral

border of pectoralis major to the breast. In both males and females, it gives off cutaneous branches that pass around the lateral border of pectoralis major to supply the skin in this region.

Subscapular artery

The subscapular artery is the largest branch of the axillary artery (see Figs 48.31, 48.47). It usually arises from the third part of the axillary artery at the distal (inferior) border of subscapularis, which it follows to the inferior scapular angle, where it anastomoses with the lateral thoracic and intercostal arteries and the deep branch of the transverse cervical artery. It supplies adjacent muscles and the thoracic wall. It is accompanied distally by the nerve to latissimus dorsi. Approximately 4 cm from its origin, the subscapular artery divides into the circumflex scapular and thoracodorsal arteries.

Circumflex scapular artery The circumflex scapular artery, the larger of the two terminal branches of the subscapular artery, curves backwards around the lateral border of the scapula, traversing a triangular space between subscapularis above, teres major below and the long head of the triceps laterally (see Fig. 48.33). It enters the infraspinous fossa under teres minor and then divides. One branch (infrascapular) enters the subscapular fossa deep to subscapularis, and anastomoses with the suprascapular and dorsal scapular arteries (or deep branch of the transverse cervical artery). The other branch continues along the lateral border of the scapula between teres major and minor, then, dorsal to the inferior angle, anastomoses with the dorsal scapular artery. Small branches supply the posterior part of deltoid and the long head of triceps, and anastomose with an ascending branch of the profunda brachii artery.

Thoracodorsal artery The other terminal branch of the subscapular artery, the thoracodorsal artery, follows the lateral margin of the scapula, posterior to the lateral thoracic artery, between latissimus dorsi and serratus anterior. Before entering the deep surface of latissimus dorsi, it supplies teres major and the intercostals, and sends one or two branches to serratus anterior. It enters latissimus dorsi muscle with the thoracodorsal nerve; this constitutes the principal neurovascular pedicle to the muscle. It provides numerous musculocutaneous perforators that supply the skin over the superior part of latissimus dorsi. The intramuscular portion of the artery anastomoses with intercostal arteries and lumbar perforating arteries.

Anterior circumflex humeral artery

The anterior circumflex humeral artery arises from the lateral side of the axillary artery at the distal border of subscapularis (see Fig. 48.31). It runs horizontally behind coracobrachialis and the short head of biceps, anterior to the surgical neck of the humerus. Reaching the intertubercular sulcus, it sends an ascending branch to supply the humeral head and shoulder joint (Brooks et al 1993). It continues laterally under the long head of biceps and deltoid, and anastomoses with the posterior circumflex humeral artery.

Posterior circumflex humeral artery

The posterior circumflex humeral artery is larger than the anterior (see Fig. 48.31). It branches from the third part of the axillary artery at the distal border of subscapularis and runs backwards with the axillary nerve through a quadrangular space (see Fig. 48.22B), which is bounded by subscapularis, the capsule of the shoulder joint and teres minor above, teres major below, the long head of triceps medially, and the surgical neck of the humerus laterally. It curves round the humeral neck and supplies the shoulder joint, deltoid, teres major and minor, and long and lateral heads of triceps (Gerber et al 1990). It gives off a descending branch that anastomoses with the deltoid branch of the profunda brachii artery and with the anterior circumflex humeral and acromial branches of the suprascapular and thoraco-acromial arteries.

Variants

The branches of the axillary artery exhibit considerable variation. An alar thoracic artery, often from the second part, may supply fat and lymph nodes in the axilla. The lateral thoracic artery may be absent, in which case it is replaced by lateral perforating branches of the intercostal arteries; alternatively, it may become, or give off, a direct cutaneous vessel (named the superficial thoracic artery by some sources) that supplies the skin over the lateral border of pectoralis major. In up to one-third of cases, the subscapular artery can arise from a common trunk with the posterior circumflex humeral artery. Occasionally, the subscapular, circumflex humeral and profunda brachii arteries arise in common; in this case, branches of the brachial plexus surround this common vessel instead of the axillary artery. The posterior circumflex

Two significant cutaneous branches arise from the circumflex scapular artery as it emerges through the upper triangular space. The superior, or horizontal, branch is a direct cutaneous artery that passes medially at the level of the deep fascia parallel with the spine of the scapula; it supplies a band of skin overlying the spine of the scapula. The lower branch (parascapular branch) is also a direct cutaneous vessel that passes in an inferomedial direction, again at the level of the deep fascia, and supplies an area of skin overlying the lateral border of the scapula. Both of these cutaneous vessels provide the anatomical basis of skin flaps that can be surgically raised in this region (scapular flap based on the horizontal branch and parascapular flap based on the lower, parascapular branch) to reconstruct areas of missing tissue elsewhere in the body.

humeral artery may arise from the profunda brachii artery and pass back below teres major, instead of passing through the quadrangular space. Sometimes, the axillary artery divides into radial and ulnar arteries (anomalous 'high division'), and is, occasionally, the source of the anterior interosseous artery.

Potential vascular anastomoses around the shoulder girdle: clinical relevance

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Angiomyosomes

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Brachial artery

The brachial artery, a continuation of the axillary, begins at the distal (inferior) border of the tendon of teres major and ends about a centimetre distal to the elbow joint (at the level of the neck of the radius) by dividing into the radial and ulnar arteries (see Figs 48.33, 48.44, 49.21, 49.22). At first, it is medial to the humerus, but gradually spirals anterior to it until it lies midway between the humeral epicondyles. Its pulsation can be felt throughout the arm.

Relations The brachial artery is wholly superficial, covered anteriorly only by skin and superficial and deep fasciae. The bicipital aponeurosis crosses it anteriorly at the elbow, separating it from the median cubital vein; the median nerve crosses it lateromedially near the distal attachment of coracobrachialis. Posterior to the artery are the long head of triceps, separated by the radial nerve and profunda brachii artery, and then successively by the medial head of triceps, the attachment

of coracobrachialis and brachialis. Proximally, the median nerve and coracobrachialis lie laterally and the medial cutaneous nerve of the forearm and ulnar nerve lie medially. Distally, biceps brachii overlaps the artery laterally and the median nerve and basilic vein lie medially. The artery is accompanied by two venae comitantes, connected by transverse and oblique branches. At the elbow, the brachial artery sinks deeply into the triangular intermuscular cubital fossa and its further course is described on page 856.

Variants The brachial artery, with the median nerve, may diverge from the medial border of biceps brachii and descend towards the medial humeral epicondyle, usually behind a supracondylar process from which a fibrous arch crosses the artery, and which then runs behind or through pronator teres to the elbow. Occasionally, the artery divides proximally into two trunks that reunite. Frequently, it divides more proximally than usual into radial, ulnar and common interosseous arteries. Most often, the radial branches arise proximally, leaving a common trunk for the ulnar and common interosseous arteries. Sometimes, the ulnar artery arises proximally and the radial and common interosseous arteries form the other division, or the common interosseous may also arise proximally. Slender vasa aberrantia may connect the brachial artery to the axillary artery or to one of the forearm arteries, usually the radial. The brachial artery may be crossed by muscular or tendinous slips from coracobrachialis, biceps brachii, brachialis or pronator teres. Rarely, the median nerve crosses posterior, rather than anterior, to the brachial artery near the insertion of coracobrachialis.

Branches The branches of the brachial artery are the profunda brachii, nutrient, superior, middle and inferior ulnar collateral, deltoid, muscular, radial and ulnar arteries.

Profunda brachii artery The profunda brachii (Fig. 48.33; see Figs 48.29, 48.31) is a large branch from the posteromedial aspect of the

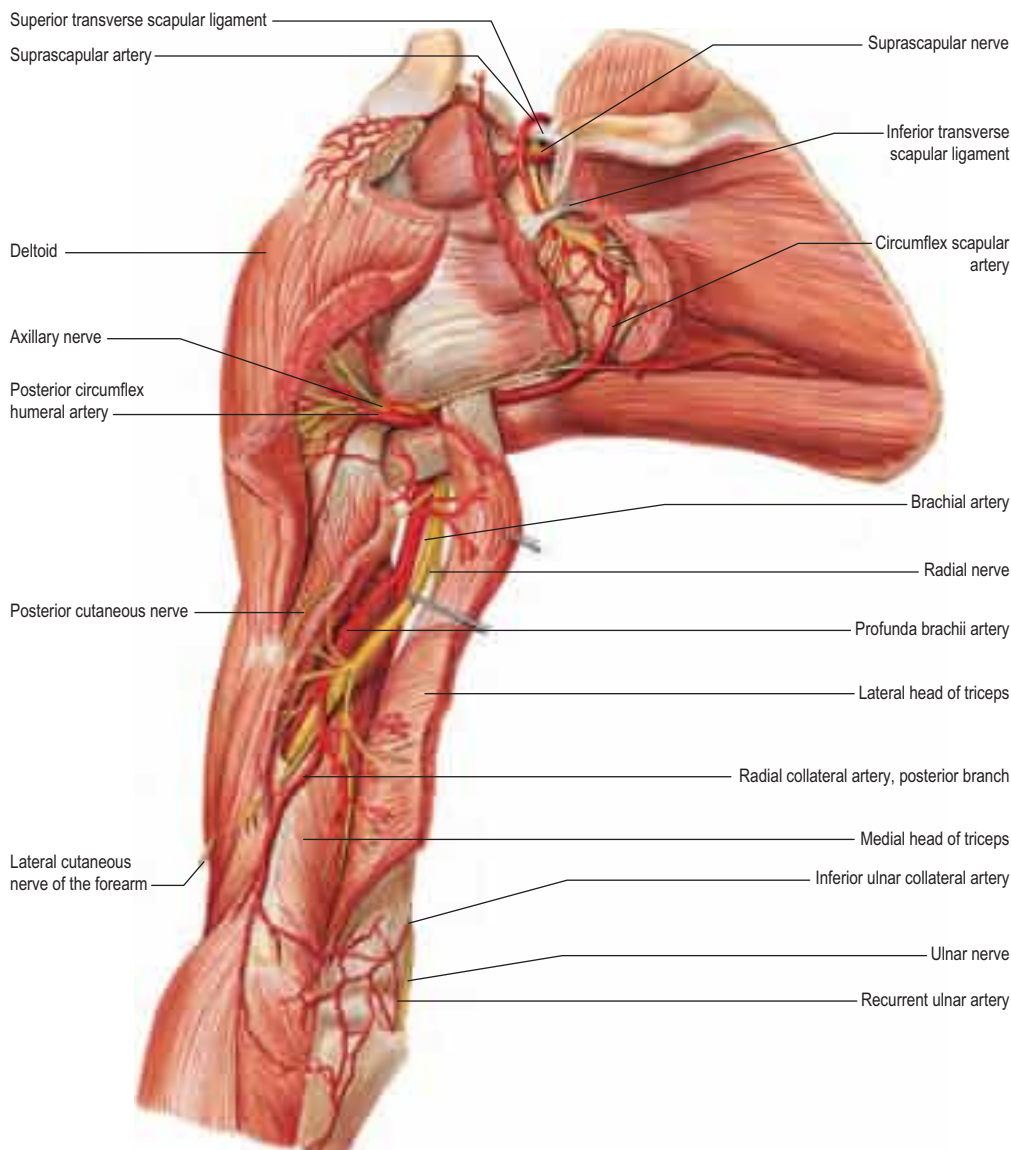


Fig. 48.33 Muscles, vessels and nerves of the left upper arm, posterior aspect. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

The potential vascular anastomoses around the shoulder girdle are numerous, so that collateral perfusion of the upper limb is possible in axillary artery or vein occlusion. In acute occlusion of the axillary artery, these potential anastomoses are not reliable; clinicians should not rely on anastomotic perfusion of the acutely ischaemic limb. The axillary artery is most 'at risk' of occlusion due to intimal separation and dissection in its second part, behind the pectoralis minor. This injury is associated with infraclavicular brachial plexus traction lesions ([Birch 2011](#)). Although magnetic resonance imaging (MRI) with angiography is recommended for the diagnosis of the poorly perfused limb after distraction trauma, the site of intimal rupture is so consistent as to make urgent surgical exploration without imaging a reasonable clinical decision.

The axillary artery supplies the muscles of the ventral compartments of the shoulder, including the pectoral–deltoid muscle sheet, the scapulo-humeral muscles (other than the posterosuperior rotator cuff) and the shoulder joint, including the scapula (with the serratus muscles) and proximal humerus. Its terminal branch, the brachial artery, supplies the muscles of the anterior compartment of the arm (for the branches of the brachial artery at the elbow and in the forearm, see [pages 856–858](#)); the profunda brachii artery, the major proximal branch of the brachial artery, supplies the muscles of the posterior compartment of the arm and the shaft of the humerus.

SHOULDER GIRDLE AND ARM

brachial artery, distal to *teres major*. It follows the radial nerve closely, at first posteriorly between the long and medial heads of *triceps*, then in the spiral groove covered by the lateral head of *triceps*. It supplies muscular branches, and the nutrient artery of the humerus, and finally divides into terminal radial and middle collateral branches (see [Fig. 48.33](#)). The radial collateral branch pierces the lateral intermuscular septum to reach the anterior aspect of the epicondyle of the humerus in the groove between *brachioradialis* and *brachialis*, and takes part in the anastomosis around the elbow. The middle collateral branch runs posterior to the septum and epicondyle.

The profunda brachii may originate from a common origin with the posterior circumflex humeral artery, from the axillary artery proximal to the tendon of *latissimus dorsi*, or from the distal portion of the axillary artery.

Middle collateral (posterior descending) branch The middle collateral artery is the larger terminal branch of the profunda brachii artery. It arises posterior to the humerus and descends along the posterior surface of the lateral intermuscular septum to the elbow. Proximally, the artery lies between *brachialis* (anteriorly) and the lateral head of *triceps* (posteriorly). Distally, it lies between *brachioradialis* (anteriorly) and the lateral head of *triceps* (posteriorly). It may pierce the deep fascia and become cutaneous, or remain deep to the fascia until it anastomoses with the interosseous recurrent artery behind the lateral epicondyle; it gives off about five small fasciocutaneous perforators and often has a small branch that accompanies the nerve to *anconeus*.

Radial collateral (anterior descending) branch The radial collateral artery accompanies the radial nerve through the lateral intermuscular septum, descending between *brachialis* and *brachioradialis* anterior to the lateral epicondyle and anastomosing with the radial recurrent artery (see [Figs 49.29, 49.31](#)). It supplies *brachialis*, *brachioradialis*, the radial nerve and a few fasciocutaneous perforators.

Nutrient artery of humerus The nutrient artery of the humerus arises near the mid-level of the upper arm and enters the nutrient canal near the attachment of *coracobrachialis*, posterior to the deltoid tuberosity; it is directed distally.

Superior ulnar collateral artery The superior ulnar collateral artery arises a little distal to the mid-level of the upper arm, usually from the brachial artery, but often as a branch from the profunda brachii (see [Fig. 48.44](#)). It accompanies the ulnar nerve, piercing the medial intermuscular septum to descend in the posterior compartment and supply the medial head of *triceps*. It passes between the medial epicondyle and olecranon, ending deep to *flexor carpi ulnaris* by anastomosing with the posterior ulnar recurrent and inferior collateral arteries. A branch sometimes passes anterior to the medial epicondyle and anastomoses with the anterior ulnar recurrent artery.

Middle ulnar collateral artery If present, the middle ulnar collateral artery arises from the brachial artery between the superior and inferior ulnar collateral arteries. It passes anterior to the medial epicondyle and anastomoses with the anterior ulnar recurrent artery. It supplies *triceps* and sends small fasciocutaneous perforators to the skin.

Inferior ulnar collateral (supratrochlear) artery The inferior ulnar collateral artery begins approximately 5 cm proximal to the elbow, passes medially between the median nerve and *brachialis*, and, piercing the medial intermuscular septum, curls round the humerus between *triceps* and bone (see [Fig. 48.33](#)). By its junction with the middle collateral branch of the profunda brachii artery, it forms an arch proximal to the olecranon fossa. As it lies on *brachialis*, it gives off branches that descend anterior to the medial epicondyle to anastomose with the anterior ulnar recurrent artery. Behind the epicondyle, a branch anastomoses with the superior ulnar collateral and posterior ulnar recurrent arteries.

Muscular branches Muscular branches are distributed to *coracobrachialis*, *biceps* and *brachialis*.

Deltoid (ascending) branch The deltoid branch ascends between the lateral and long heads of *triceps*, and anastomoses with a descending branch of the posterior humeral circumflex artery.

Radial and ulnar arteries The radial and ulnar arteries are described in detail on [page 856](#) and [857](#), respectively. For an account of the anastomosis about the elbow, see [page 858](#).

VEINS

The deep veins have numerous anastomoses with each other and with the superficial veins.

Axillary vein

The axillary vein is the continuation of the basilic vein (see [Fig. 46.4](#)). It begins at the lower border of *teres major* and ascends to the outer border of the first rib, where it becomes the subclavian vein. It is joined by the brachial vein near *subscapularis*, and by the cephalic vein near its costal end; other tributaries follow the axillary arterial branches. It lies medial to the axillary artery, which it partly overlaps. The medial pectoral nerve, medial cord of the brachial plexus, ulnar nerve and medial cutaneous nerve of the forearm lie between the artery and the vein. The medial cutaneous nerve of the arm is medial to the vein; the lateral group of axillary lymph nodes is posteromedial. There are a pair of valves near its distal end, and valves also occur near the ends of the cephalic and subscapular veins.

Subclavian vein

The subclavian vein is a continuation of the axillary vein (see [Fig. 46.4](#)). It extends from the outer border of the first rib to the medial border of *scalenus anterior*, where it joins the internal jugular to form the brachiocephalic vein. The clavicle and *subclavius* are anterior; the subclavian artery is posterolateral, separated by *scalenus anterior* and the phrenic nerve; and the first rib and pleura are inferior. The vein usually has a pair of valves about 2 cm from its end. Its tributaries are the external jugular, dorsal scapular and (sometimes) anterior jugular veins, and, occasionally, a small branch from the cephalic vein that ascends anterior to the clavicle. At its junction with the internal jugular vein, the left subclavian receives the thoracic duct, and the right subclavian vein receives the right lymphatic duct (see [Fig. 29.16](#)).

Brachial veins

The brachial veins flank the brachial artery, as *venae comitantes* with tributaries similar to the arterial branches; they join the axillary vein near the lower margin of *subscapularis*. The medial branch often joins the basilic vein before it becomes the axillary vein.

Superficial veins

The cephalic vein ascends anterior to the elbow, superficial to a groove between *brachioradialis* and *biceps brachii*, crosses superficial to the lateral cutaneous nerve of the forearm, and ascends lateral to *biceps* and between *pectoralis major* and *deltoid*, where it adjoins the deltoid branch of the thoraco-acromial artery ([Fig. 48.34](#)). Entering the infraclavicular fossa to pass posterior to the clavicular head of *pectoralis major*, it pierces the clavipectoral fascia, crosses the axillary artery and joins the axillary vein just below clavicular level. It may connect with the external jugular vein by a branch anterior to the clavicle. Sometimes, the median cubital vein is large, transferring most blood from the cephalic to the basilic vein, in which case the proximal cephalic vein is either absent or much diminished.

LYMPHATIC DRAINAGE

Lymphatics follow the usual pattern, in that superficial lymphatics follow the veins, and deep lymphatics follow the arteries. For details of the infraclavicular nodes, see [Fig. 48.45](#) and [page 834](#).

INNERVATION

BRACHIAL PLEXUS

For an overview of the brachial plexus, see [pages 781-786](#) and [Figure 46.7](#).

In the axilla, the lateral and posterior cords of the brachial plexus are anterior and lateral to the first part of the axillary artery, respectively, and the medial cord is behind the artery. The cords assume their appropriate relations around the second part of the artery, deep to *pectoralis major*. In the lower axilla, the cords divide into nerves that supply the upper limb ([Fig. 48.35](#)). Other than the medial root of the median

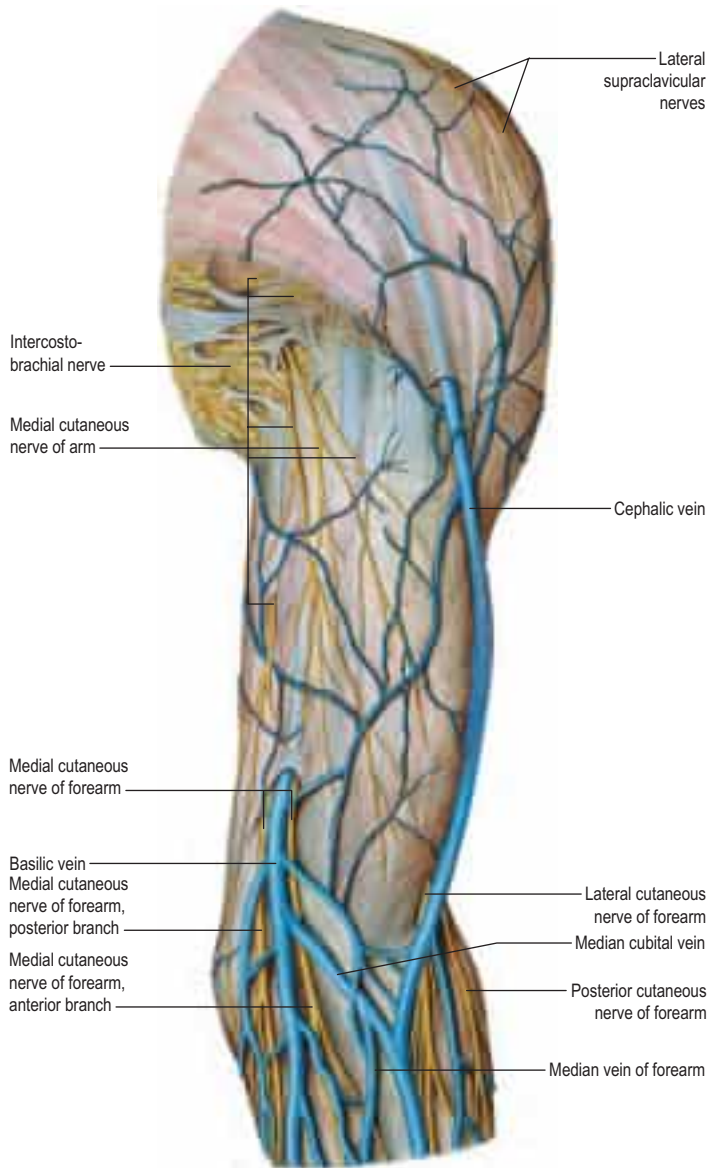


Fig. 48.34 Superficial veins and nerves, left arm, anterior (flexor) aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

nerve, these nerves are related to the third part of the artery in the same way as their cords are related to the second part, i.e. branches of the lateral cord are lateral to the artery, branches of the medial cord are medial, and branches of the posterior cord are posterior.

Suprascapular nerve

The suprascapular nerve is a large branch of the upper trunk (C5, 6) (see Figs 29.17, 48.35). It runs laterally, deep to trapezius and omohyoid, and enters the suprascapular fossa through the suprascapular notch inferior to the superior transverse scapular ligament. It runs deep to supraspinatus, supplies it, and curves round the lateral border of the spine of the scapula with the suprascapular artery to reach the infraspinous fossa, where it gives two branches to infraspinatus and articular rami to the shoulder and acromioclavicular joints. The suprascapular nerve rarely has a cutaneous branch. When present, it pierces deltoid close to the tip of the acromion and supplies the skin of the proximal third of the arm within the territory of the axillary nerve.

Lateral pectoral nerve

The lateral pectoral nerve (C5, 6) is larger than the medial, and may arise either from the anterior divisions of the upper and middle trunks, or by a single branch from the lateral cord, just above or deep to the clavicle. It crosses anterior to the axillary artery and vein, pierces the clavipectoral fascia and supplies the deep surface of pectoralis major. It sends a branch to the medial pectoral nerve, forming a loop in front of the first part of the axillary artery to supply some fibres to pectoralis minor.

Medial pectoral nerve

The medial pectoral nerve (C8, T1) branches from the medial cord while the latter lies posterior to the axillary artery and curves forwards between the axillary artery and vein. Anterior to the artery, it joins a ramus of the lateral pectoral nerve and enters the deep surface of pectoralis minor, which it supplies. Two or three branches pierce pectoralis minor and others may pass round its inferior border to end in pectoralis major.

Upper (superior) subscapular nerve

The superior subscapular nerve is smaller than the inferior. It arises from the posterior cord (C5, 6), enters subscapularis at a high level, and is frequently double.

Lower (inferior) subscapular nerve

The inferior subscapular nerve arises from the posterior cord (C5, 6). It supplies the lower part of subscapularis and ends in teres major, which is sometimes supplied by a separate branch.

Thoracodorsal nerve

The thoracodorsal nerve arises from the posterior cord (C6–8) between the subscapular nerves. It accompanies the subscapular artery along the posterior axillary wall and supplies latissimus dorsi, reaching its distal border (see Fig. 48.29).

Axillary nerve

The axillary nerve arises from the posterior cord (C5, 6) (see Figs 48.33, 48.35, 46.8). It is, at first, lateral to the radial nerve, posterior to the axillary artery and anterior to subscapularis. At the lower border of subscapularis, it curves back inferior to the humeroscapular articular capsule and, with the posterior circumflex humeral vessels, traverses a quadrangular space bounded above by subscapularis (anteriorly) and teres minor (posteriorly), below by teres major, medially by the long head of triceps, and laterally by the surgical neck of the humerus. It divides in the space into anterior and posterior branches. The anterior branch curves round the neck of the humerus with the posterior circumflex humeral vessels, deep to deltoid, to reach the anterior part of the muscle. At the level of the anterior border of the acromion, the nerve often consists of two or more thin filaments that supply the anterior deltoid; it gives off a few small cutaneous branches that pierce deltoid and ramify in the skin over its lower part. The posterior branch courses medially and posteriorly along the attachment of the lateral head of triceps, inferior to the glenoid rim. It usually lies medial to the anterior branch in the quadrangular space. It gives off the nerve to teres minor and the upper lateral cutaneous nerve of the arm at the lateral edge of the attachment of the long head of triceps. The nerve to teres minor enters the muscle on its inferior surface. The posterior branch frequently supplies the posterior aspect of deltoid, usually via a separate branch from the main stem, but occasionally from the superior lateral cutaneous nerve of the arm. However, the posterior part of deltoid has a more consistent supply from the anterior branch of the axillary nerve, which should be remembered when performing a posterior deltoid-splitting approach to the shoulder. The upper lateral cutaneous nerve of the arm pierces the deep fascia at the medial border of the posterior aspect of deltoid, and supplies the skin over the lower part of deltoid and the upper part of the long head of triceps. The posterior branch is intimately related to the inferior aspects of the glenoid and shoulder joint capsule. There is often an enlargement or pseudoganglion on the branch to teres minor. The axillary trunk supplies a branch to the shoulder joint below subscapularis.

Injuries to the shoulder girdle involving the accessory, long thoracic, axillary and suprascapular nerves



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Musculocutaneous nerve

The musculocutaneous nerve arises from the lateral cord (C5–7), opposite the lower border of pectoralis minor (Fig. 48.44; see Figs 46.13,

Diagnosis of injuries to the accessory, long thoracic, axillary and supra-scapular nerves is by no means straightforward; serious mistakes are most common in the diagnosis of severe injuries about the shoulder girdle. The delay before diagnosis is inexplicable despite the reliability of precise but elementary clinical examination. Matters are not helped by the persistence with which incorrect and discredited ideas about what muscles do continue to be taught. Many nerve injuries in this region are painful and, too often, a patient is passed to a pain clinic without a diagnosis.

Inferior scapulohumeral angle (ISHA)

Assessing the inferior scapulohumeral angle (ISHA) informs the clinician about function at the thoracoscaphal and shoulder joints (Fig. 48.36). The ISHA is subtended by the long axis of the humerus and the lateral border of the scapula; the apex of the angle is centred over the glenohumeral joint. Stiffness of the joints is detected by measuring *passive* range, which is normally between 170° and 180°. The excursion between the humerus and the scapula in a healthy adult is between 130° and 140°: that is, the difference between the ISHA at rest and in full elevation. It is important to remember that full elevation is possible

after successful thoracoscaphal arthrodesis in scapulohumeral dystrophy (Copeland and Howard 1978). This range of movement occurs at the shoulder joints, mainly the glenohumeral joint, with some involvement of the acromioclavicular and sternoclavicular joints. Successful glenohumeral arthrodesis enables a range of elevation of 30–40°; this is provided by the thoracoscaphal joint.

Accessory nerve

The integrity of the accessory nerve is fundamental to thoracoscaphal function and essential for scapulohumeral rhythm. The nerve is usually divided at the apex of the posterior triangle, usually by surgeons, and is a crippling injury. Typically, there is immediate pain, then dropping of the shoulder and a remarkable loss of function. The average ISHA is about 50°. The pain is characteristic: often severe, deep-seated, dull, constant and boring in nature, and felt around the scapula and the posterior aspect of the neck and shoulder. The posture of the scapula is also characteristic: because of the loss of the great suspensory muscle, trapezius, the scapula drops down and away from the spinal column (Fig. 48.37). There is often winging of the scapula, which is easily distinguished from that caused by paralysis of serratus anterior by the

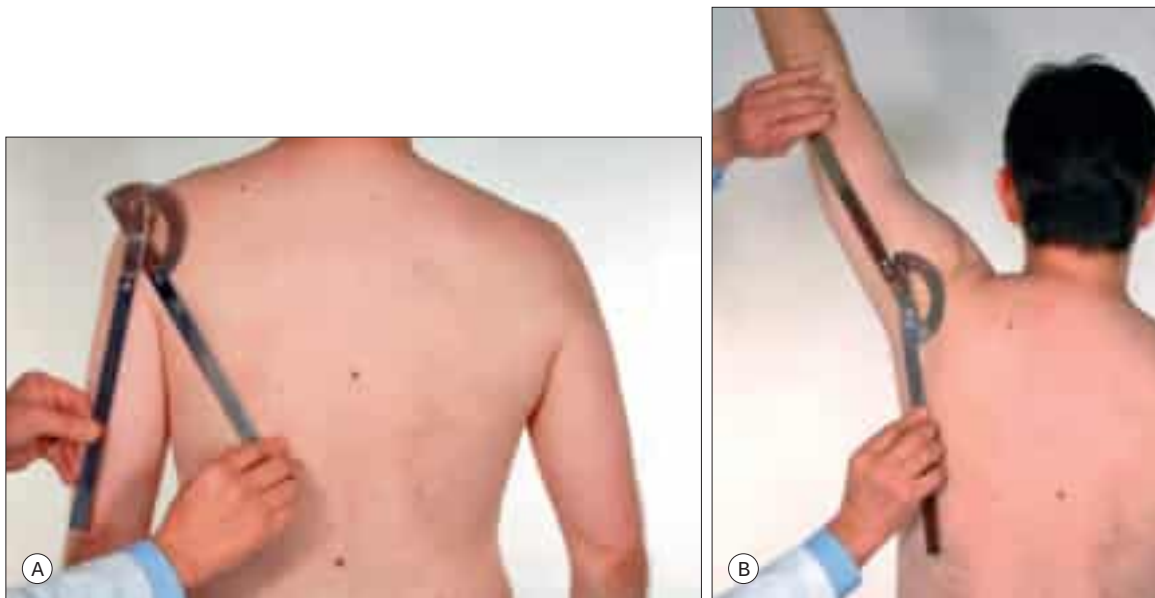


Fig. 48.36 The active inferior scapulohumeral angle (ISHA) in a normal shoulder is about 170°. **A**, At rest. **B**, Arm abducted. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

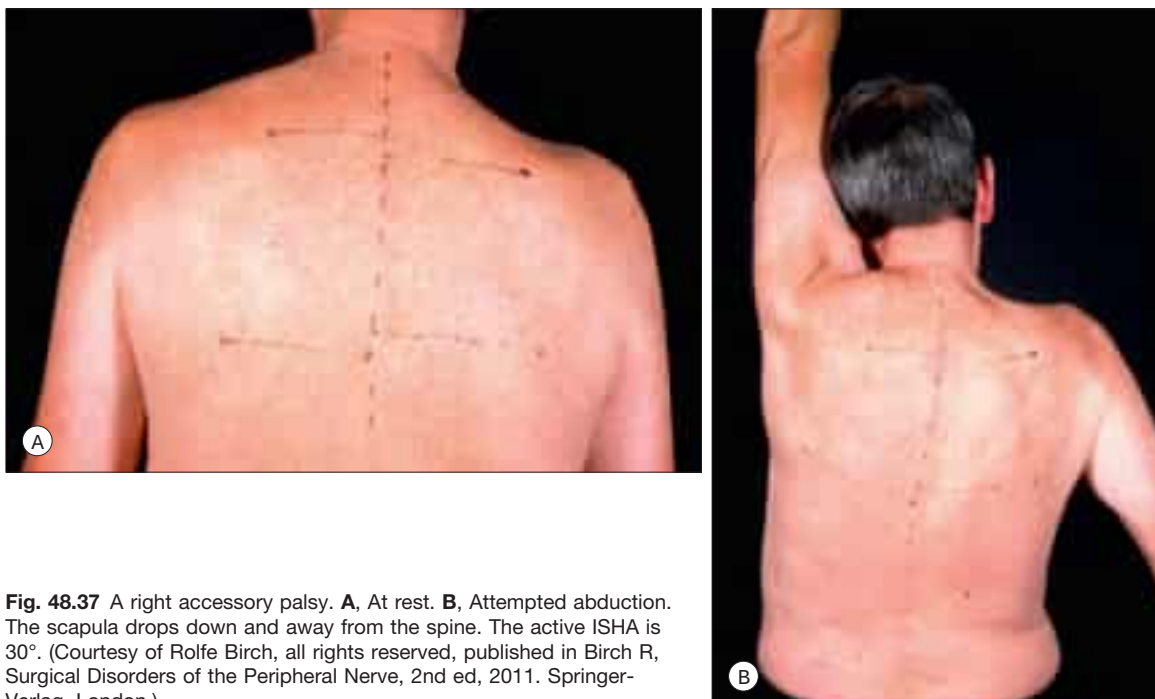


Fig. 48.37 A right accessory palsy. **A**, At rest. **B**, Attempted abduction. The scapula drops down and away from the spine. The active ISHA is 30°. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

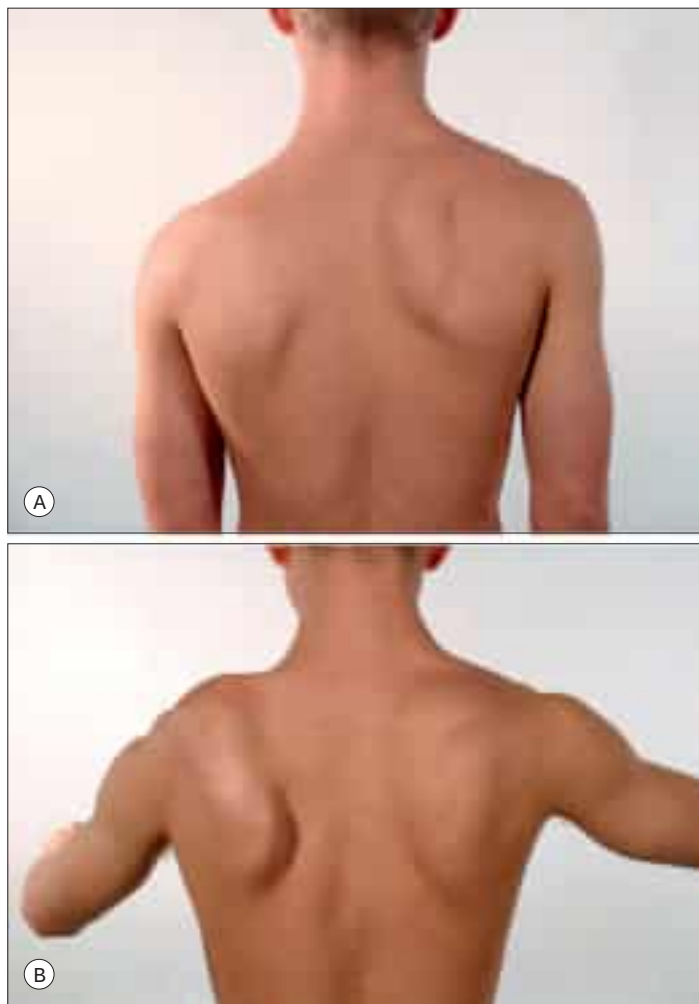


Fig. 48.38 A left accessory palsy. Scapular winging, without prominence of the lower fibres of trapezius, in a case where there is some early recovery into the upper fibres after repair of the accessory nerve. **A**, At rest, the scapula is displaced downwards and away from the spine. **B**, Arm abducted. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

posture of the scapula (Fig. 48.38). It is fortunate that repair of the nerve is usually successful, even when repair is performed as late as 4 years after injury, with early relief of pain long before there is re-innervation of muscle (Camp and Birch 2011). The nerve appears to be relatively immune to the harmful effects of delayed repair.

Long thoracic nerve (nerve to serratus anterior)

The long thoracic nerve is frequently injured during operations around the first rib, in the posterior triangle and in the axilla, and by misplaced intercostal drains. It is particularly susceptible to involvement in neuralgic amyotrophy. Serratus palsy is usual in preganglionic injury to C5, 6 and 7. Pain and loss of function are usual and are only slightly less than those seen after accessory nerve palsy. The signs are characteristic. The active ISHA is, on average, 130°; this is the only nerve lesion in which the active ISHA actually exceeds the total range of active abduction. The scapula is pulled upwards and towards the spine by the unopposed action of trapezius (Fig. 48.39). This action is increased by the weight of the limb, which contributes to the winging and is most notable in protraction.

Dislocation of the shoulder: the suprascapular and axillary nerves

Although most nerve lesions in uncomplicated anterior dislocations are benign, it is well to remember the risk to adjacent vessels, more so in high-energy transfer injuries, in older patients and in those with peripheral vascular disease. The consequences of delay in treatment are severe. It is very hard to examine muscles in a patient with a fracture dislocation of the shoulder. Sensory loss is inconsistent. Consider, for example, a patient after successful reduction of a dislocated shoulder, whose arm is supported in a sling. There are three simple tests: initiation of abduction shows that the suprascapular nerve is intact and that the rotator cuff is not ruptured; abnormal sensibility in the skin over deltoid indicates a lesion of the axillary nerve; and gentle extension of the shoulder on the part of the patient, supported as it is in a sling, means that the examiner can palpate activity in the posterior fibres of deltoid (Figs 48.40–48.41).

The diagnosis of nerve injury is by no means straightforward. All too often, rupture of the axillary nerve is not recognized until deltoid is atrophied, long past the ideal time for repair. The widely held view that deltoid is the abductor of the shoulder is erroneous. Deltoid provides power and stamina, and abduction is initiated and maintained by the suprascapular nerve, acting through supra- and infraspinatus, their tendons and attachments. Disruption of any component of this abductor mechanism interferes with or precludes abduction. Wynn Parry (1981) treated 145 patients in whom paralysis was confined to deltoid. The range of abduction was full or nearly so, and a system of training enabled most patients to return to full military duties. He wrote:

It must be stressed that these movements providing full abduction and elevation are not trick movements, in the sense usually associated with this word; all the muscles involved normally help to abduct the shoulder. The scapulohumeral rhythm is quite normal and in the later stages of re-education the patient does not even need to rotate the humerus externally to initiate movement.

Seddon (1975) was a little more cautious, finding abduction to about 155° was more usual. The ISHA is diminished by about 20° in uncomplicated ruptures of the axillary nerve when there is no stiffness of the shoulder.

Suprascapular nerve

The suprascapular nerve is essential for abduction and lateral rotation at the glenohumeral joint. When the nerve or the rotator cuff is torn, or both are injured, the ISHA is so reduced that the patient is unable to move the arm away from the body. At rest, the head of the humerus drops inferiorly and migrates proximally in attempted abduction (see Fig. 48.41). In isolated cuff tears, the ISHA opens to about 30° or even more. In isolated axillary palsy, the ISHA is sometimes normal, or, more often, reduced by about 20° (Figs 48.42, 48.43).

Axillary nerve

The axillary nerve innervates deltoid. Although a good range of elevation is usually maintained by an intact abductor mechanism, strength and stamina are greatly reduced. Early diagnosis of rupture of the axillary nerve can prove difficult; the nerve is deeply seated, which means that Tinell's sign is rarely detectable. Active movement is restricted by pain and by the restraint of any sling. The loss of sensation varies and there may not be complete anaesthesia. Patients with an intact suprascapular nerve and an intact rotator cuff are able to initiate abduction and many can fully elevate the arm, or nearly so. The power of forward flexion and of abduction is reduced to, at most, 30–40% of normal, whereas extension, with the shoulder at 90° of abduction, is 10% or less. One useful test is to palpate the posterior third of deltoid with the fingers and ask the patient to move the elbow gently backwards. It should be possible to detect activity in the muscle when the arm is supported in a sling or by the examiner's arm.

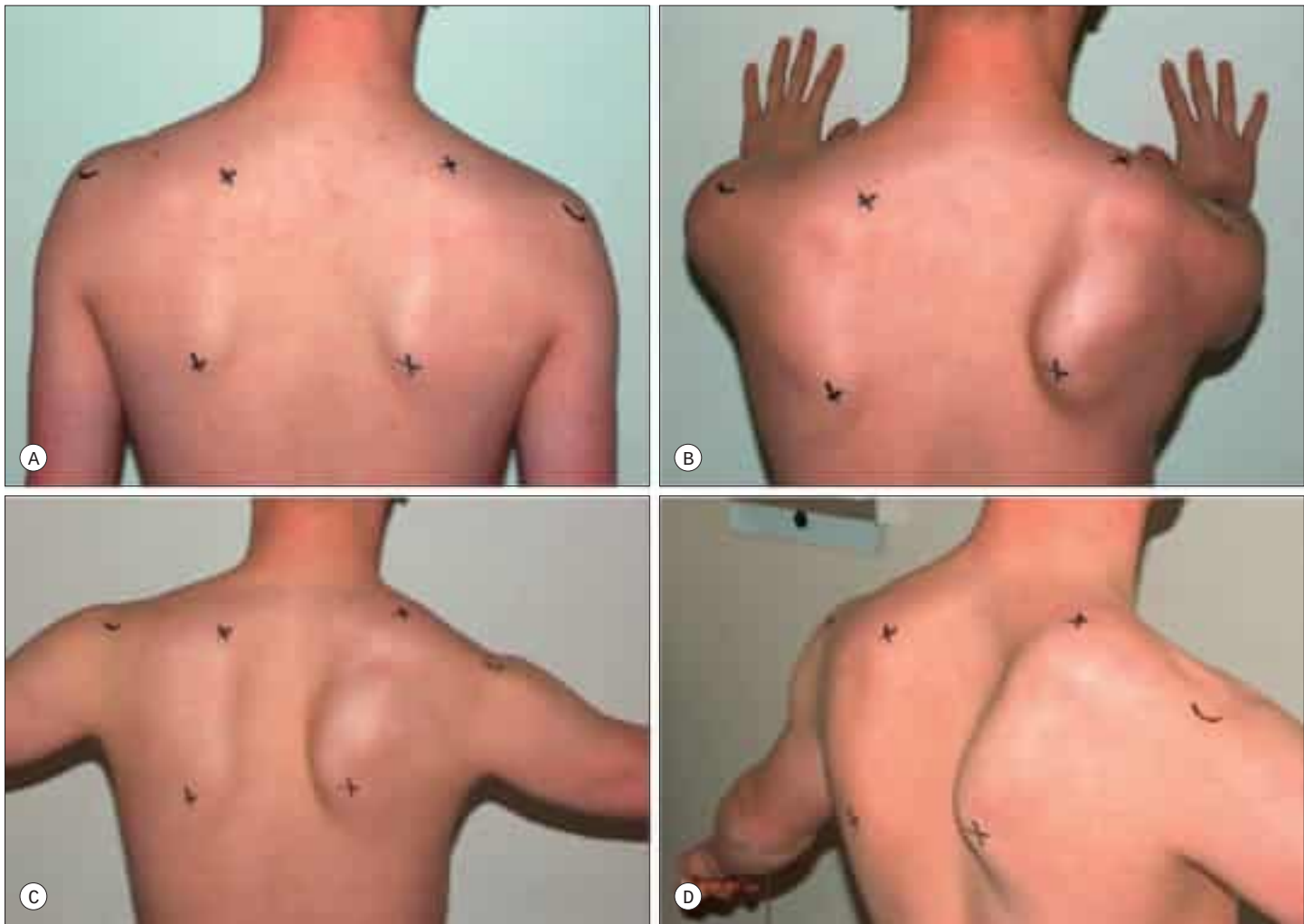


Fig. 48.39 Scapular winging in long thoracic palsy. This is easily distinguishable from the winging provoked by accessory palsy by the position of the scapula, which is drawn upwards and towards the spine by the unopposed action of trapezius, levator scapulae and the rhomboids. **A**, Arms at rest. **B**, Hands pressed against the wall. **C**, Attempted abduction. **D**, Oblique view. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 48.40 Initiation of abduction, with opening of the active inferior scapulohumeral angle (ISHA), was the first sign of recovery into supraspinatus after repair of the suprascapular nerve. The lesion of the axillary nerve was irreparable. **A**, At rest. **B**, Attempted abduction. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 48.41 Rupture of the rotator cuff, with lesions of the suprascapular and axillary nerves from fracture/dislocation of the shoulder. **A**, The range of elevation at the right shoulder in a 74-year-old man, in whom there was clear evidence of recovery for both of the nerves. **B–C**, This shipwright held on to a cable to rescue a man from the River Thames. The weight of the man and the force of the current were such that he felt the muscles tearing in his right shoulder; then he felt the head of the humerus pulling out from the socket and his arm went dead. Both nerves were recovering. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

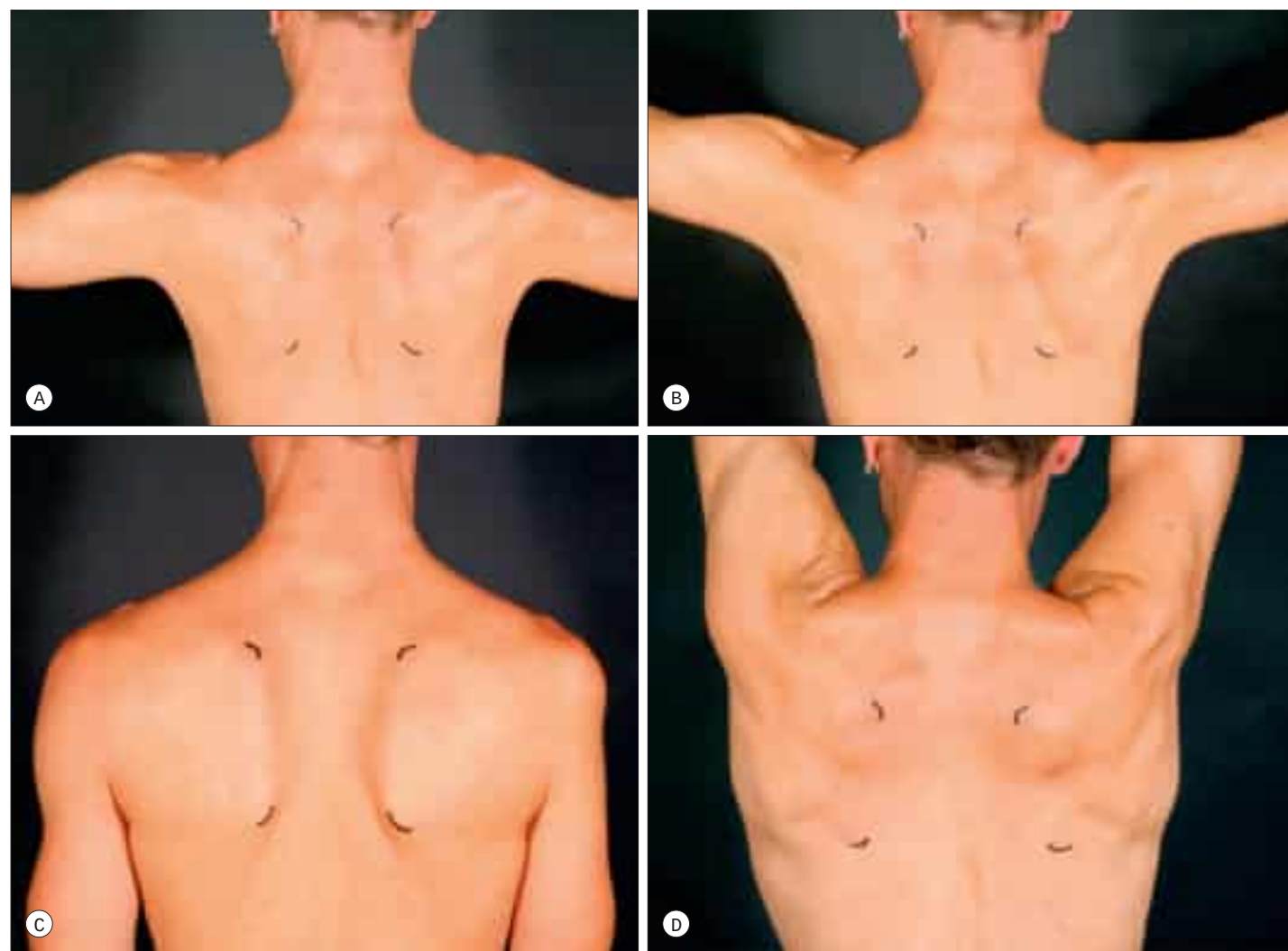


Fig. 48.42 Movement of the scapula in a case of rupture of the right axillary nerve. **D**, The active inferior scapulohumeral angle (ISHA) on the side of the injury is reduced by 20°, showing that 20° of the range of elevation is provided by extra movement at the thoracoscapular joint. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

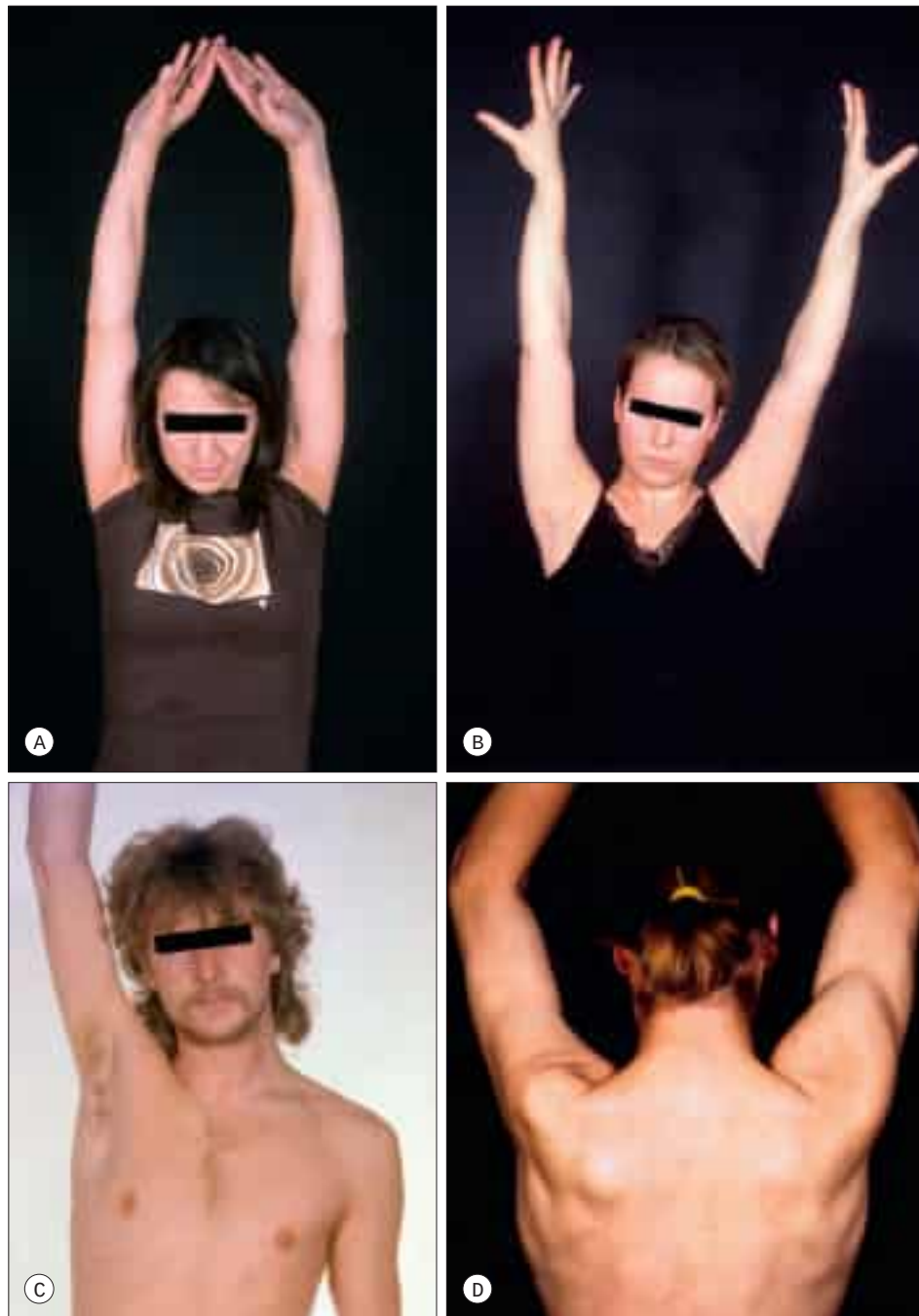


Fig. 48.43 A–D, Elevation of the upper limb in full medial rotation by supraspinatus in the absence of deltoid in four cases of proven rupture of the axillary nerve. Note the activity in the clavicular head of pectoralis major in C. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

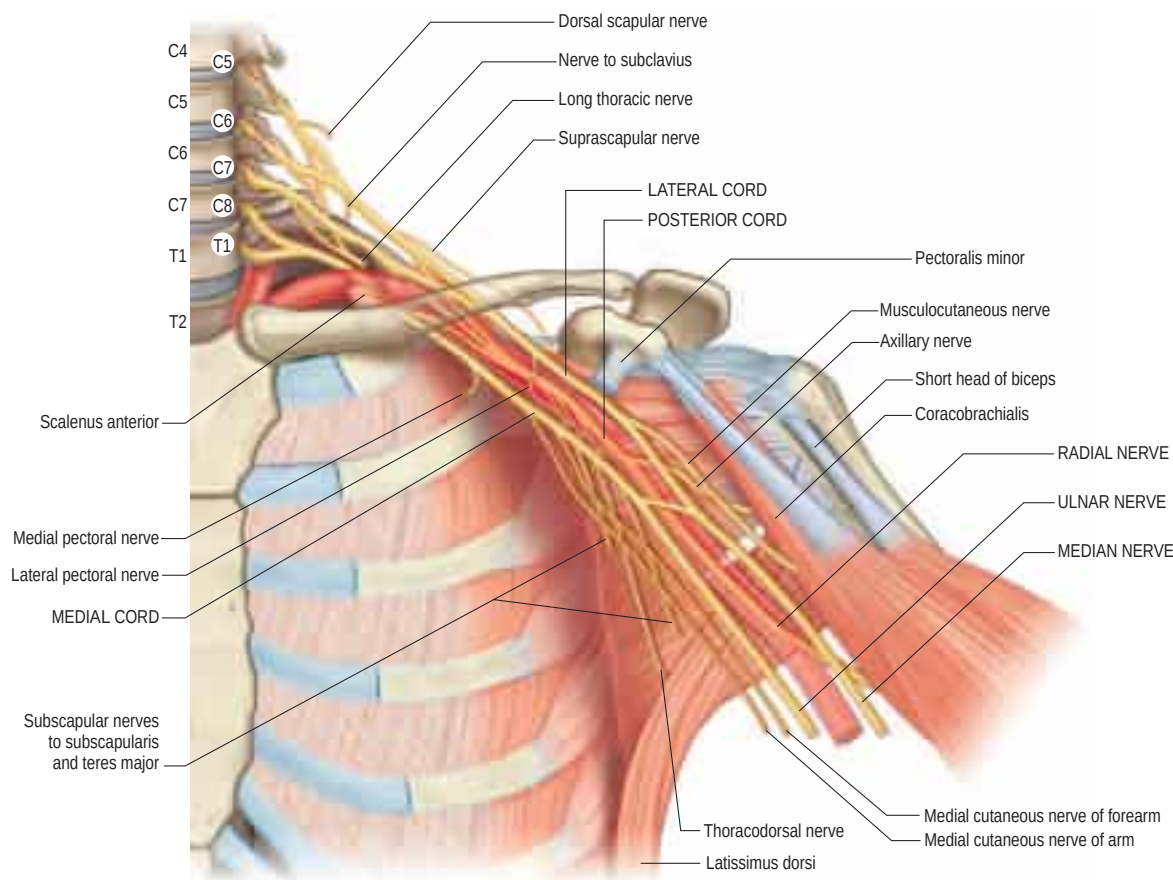


Fig. 48.35 The brachial plexus and its major branches. The axillary artery, shown in red outline, runs anterior to the radial nerve. The vertebral levels are labelled C4–T2; the spinal nerves are labelled C5–T1 (white circles).

48.45). It pierces coracobrachialis a variable distance below the tip of the coracoid process and then descends laterally between biceps brachii and brachialis to the lateral side of the arm. The first branch, to coracobrachialis, may pass into the muscle no more than 2 cm distal to the tip of the coracoid; fibres from C7 may branch directly from the lateral cord. Branches to biceps brachii and brachialis leave after the musculocutaneous nerve has pierced coracobrachialis. There are usually two branches to brachialis; the lower branch also supplies the anterior aspect of the elbow joint. The nerve to biceps brachii is consistently accompanied by a vascular pedicle. The nerve now inclines lateral and deep, into the plane between brachialis posteriorly and biceps anteriorly, where it supplies a small branch to the humerus that enters the shaft with the nutrient artery. Below the branches to brachialis, the musculocutaneous nerve continues as the lateral cutaneous nerve of the forearm.

The musculocutaneous nerve has frequent variations. It may run behind coracobrachialis or adhere for some distance to the median nerve and pass behind biceps brachii. Some fibres of the median nerve may run in the musculocutaneous nerve, leaving it to join their proper trunk; less frequently, the reverse occurs, and the median nerve sends a branch to the musculocutaneous nerve. Occasionally, it supplies pronator teres and may replace radial branches to the dorsal surface of the thumb.

Medial cutaneous nerve of the arm

The medial cutaneous nerve of the arm (C8, T1) is the smallest and most medial branch of the brachial plexus, arising from the medial cord (see Fig. 48.35). It crosses the axilla, either anterior or posterior to the axillary vein, then passes medial to the axillary vein, communicates with the intercostobrachial nerve, and descends medial to the brachial artery and basilic vein. It pierces the deep fascia at the midpoint of the upper arm to supply the skin over the medial aspect of the distal third of the upper arm (see Fig. 48.34). Sometimes, the medial cutaneous nerve of the arm and the intercostobrachial nerve are connected in a plexiform manner in the axilla. The intercostobrachial nerve may be large and reinforced by part of the lateral cutaneous branch of the third intercostal nerve. It then replaces the medial cutaneous nerve of the arm and receives a connection representing the latter from the brachial plexus (occasionally, this connection is absent). It connects with the posterior branch of the medial cutaneous nerve of the forearm.

Medial cutaneous nerve of the forearm

The medial cutaneous nerve of the forearm comes from the medial cord (T1) (see Fig. 48.34). It lies initially between the axillary artery and vein, and gives off a ramus that pierces the deep fascia to supply the skin over the biceps, almost to the elbow. The nerve descends medial to the brachial artery, pierces the deep fascia with the basilic vein midway in the arm, and divides into anterior and posterior branches. Its further course is described on page 860.

Median nerve

The median nerve has two roots from the lateral (C5, 6, 7) and medial (C8, T1) cords, which embrace the third part of the axillary artery and unite anterior or lateral to it (see Figs 48.29, 48.44, 46.7, 46.10). Some fibres from C7 often leave the lateral root in the lower part of the axilla; they pass distomedially posterior to the medial root, and usually anterior to the axillary artery, to join the ulnar nerve. Clinically, they are believed to be mainly motor and to supply flexor carpi ulnaris. If the lateral root is small, the musculocutaneous flexor (C5, 6, 7) connects with the median nerve in the arm. It gives off vascular branches to the brachial artery. The further course of the median nerve is described on pages 858 and 891.

Ulnar nerve

The ulnar nerve arises from the medial cord (C8, T1) but often receives fibres from the ventral ramus of C7. It usually has no branches in the arm, but occasionally may give a branch to the medial head of triceps (see Figs 48.29, 48.44, 46.12). The nerve runs distally through the axilla, medial to the axillary artery and between it and the vein, and continues distally medial to the brachial artery as far as the midarm. Here, it pierces the medial intermuscular septum, inclining medially as it descends anterior to the medial head of triceps to the interval between the medial epicondyle and the olecranon. The further course of the ulnar nerve is described on pages 859 and 893.

Radial nerve

The radial nerve is the largest branch of the brachial plexus. It arises from the posterior cord (C5, 6, 7, 8, occasionally T1) (see Figs 48.33,

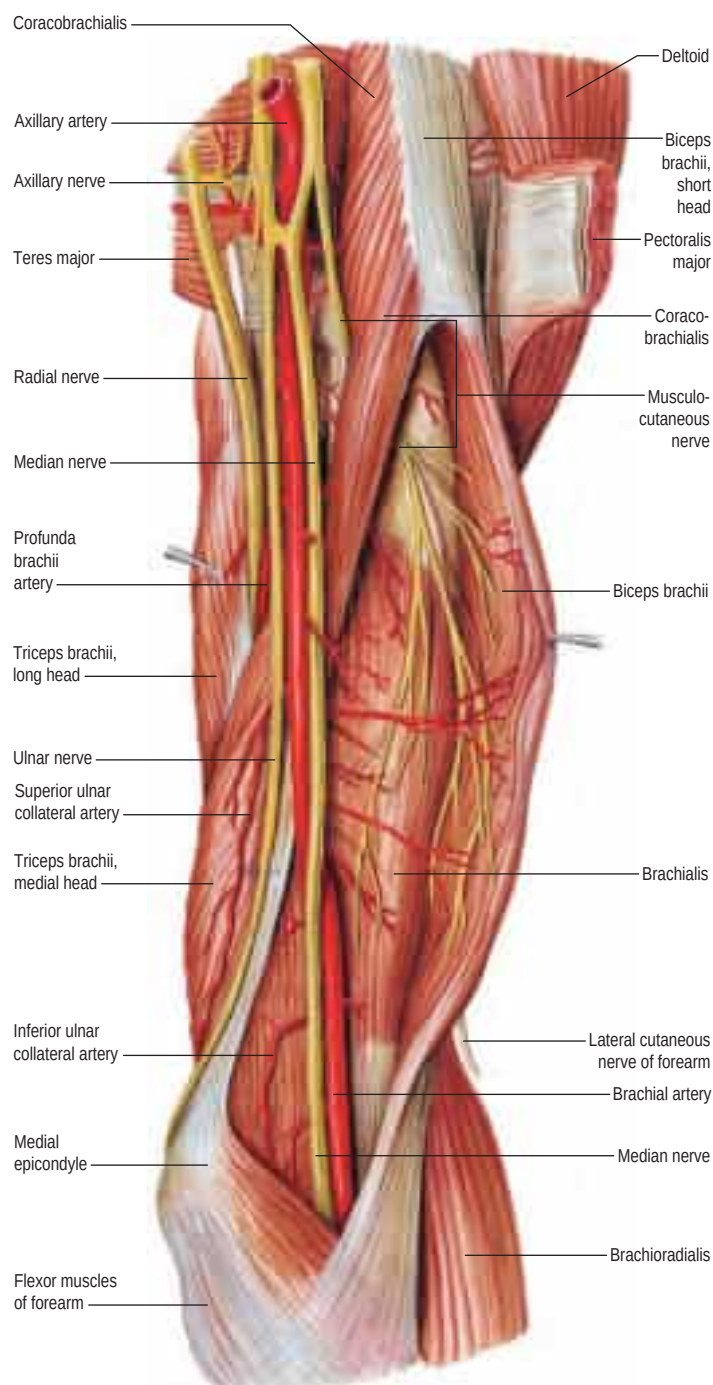


Fig. 48.44 The left upper arm dissected to show the musculocutaneous nerve. Note also the changing relationship of the brachial artery and median nerve. Anterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

48.45, 46.9) and descends behind the third part of the axillary artery and the upper part of the brachial artery, anterior to subscapularis above and the tendons of latissimus dorsi and teres major below, giving branches to the medial head of triceps at this level (see Fig. 48.33). With the profunda brachii artery, it inclines dorsally, passing through the lower triangular space below the lower border of teres major, between the long head of triceps and the humerus. Here, it supplies the long head of triceps and gives rise to the posterior cutaneous nerve of the arm, which supplies the skin along the posterior surface of the upper arm. It then spirals obliquely across the back of the humerus, lying posterior to the uppermost fibres of the medial head of triceps, which separate the nerve from the bone in the first part of the spiral groove. It gives off a muscular branch to the lateral head of triceps and a branch that passes through the medial head of triceps to innervate anconeus. On reaching the lateral side of the humerus, the radial nerve pierces the lateral intermuscular septum to enter the anterior compartment of the arm. The further course of the radial nerve is described on pages 860 and 894.

The branches of the radial nerve in the upper arm are: muscular, cutaneous, articular and superficial terminal and posterior interosseous.

Muscular branches

Muscular branches supply triceps, anconeus, brachioradialis, extensor carpi radialis longus and brachialis in medial, posterior and lateral groups. Medial muscular branches arise from the radial nerve on the medial side of the arm. They supply the medial and long heads of triceps. The branch to the medial head is a long, slender filament that lies close to the ulnar nerve as far as the distal third of the arm, and is often termed the ulnar collateral nerve. A large posterior muscular branch arises from the nerve as it lies in the humeral groove. It divides to supply the medial and lateral heads of triceps and anconeus. The branch to anconeus is a long nerve that descends in the medial head of triceps and partially supplies it; the nerve is accompanied by the middle collateral branch of the profunda brachii artery and passes behind the elbow joint to end in anconeus. Lateral muscular branches arise in front of the lateral intermuscular septum and supply the lateral part of brachialis, brachioradialis and extensor carpi radialis longus.

Cutaneous branches

Cutaneous branches are the posterior and lower lateral cutaneous nerves of the arm and the posterior cutaneous nerve of the forearm.

Posterior cutaneous nerve of the arm The small posterior cutaneous nerve of the arm arises in the axilla and passes medially to supply the skin on the dorsal surface of the arm nearly as far as the olecranon. It crosses posterior to, and communicates with, the intercostobrachial nerve.

Lower lateral cutaneous nerve of the arm The lower lateral cutaneous nerve of the arm perforates the lateral head of triceps distal to the deltoid tuberosity, passes to the front of the elbow close to the cephalic vein, and supplies the skin of the lateral part of the lower half of the arm.

Posterior cutaneous nerve of the forearm The posterior cutaneous nerve of the forearm arises with the lower lateral cutaneous nerve of the arm. Perforating the lateral head of triceps, it descends first lateral in the arm, then along the dorsum of the forearm to the wrist, supplying the skin in its course and joining, near its end, with dorsal branches of the lateral cutaneous nerve of the forearm.

AXILLA

BOUNDARIES

The axilla is a pyramidal region between the upper thoracic wall and the arm (Figs 48.45–48.46; see Fig. 48.35). Its blunt apex continues into the root of the neck (cervico-axillary canal) between the external border of the first rib, superior border of the scapula, posterior surface of the clavicle, and the medial aspect of the coracoid process. Its triangular base, which is virtual, can be imagined as facing downwards; it is broad at the chest and narrow at the arm, and corresponds to the skin and a thick layer of axillary fascia between the inferior borders of pectoralis major anteriorly and latissimus dorsi posteriorly. It is convex upwards, conforming to the concavity of the armpit.

The anterior wall is formed by pectorales major and minor, the former covering the whole wall, and the latter its intermediate cranial part, with a clear fascial plane between the two muscles. The interval between the upper border of pectoralis minor and clavicle is occupied by the clavipectoral fascia. The posterior wall is formed by subscapularis superiorly and teres major and latissimus dorsi inferiorly (Fig. 48.47). The medial 'wall' is convex laterally and is composed of the first four ribs and their associated intercostal muscles, together with the upper part of serratus anterior. The anterior and posterior walls converge laterally; the 'wall' is narrow and consists of the humeral intertubercular sulcus. The lateral angle lodges coracobrachialis and biceps.

CONTENTS

The axilla contains the axillary vessels, the infraclavicular part of the brachial plexus and its branches, lateral branches of some intercostal nerves, many lymph nodes and vessels, loose adipose areolar tissue and, in many instances, the 'axillary tail' of the breast. The axillary vessels and brachial plexus run from the apex to the base along the lateral wall, nearer to the anterior wall; the axillary vein is anteromedial to the artery.

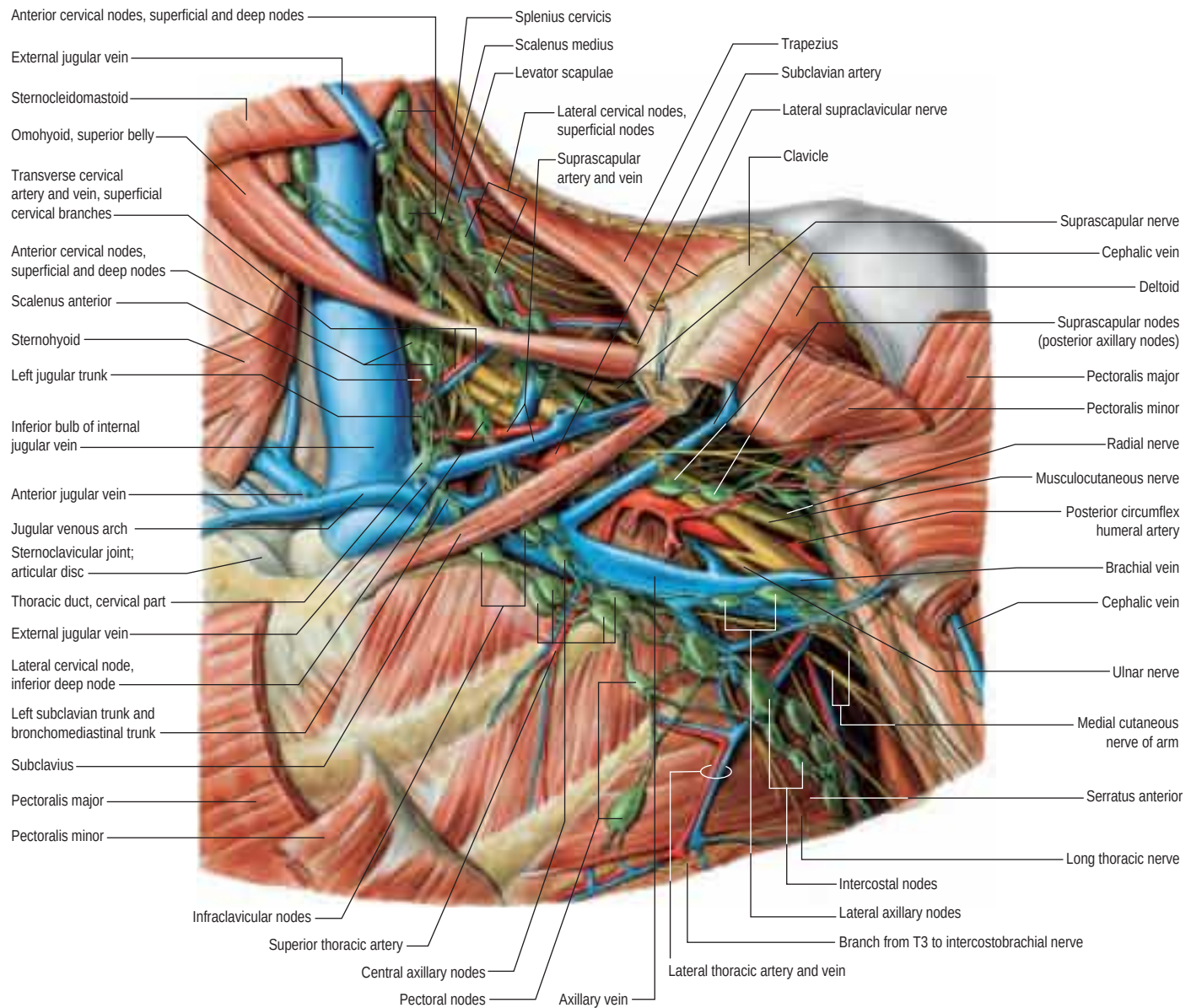


Fig. 48.45 The left axillary fossa and the deep aspect of the thoracic inlet, anterior aspect. The clavicle has been partially removed and the pectoral muscles have been reflected. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

The obliquity of the upper ribs means that the neurovascular bundle, after it emerges from behind the clavicle, crosses the first intercostal space; its relations are, therefore, different at upper and lower levels. Thoracic branches of the axillary artery are in contact with the pectoral muscles; the lateral thoracic artery reaches the thoracic wall along the lateral margin of pectoralis minor. Subscapular vessels descend on the posterior wall at the lower margin of subscapularis. The subscapular and thoracodorsal nerves cross the anterior surface of latissimus dorsi at different inclinations. Circumflex scapular vessels wind round the lateral border of the scapula; posterior circumflex humeral vessels and the axillary nerve curve back and laterally around the surgical neck of the humerus (see Figs 48.47, 48.31).

No large vessel lies on the medial 'wall', which is crossed proximally only by small branches of the superior thoracic artery. The long thoracic nerve descends on serratus anterior and the intercostobrachial nerve perforates the upper anterior part of this wall, crossing the axilla to its lateral 'wall'.

AXILLARY LYMPH NODES

There are between 20 and 30 axillary nodes, which may be divided into five not wholly distinct groups, namely: lateral, anterior (pectoral), posterior (subscapular), central and apical (see Fig. 53.24). Four of the groups are intermediary; only the apical group is terminal. Collectively, they drain the entire upper limb, breast and trunk above the umbilicus.

The lateral group of four to six nodes is posteromedial to the axillary vein; its afferents drain the whole limb, except the vessels that accompany the cephalic vein (Fig. 48.48, see Fig. 53.24). Efferent vessels pass

partly to the central and apical axillary groups, and partly to the inferior deep cervical nodes. The anterior group of four or five nodes spreads along the inferior border of pectoralis minor near the lateral thoracic vessels. Their afferents drain the skin and muscles of the supra-umbilical anterolateral body wall and breast, and efferents pass partly to the central and partly to the apical axillary nodes. The posterior group of six or seven nodes lie on the inferior margin of the posterior axillary wall, along the subscapular vessels. Their afferents drain the skin and superficial muscles of the inferior posterior region of the neck and the dorsal aspect of the trunk down to the iliac crest; efferents pass to the apical and central axillary nodes. A central group of three or four large nodes embedded in axillary fat receives afferents from all preceding groups, and their efferents drain to the apical nodes. An apical group of six to twelve nodes is partly posterior to the superior part of pectoralis minor and partly above its superior border, extending to the apex of the axilla medial to the axillary vein. The only direct territorial afferents are those that accompany the cephalic vein and/or some that drain the upper peripheral region of the breast; the group drains all the other axillary nodes. Their efferents unite as the jugulosubclavian venous junction, the subclavian vein, the jugular lymphatic trunk or (occasionally) to a right lymphatic duct; the left trunk usually ends in the thoracic duct (see Fig. 29.16). A few efferents from apical nodes usually reach the inferior deep cervical nodes.

One or two infraclavicular nodes appear beside the cephalic vein in the groove between pectoralis major and deltoid, just inferior to the clavicle. Their efferents pass through the clavipectoral fascia to the apical axillary nodes. Occasionally, some pass anterior to the clavicle to the inferior deep cervical (supraclavicular) nodes.

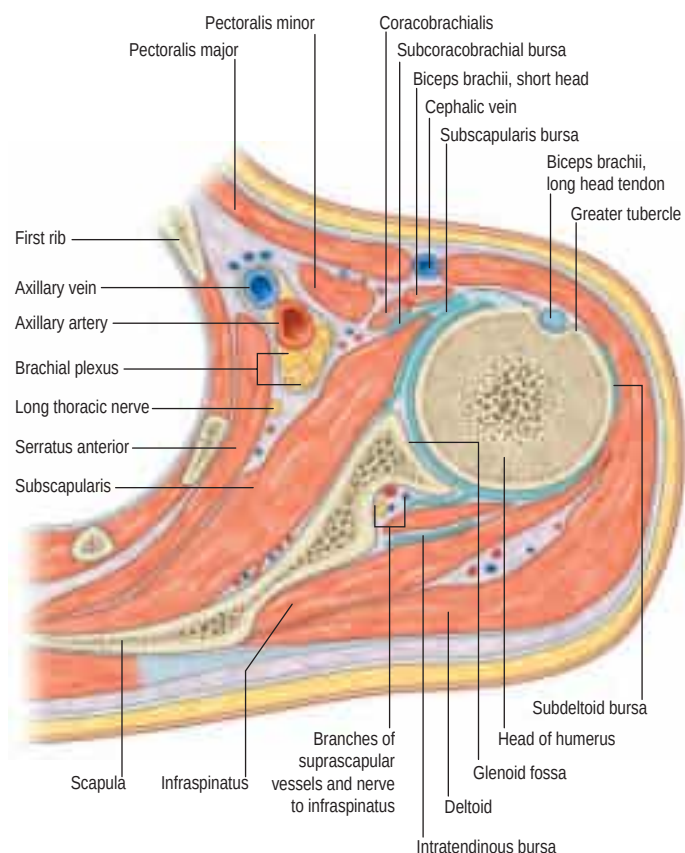


Fig. 48.46 A transverse section through the left shoulder joint and axilla, viewed from below. Compare with [Figure 48.23A](#).

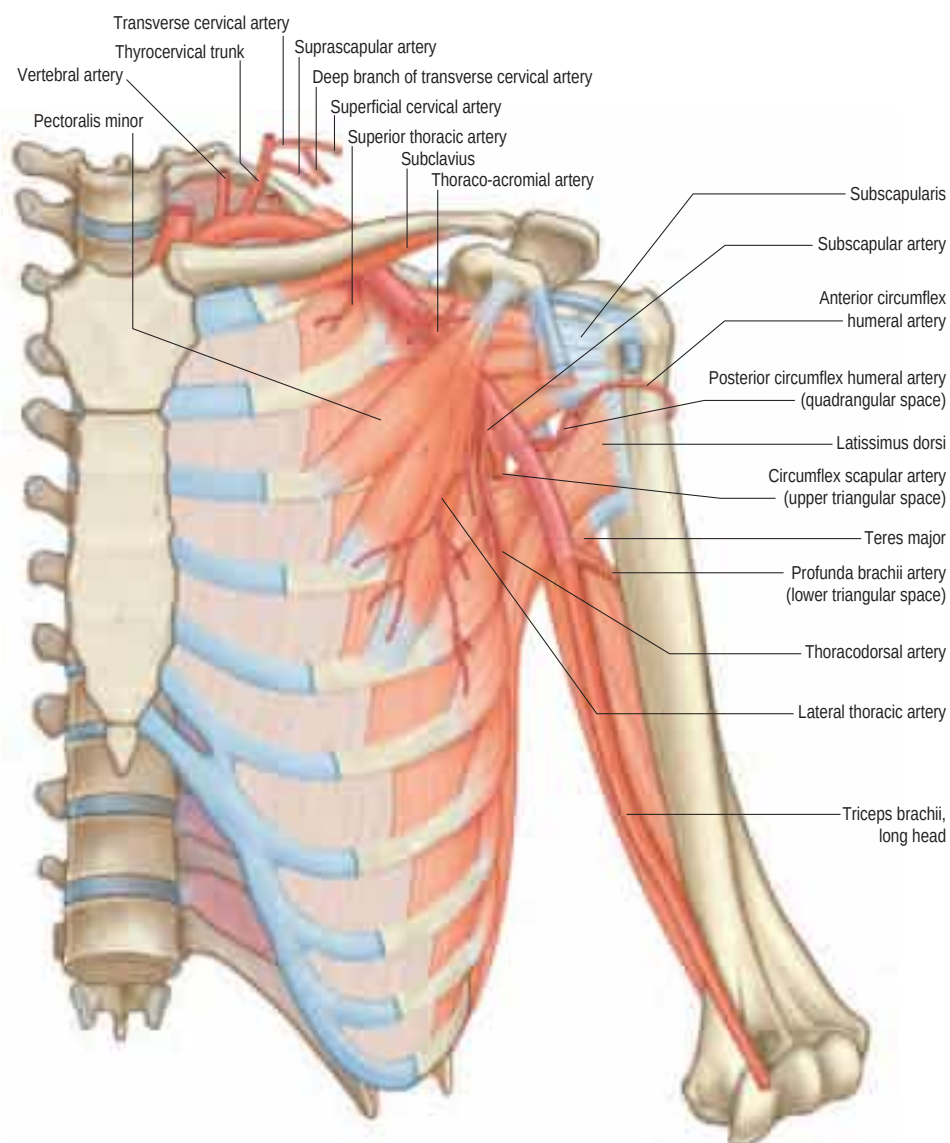


Fig. 48.47 The left axillary artery and its branches. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Bonus e-book images

Fig. 48.36 The active inferior scapulohumeral angle (ISHA) in a normal shoulder is about 170°.

Fig. 48.37 A right accessory palsy.

Fig. 48.38 A left accessory palsy.

Fig. 48.39 Scapular winging in long thoracic palsy.

Fig. 48.40 Initiation of abduction, with opening of the active inferior scapulohumeral angle (ISHA), was the first sign of recovery into supraspinatus after repair of the suprascapular nerve.

Fig. 48.41 Rupture of the rotator cuff, with lesions of the suprascapular and axillary nerves from fracture/dislocation of the shoulder.

Fig. 48.42 Movement of the scapula in a case of rupture of the right axillary nerve.

Fig. 48.43 Elevation of the upper limb in full medial rotation by supraspinatus in the absence of deltoid in four cases of proven rupture of the axillary nerve.

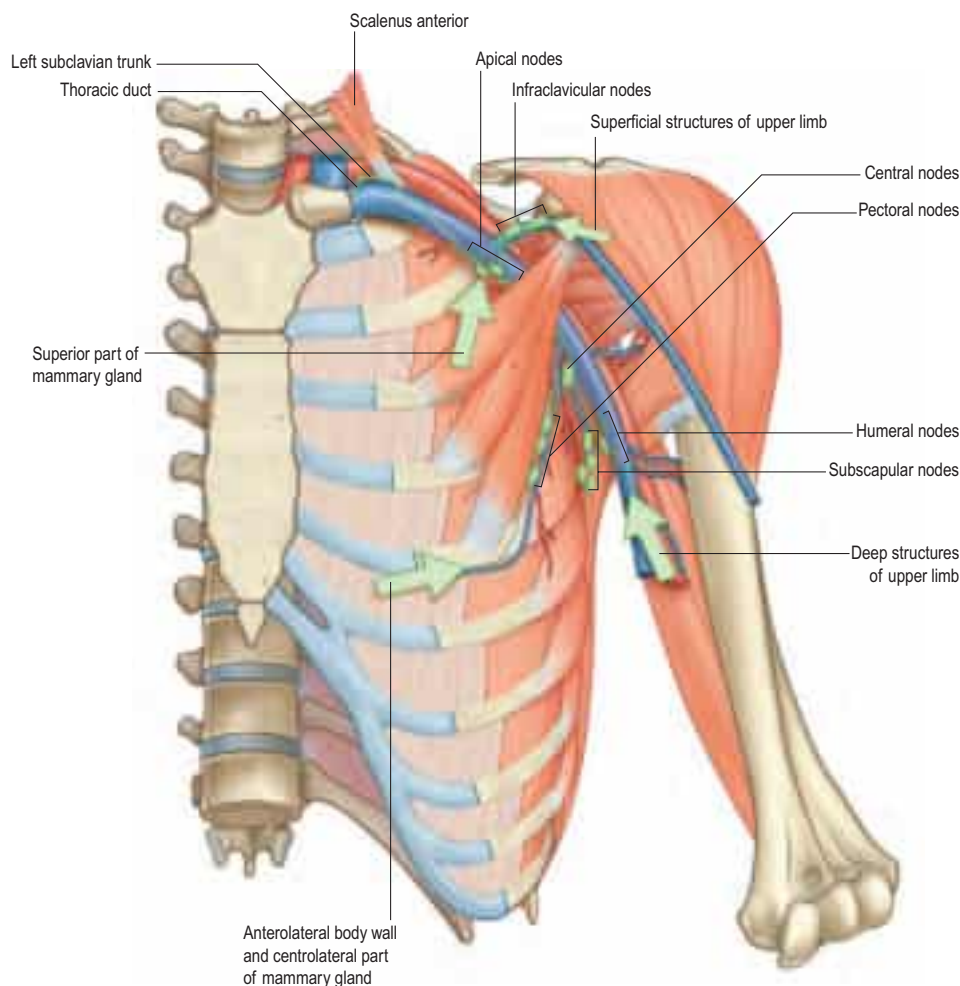


Fig. 48.48 The lymph nodes and vessels associated with the left axilla. Arrows indicate the direction of flow. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

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